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Review

Alzheimer's disease diagnosis from single and multimodal data using machine and deep learning models: Achievements and future directions

Ahmed Elazab [a,b], Changmiao Wang [c], Mohammed Abdelaziz [a], Jian Zhang [d], Jason Gu [e], Juan M. Gorriz [f], Yudong Zhang [g], Chunqi Chang [a,*]

[a] School of Biomedical Engineering, Shenzhen University Medical School, Shenzhen University, Shenzhen 518060, China
[b] Computer Science Department, Misr Higher Institute for Commerce and Computers, Mansoura 35516, Egypt
[c] Medical Big Data Lab, Shenzhen Research Institute of Big Data, Shenzhen 518172, China
[d] School of Pharmacy, Shenzhen University Medical School, Shenzhen University, Shenzhen, Guangdong 518055, China
[e] Department of Electrical & Computer Engineering, Dalhousie University, Halifax, NS B3H4R2, Canada
[f] Data Science and Computational Intelligence Institute, University of Granada, Spain
[g] School of Computing and Mathematical Sciences, University of Leicester, Leicester, UK

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ABSTRACT

Alzheimer's Disease (AD) is the most prevalent and rapidly progressing neurodegenerative disorder among the elderly and is a leading cause of dementia. AD results in significant suffering for patients, impairing their ability to perform simple daily tasks and ultimately leading to death. Despite being identified over 100 years ago, AD is still not well understood, and considerable research efforts are ongoing to achieve early diagnosis, effective treatment, and prognosis. Unfortunately, no effective drug currently exists to cure or reverse the disease. Instead, current efforts focus on the early prediction of disease onset to provide timely treatments and delay its progression, particularly during the prodromal stage known as mild cognitive impairment (MCI). This requires measurable indicators (biomarkers) to assess subjects through brain imaging, cerebrospinal fluid analysis, genetic analysis, and/or blood testing. Recent advances in machine learning and deep learning techniques have significantly improved the efficiency of computer diagnostic tools in identifying disease-related biomarkers from single or multimodal data. In this paper, we first present the main framework for building disease diagnosis models, providing a detailed description of each step, followed by an overview of the different machine learning and deep learning models used in each phase. Next, we discuss the various criteria for constructing AD diagnosis models and explore various multimodal data fusion techniques. We then review the latest studies employing both machine learning and deep learning methods with single-modal or multi-modal biomarkers. Finally, we discuss existing challenges, potential solutions, and future research directions to further enhance the diagnostic performance of AD.

1. Introduction

Alzheimer's Disease (AD) is a quite prevalent irreversible neurodegenerative disorder among elderly and is considered the main cause of dementia. Consequences for AD are very much affecting patients as well as caregivers and society. It commonly causes several sufferings for patients and deficiencies in performing simple daily tasks and eventually leads to death. Advances in health care technologies, made the pace of population ageing faster than before. According to the WHO, the world's population over 60 years will increase from 12 % to 22 %, out of them 80 % in low- and middle-income countries.[1] In 2020, about 50 million subjects were diagnosed with dementia and this number is anticipated to reach 152 million by 2050 (Vitali et al., 2021). Most elderly above 65 years old are likely diagnosed with AD, subjective memory concern (SMC), or mild cognitive impairment (MCI). SMC refers to the stage where individuals notice memory issues but these issues are not yet severe enough to be detected by objective cognitive tests. MCI, on the other hand, represents a more advanced stage where cognitive decline is

* Corresponding author.
E-mail addresses: ahmedelazab@szu.edu.cn (A. Elazab), wangcm@sribd.cn (C. Wang), Jason.gu@dal.ca (J. Gu), gorriz@ugr.es (J.M. Gorriz), yudong.zhang@le.ac.uk (Y. Zhang), cqchang@szu.edu.cn (C. Chang).

[1] https://www.who.int/news-room/fact-sheets/detail/ageing-and-health.

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measurable and noticeable but not severe enough to interfere significantly with daily life and it can be either progressive (pMCI) or stable (sMCI). The estimated yearly conversion of MCI to AD varies in literature. A study reported that about 28.9 %–33.6 % (Mitchell and Shiri-Feshki, 2009) of MCI eventually converts to AD which is relatively high. Hence, early prediction of MCI and timely treatment can reduce the progression risk to AD. Unfortunately, to date, there is no effective treatment to revert AD and MCI except some drugs that treat the symptoms, yet they come with some side effects (Alzheimer's disease facts and figures, 2022). Additionally, the AD is still poorly understood and there is no clear consent on disease causes among researchers. Major efforts are thus focused on delaying the disease onset or early predicting the disease status to provide timely treatments and delay its progression as long as possible. Specially, the caregiving costs after the disease onset tremendously increase which are unaffordable for many patients, particularly in middle- and low-level-income countries. Thus, collaborative efforts are being exerted to detect the early-onset of the disease and distinguish it from normal cognitive (NC) using machine learning (ML) and deep learning (DL) methods to analyze patients' data.

In clinical practice, there are different techniques to assess AD and provide close view of the brain status. The most common technique is brain imaging as it can provide a clear overview of the brain, come with rich information (e.g., brain volume and shape), and highlight the abnormal brain tissues/regions. Indeed, non-invasive neuroimaging techniques remain the gold standard in AD diagnosis and are widely available in clinics which makes it accounting most of studies in the literature. Neuroimaging can be broadly classified as structural/anatomical, functional, or molecular. The structural neuroimaging includes magnetic resonance imaging (MRI) and computed tomography (CT) which provide anatomical information of brain tissue volume and shape. This modality comes with high spatial resolution and can greatly distinguish the changes of anatomical brain regions related to a specific disease and has been widely used to study different neurodegenerative diseases. On the other hand, functional neuroimaging reveals the activities of brain cells/regions based on the levels of sugar or oxygen consumption (blood oxygen level dependent response, BOLD response). Functional MRI (fMRI) and fluorodeoxyglucose (FDG) positron emission tomography (PET) are other techniques for functional neuroimaging and have been widely used in the literature to diagnose AD and cognitive decline. On the one hand, fMRI can measure brain activity by detecting changes associated with blood flow that increases when a specific brain region is active and decreases otherwise. Using fMRI, brain functional connectivity network (FCN) can be analyzed to detect and understand the functional interactions among brain ROIs (Wang et al., 2021; Zhang et al., 2021). On the other hand, the PET (FDG) imaging can detect alterations in brain metabolism much earlier than structural imaging techniques allowing for early detection of AD. In addition, PET FDG imaging helps in differentiating AD from other causes of cognitive impairment by assessing patterns of glucose metabolism. AD typically exhibits a characteristic pattern of reduced glucose metabolism in specific brain regions, such as the posterior cingulate cortex and temporoparietal association areas (Marcus et al., 2014; Levin, 2021).

Molecular imaging is also another promising modality that can help detecting AD before its onset and monitoring the disease progression (Chételat, 2020). It is likely that, in most of cases, the functional alterations due to the neurodegeneration precedes the brain atrophy and before the appearance of any cognitive symptom (Jack, 2012; Valotassiou, 2018). Therefore, studying these alterations at molecular level using molecular imaging is quite significant for early diagnosis of AD. Molecular imaging mainly uses some targeted radiotracers with PET (amyloid, tau) or single photon emission computed tomography (SPECT) to measure abnormal deposits/accumulation of beta amyloid/tau proteins. SPECT proved good potential to detect the functional alterations in various neurodegenerative diseases, particularly AD (Ferrando and Damian, 2021).

Finally, diffusion neuroimaging (diffusion tensor imaging, DTI) is another type of MRI that measures the random motion of water molecules in brain tissue, which can be used to generate images of the brain's white matter tracts. White matter tracts are bundles of nerve fibers that connect different areas of the brain. In healthy white matter tracts, water molecules primarily move along the length of the tract, a phenomenon known as anisotropic diffusion. In damaged or diseased white matter, the movement of water molecules may be less restricted and more random, a phenomenon known as isotropic diffusion (Amlien and Fjell, 2014). DTI can visualize and measure these differences in water molecule movement, providing a picture of the health of the brain's white matter tracts. Measuring some DTI-related measurements (e.g., fraction anisotropy and mean diffusivity) proved to boost the AD diagnosis (Maggipinto, 2017; Xiao et al., 2022).

Cerebrospinal fluid (CSF) can provide important biomarkers that reflect brain status. The basic role CSF is to cushion and bath the brain as well as the spinal cord, however, the pathophysiological changes in the CSF help to assess the early stages of AD. The biomarkers Amyloidβ 42 (Aβ42), total Tau (T-tau), and Tau phosphorylated at threonine 181 (P-tau181) have proven diagnostic accuracy for MCI and AD (Hansson et al., 2019; Lee et al., 2019). CSF can be considered as an individual biomarker for AD/MCI diagnosis; yet, it provides better diagnostic accuracy when combined as an auxiliary feature with neuroimaging biomarkers (Shi et al., 2019). Therefore, many studies in the literature adopted the CSF as individual modality with comparable performance, e.g. (Palmqvist, 2015).

Recent studies believed that genes are playing a major role in AD with late-onset and early-onset showing heritability of 58–79 % and 90 %, respectively (Sims et al., 2020). In addition, the strong genetic component whose characterization became a critical part to understand the pathophysiological processes of the disease (Bellenguez et al., 2020). It is also proven that genetic mutations can cause disease, which if inherited, can also cause a certain disease. Generally, genetic variants may increase/decrease the risk of developing a specific disease that is known as a genetic risk factor. Identifying the genetic variants or single-nucleotide polymorphisms (SNPs) is deemed one of the effective ways to predict, treat, or even prevent AD. Therefore, genome wide association study of big number of traits can provide a cross-phenotype analysis and robust platform to understand the etiology of AD. Nevertheless, a recent study by Bellenguez *et al.* (Bellenguez et al., 2020) suggests larger and more diverse genetic studies are required as the current knowledge of the genetics of AD is far from complete. Another commonly used way in the literature is to use genotype data as a complementary modality with neuroimaging modality to boost the disease prediction performance as in (Zhu et al., 2017; Zhou et al., 2019).

Finally, in clinical practice, physicians may use some tests/scores to assess the cognitive abilities of patients. The most common tests are as follows. The Mini-Mental State Exam (MMSE) which is a 0–30-point test to assess orientation to time and place, recall, attention, comprehension and language, and ability to solve problems (Folstein et al., 1975). The Montreal Cognitive Assessment (MoCA) is also a 30-point test that was developed to overcome the difficulties of MMSE and discriminate normal cognition, MCI, and early onset dementia (Nasreddine, Apr 2005). Clinical dementia rating-global (CDR-GLOB) and its upgraded version Clinical dementia Rating-Sum Of Boxes (CDR-SOB) are both used for staging dementia severity. The former ranges from 0 to 3 while the later ranges from 0 to 18. The CDR-SOB combines both scores in a single score with wider range that is simpler, easy to calculate, and has an increasing precision for tracking changes across time (O'Bryant, 2008). Lastly, the Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS-Cog) which was proposed to assess the cognitive and non-cognitive dysfunction severity with range 0–70 (Kueper et al., 2018).

Recent progresses in ML and DL techniques have greatly achieved the state-of-the-art performance in many medical image computing tasks. Using these techniques, we can efficiently learn the effective feature representation from single or multimodal data towards identifying the highly-related biomarkers that help to detect, classify, and diagnose different brain diseases. Many studies have been proposed in the literature for AD diagnosis and classification (Ebrahimighahnavieh et al., 2020; Fathi et al., 2022; Khojaste-Sarakhsi et al., 2022; Shoeibi, 2023; Illakiya and Karthik, 2023; Xu et al., 2023). In this paper, we aim to provide comprehensive overview of the state-of-the-art methods in AD diagnosis from different patients' data. Specifically, we first recall the most common standard ML and DL methods used in AD diagnosis. Then, we review the up-to-date studies that used either single or multimodal data. Finally, we discuss the open research problems and provide prospective insights for further research directions in this field.

The rest of this paper is organized as follows. In Section 2 we first discuss the data collection and preprocessing steps. Then, we present the different ML methods used for feature selection and prediction followed by the DL models in Sections 3 and 4, respectively. We further discuss in Section 5 the different criteria for selection, implementation, and performance evaluation of AD diagnosis models. Next, we explore the different data fusion techniques for multimodal studies in Section 6. Afterward, we summarize and analyze the recently published studies that use ML and DL models in Sections 7 and 8, respectively. Finally, In Sections 9 and 10, we discuss the current challenges together with potential solutions and highlight some concluding points, respectively.

2. Data collection and preprocessing

Typically, AD computer-aided diagnosis and detection models passes through a set of stages (Fig. 1). The first stage is data collection which is the process of acquiring brain signals using different devices, depending on the type of biomarkers. After data acquisition, preprocessing is an indispensable to enhance the data representation by removing outliers and irrelevant information, bias correction, and normalization. Indeed, preprocessing should be dealt with much care to ensure the performance reliability and accuracy of the subsequent steps and the model, in general. Below, we discuss the data collection and preprocessing steps in more details.

2.1. Data collection

Data acquisition is the first step in any computer-aided diagnosis system. However, collecting clinical data for AD research can be quite costly due to the machinery, operating, and data annotation fees, particularly neuroimaging data. Besides, some studies require longitudinal data which takes long time to acquire (years). Therefore, many studies adopted the publicly available repositories. The biggest and widely used resource is the Alzheimer's Disease Neuroimaging Initiative (ADNI) which is an ongoing initiative started in 2004 and lead by Dr. Michael W. Weiner to develop clinical, imaging, genetic, and biochemical biomarkers for the early detection and tracking of AD progression. This dataset comes with plenty of subjects with their demographic information, modalities, and clinical information, e.g., MRI, PET (FDG, amyloid, tau), fMRI, DTI, CSF, genetic, and cognitive scores. In addition to ADNI dataset, there are few more common resources we summarize them in Table 1.

2.2. Data preprocessing and feature extraction

Data preprocessing is a key step before feature extraction to extract meaningful features/biomarkers that describe the brain status. Data preprocessing steps vary based on the modality being used. Below, we summarize the basic steps involved in preprocessing the common data modalities used in AD research.

The preprocessing steps of MRI images are standardized in many studies in the literature (Lorenzini, 2022; Zhou et al., 2019). These steps start with correcting the anterior commissure-posterior commissure (AC-PC), the intensity inhomogeneity correction, then skull stripping, and cerebellum removal. Afterward, brain tissue segmentation into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) is performed. Then, regions of interest (ROIs) are projected by registering subject's image to a brain atlas space. Eventually, GM volumes of all ROIs of every subject are computed and then normalized with the intracranial volume to obtain the final feature representation. For PET images, they are first registered to their T1 MRI counterparts and then compute the average PET intensity value of each ROI. This procedure will yield the same number of features extracted from the MRI scan. Preprocessing the SPECT images follows the one of PET images. However, SPECT images usually have low resolution and signal-to-noise ratio. Firstly, registration is required to align all SPECT scans to a reference space (MNI atlas) then averaging all available scans (Merhof, 2011). Following registration, the voxels' intensity values are commonly smoothed using Gaussian filter with filter width 6 ~ 12 mm. Thereafter, intensity values are normalized to a new reference range that avoids any potential bias to a particular region which is not affected by the disease.

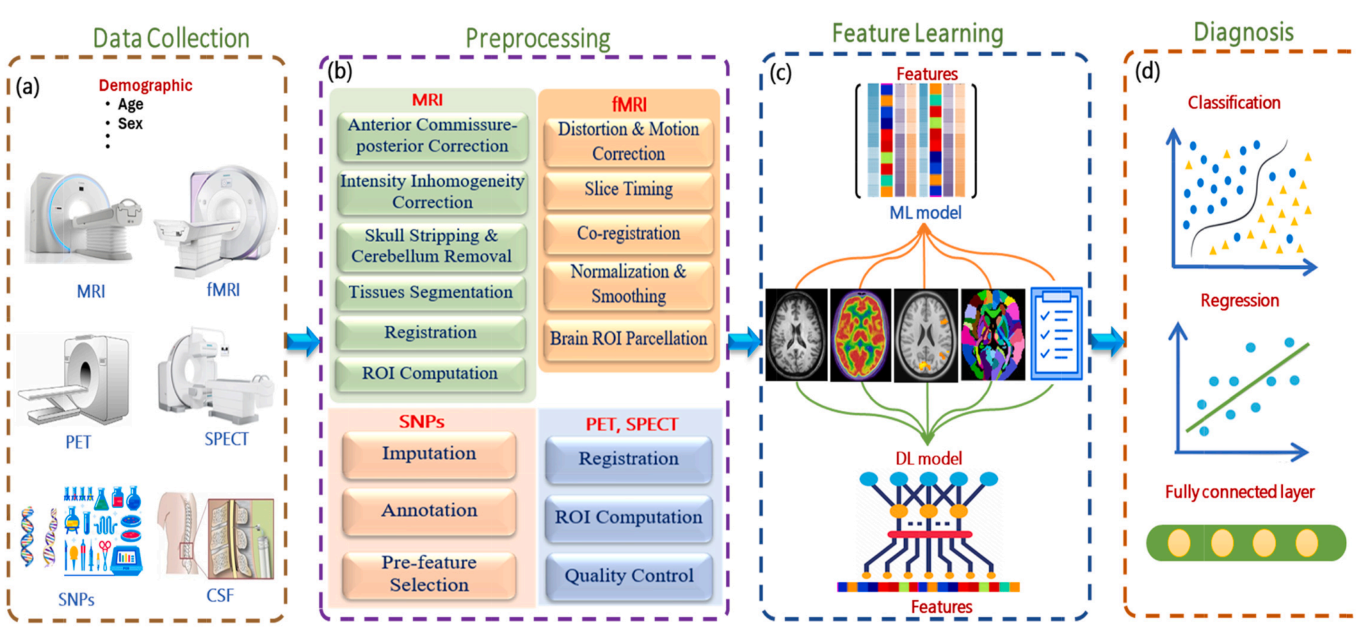

Fig. 1. Illustration of the different stages involved in building AI-based models for AD diagnosis.

Table 1
Most commonly and publicly available data repositories of AD.

Dataset	# Subjects & Classes	Modalities	Website
Alzheimer's Disease Neuroimaging Initiative (ADNI)	• ADNI-1: NC (200), MCI (400), AD (200) ADNI-GO: ADNI-1 + 200 (EMCI) ADNI-2: ADNI-1 + ADNI-GO+NC (150), EMCI (150), LMCI (150) ADNI-3: ADNI-1 + ADNI-GO+ADNI-2 + NC (133), MCI (151), AD (87)	MRI, PET (FDG, amyloid, tau), fMRI, DTI, CSF, genetic, clinical scores	https://adni.loni.usc.edu/
The Australian Imaging, Biomarker & Lifestyle Flagship Study of Ageing (AIBL)	• NC (768), 133 (MCI), and 211 (AD)	MRI, PET (amyloid), clinical scores	https://aibl.csiro.au/
Open Access Series of Imaging Studies (OASIS)	• OASIS-1: Non-demented (316), Demented (100) OASIS-2: Non-demented (77), Demented (64), Converted (10) OASIS-3: NC (755), various stages of cognitive decline (622) OASIS-4: 663 subjects	MRI, PET, CT, rs-fMRI, Cognitive scores, CSF, neurospychometric	https://www.oasis-brains.org/
National Health and Aging Trends Study	• 8334 subjects with different cognitive stages.	Clinical Data, Cognitive Tests	https://www.nhats.org/
Minimal Interval Resonance Imaging in Alzheimer's Disease	• NC (23), AD (46)	MRI	https://www.nitrc.org/projects/miriad/
National Alzheimer's Coordinating Center (NACC)	• 48,600 subjects with different cognitive stages.	Clinical Data, Cognitive Tests, Neuropathological Data	https://naccdata.org/

The preprocessing of fMRI data (4D) is different from the structural MRI (3D). It is typically processed using the following steps (Lorenzini, 2022). First, it is recommended to remove the first few volumes (e.g., 10) to maintain magnetization equilibrium. Then, slice timing and head-motion corrections are usually performed to, respectively, match all time points to an intermediate time point and reduce the head-motion artifacts in the fMRI time-series. Afterwards, a spatial normalization of the volumes to an atlas space and smoothing are commonly performed to the dataset before brain parcellation into ROIs using brain template (e.g., AAL). Ultimately, the mean time series of each ROI of each subject is calculated by averaging the time series over the GM voxels in that ROI.

Regarding genetic data, the genotyping process typically results in hundreds of thousands of SNPs. Therefore, preprocessing the genetic data is quite different from previous modalities. The major processing steps of genotype data are as follows (Zhou et al., 2019). Quality control is firstly performed to ignore those with minor allele frequency and low call rate. Afterwards, imputation is performed to impute the missing individuals' genotypes followed by annotation the SNPs of genes that are highly related to AD within a specific range. Finally, it is widely accepted in literature to carry out dimensionality reduction of the SNPs using statistical tests (e.g., *t*-test) as a pre-feature selection so that the eventual feature vector becomes within feasible size for the subsequent steps. To summarize, we provide in Table 2 a list of the common toolboxes used to process different modalities.

3. Traditional Machine learning models

Feature redundancy is common challenge in the extracted features as not all extracted features are useful to train a ML model. Therefore, feature selection is commonly used to identify the most discriminative features, facilitate the task of the classifier, tackle the potential problem of overfitting, and also reduce computational costs. Finally, either a classification is used to predict the class labels (categories) or regression to estimate continuous values (cognitive scores) based on modeling the relationship between independent and dependent variables. Fig. 1 shows the general framework of the AD diagnosis model. Below, we give brief explanation of the main stages.

3.1. Feature selection and regularization

The high dimensionality of medical data makes model's robustness at the stake due to the big number of features (curse of dimensionality), feature redundancy, and the existence of irrelevant features and outliers. Therefore, feature selection is widely used in literature to identify the most discriminative features which can further facilitate the task of the classifier and potentially remedy the overfitting problem (Khan and Zubair, 2022; Khaire and Dhanalakshmi, 2022). There are mainly 4 approaches for feature selection; filter, wrapper, embedded, and hybrid. However, sparse learning methods have been widely used in literature to perform feature selection from different perspectives of the underlying data and the problem in hand. In the view of the task, these methods can be either single task or multitask. In both cases, these methods achieve feature selection by minimally optimizing a constrained objective function of loss term and a regularization/penalty term. The loss term is the sum of the empirical errors of all samples while the regularization term forces the optimization function to make the optimal solution unique and hence avoid overfitting. Indeed, constructing the loss and regularization terms is significant for the performance of the model as well as its optimization. Generally, the objective function is defined as follows:

$$\mathrm{argmin}_W \mathscr{L}(Y, XW) + \mathscr{R}(\lambda, W) \tag{1}$$

where $\mathscr{L}(.)$ is the loss term of the response variable (labels) $Y \in \mathbb{R}^{m\times c}$ and the feature matrix $X = \in \mathbb{R}^{m\times n}$. While $\mathscr{R}(.)$ is the regularization term of $W \in \mathbb{R}^{n\times c}$ where $m, n,$ and$c$ are numbers of subjects, features, and classes, respectively. Finally, λ denotes a weighting parameter that controls the effect of the regularization. Note that, it is also common to use multiple regularization parameters to control different penalty factors. Based on Eq. (1), several methods have been proposed for individual and group feature selection. The former can be either linear or nonlinear while the latter can be either automatic or structural (Li et al., 2022). Several methods were proposed by introducing different loss and regularization terms. For the fundamental linear regression, $Y = XW$, the simplest solution can be obtained by minimizing the least-squares regression $\|Y - XW\|_2^2$. On the one hand, when L_1 ($\lambda_1\|W\|_1$) is used as a penalty term with $\lambda_1 > 0$, it becomes the typical LASSO model (Tibshirani, 1996). On the other hand, when L_2 ($\lambda_2\|W\|^2$) is used as penalty term with $\lambda_2 > 0$, it produces the ridge regression. It is obvious that controlling the values of λ_1 and λ_2 yields different outcomes and can reproduce other models as well. Several versions of LASSO have been proposed to attain the optimal solution and improve feature selection. For instances, adaptive LASSO (Zou, 2006), relaxed LASSO (Meinshausen, 2007), square-root LASSO (Belloni et al., 2011), uncorrelated LASSO (Chen et al., 2013), and self-parameterized Lasso (Xiao and Xu, 2021). Since individual feature selection methods do not

Table 2
Commonly used toolboxes for preprocessing multimodal data in AD research.

Toolbox	Usage	Modality	Website
3D Slicer	Visualization, processing, segmentation, registration, and analysis of medical	MRI, fMRI, CT, PET	https://www.slicer.org/
AFNI (Analysis of Functional NeuroImages)	Processing, Transformation, activation maps computation Visualization	MRI, fMRI, DTI	https://afni.nimh.nih.gov/
ANTs (Advanced Normalization Tools)	Image processing and registration	MRI, fMRI, PET	https://stnava.github.io/ANTs/
BrainVoyager	Visualization, Segmentation, Registration, Statistical voxel & pattern analysis	MRI, DTI, fMRI, EEG, MEG	https://www.brainvoyager.com/
DIPY (Diffusion imaging analysis in Python)	Spatial normalization, signal processing, statistical analysis, visualization	DTI, MRI	https://dipy.org/
DPARSF	Registration, normalization, smoothing, and FC analysis	fMRI	https://rfmri.org/DPARSF
FreeSurfer	Skull-stripping, bias field correction, registration, segmentation, cortical surface reconstruction, and parcellation	MRI, fMRI, DTI, PET	https://surfer.nmr.mgh.harvard.edu/
FSL (FMRIB Software Library)	Brain extraction, structural changes analysis, segmentation, registration, tractography	MRI, fMRI, DTI	https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/
GRETNA	FC matrix Construction, Network analysis, Metric comparisons	fMRI	https://www.nitrc.org/projects/gretna
HAMMER	Registration	MRI	https://www.med.unc.edu/bric/ideagroup/software/fast-hammer/
Medical Imaging Toolkit (MITK) and Insight Toolkit (ITK)	Image segmentation, registration, analysis, visualization	MRI, CT, PET,	https://www.mitk.org/ https://www.itk.org/
MIPAV	Image Processing, Analysis, and Visualization	MRI, CT, PET	https://mipav.cit.nih.gov/
MIRTK (Medical Image Registration Toolkit)	Image registration, reconstruction, and segmentation	MRI, PET, fMRI, DTI	https://mirtk.github.io/
nilearn	Statistical learning, Visualization	fMRI, MRI	https://nilearn.github.io/stable/index.html
PyRadiomics	Feature extraction from medical images	MRI, CT, PET	https://pyradiomics.readthedocs.io/en/latest/
SPM (Statistical Parametric Mapping)	Registration, segmentation, Morphometry Analysis, Spatio-temporal modeling, Bayesian inference, and FC analysis	MRI, fMRI, PET, SPECT	https://www.fil.ion.ucl.ac.uk/spm/

consider the interactions among features, they may be less effective. Thus, group-based feature selection methods were proposed to capture such interactions. For instance, elastic net model (Zou and Hastie, 2005) combines both L_1 and L_2 with $\lambda_1, \lambda_2 > 0$ for group feature selection (Zou and Hastie, 2005). Elastic net was further improved by considering the correlations among features in pairwise elastic net (Lorbert et al., 2010) and the cluster elastic net (Witten et al., 2014). To encourage both sparsity of coefficients and their differences, $\lambda_1\|W\|_1$ and $\lambda_2\sum_{j=2}^{p}|W_j - W_{j-1}|$ could be used for regularization in the fused LASSO (Tibshirani et al., 2005). Trace LASSO imposed more penalty by adding the trace norm of the diagonal matrix of the all eigenvalues of W (Grave et al., 2011). For more details about different types of feature selection methods, we refer the readers to the comprehensive review on this topic by Li et al. (Li et al., 2022).

3.2. Prediction

Training a classifier or a regression model comes as the ultimate step in the computer-aided diagnosis models to, respectively, predict diagnostic labels (categories) or regress some clinical scores (cognitive scores). There are several methods in the literature to perform this task as given below.

K-Nearest Neighbors (KNN) is a simple supervised machine learning classification and regression method. It typically classifies an object based on the class labels of its K-nearest objects in the feature space. In case of regression, the average value of K-nearest objects is chosen as the predicted output. It can be also used for data interpolation and missing data imputation (Daberdaku et al., 2020). Generally, KNN utilizes a distance metric objective function to calculate the closeness of a given object to its neighbors. KNN is commonly used in literature due its simplicity and the non-linearity of the decision boundaries which leads to better performance (Song et al., 2017). Moreover, it is applicable in cases where not all classes are available during training time. Hence, it was used in many studies, either independently or combined with other methods, to identify the disease status (Lahmiri, 2023, Eroglu et al., 2022). However, it was not always showing the best performance due to its sensitivity to outliers and vulnerability to overfitting.

Logistic Regression (LR) is a supervised statistical machine learning method that it is widely used for classification (Dreiseitl and Ohno-Machado, 2002), though it can be used for regression tasks. LR adopts the “S”-shaped sigmoid function to project the real values to a probability value (between 0 and 1) which is later thresholded to attain the class label. The parameters of LR model are estimated using maximum likelihood and optimized using gradient descent algorithm. Generally, LR is simple, fast, and yields satisfactory classification accuracy. Nevertheless, it cannot represent the complex relationships among the features and it is prone to overfitting when number of samples is small. LR has been used in several AD diagnosis studies as a baseline or main classifier and attained good performance (Xiao, 2021; El-Sappagh et al., 2020; Rohini and Surendran, 2021; Dong et al., 2019; Şeker et al., 2021).

Adaptive Boosting (AdaBoost) is a common boosting technique that was proposed to enhance the performance of binary classifiers, also multiple classifiers, through ensemble learning method (*meta*-learning) (Freund and Schapire, 1996). In addition, it can be used for regression. This algorithm iteratively learns from the mistakes of the weak classifiers to improve their robustness. It adaptively assigns different weights for the samples; typically, more weight for the difficult ones that are misclassified and less for those correctly classified. The biggest advantage of AdaBoost is its flexibility to be implemented with different methods. Moreover, it does not require prior knowledge about the learner and it is susceptible to overfitting problem. Yet, it is also sensitive to noise and outliers. Therefore, data preprocessing is crucial before using it. Due to its advantages, several studies on AD diagnosis adopted the AdaBoost to perform the classification task (Khan and Zubair, 2022; Mahendran and D.R.V.P.M., 2022; Fang et al., 2020; Fan et al., 2020).

Decision Tree (DT) is another supervised machine learning method that is also used for both regression and classification. It represents the given features in an up-down structure where the top-most node is the root and the descendent ones are called leaves which can be in turn considered as parents if there exist descendent nodes from them. The internal nodes are representing the features while leaves representing the class label or the predicted value. The tree structure representation of the feature has many advantages. It not only has simple and easy way to interpret the decision but it also requires less domain knowledge or parameter settings. In addition, DT can effectively handle data with high dimensionality with logarithmic cost. However, it suffers from overfitting when tree grows larger and can be unstable in case there are small variations among the features. In literature, researchers adopted DT-based methods to predict AD and its stages from different single- and multi-modal biomarkers, e.g., (Javaid et al., 2022; Lin et al., 2019; Porter, 2018; Lin et al., 2021).

Random Forest (RF), or random decision forests, is another ensemble learning methodology that can be used for both classification and regression. As extension of decision trees, RF uses several trees in randomly selected feature spaces and generalizes their performance, complementarily (Ho, 1995). In RF, each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest (Breiman, 2001). The output of RF is usually determined based on voting the most selected label or the average predicted value by most trees in classification or regression, respectively. RF is less sensitive to noise and overfitting, hence outperforms AdaBoost (Breiman, 2001). However, the performance heavily depends on its structure. Furthermore, it lacks interpretability, which is necessary in some applications, as it is difficult to follow so many trees of the forest. RF has widely applied in literature for AD diagnosis from genetic and neuroimaging data (Song et al., 2021; Mao et al., 2020; Bi et al., 2020; Velazquez et al., 2021; Chang et al., 2021) and have showed promising performance to a great extent.

Gradient Boosted Decision Tree (GBDT) is another supervised decision tree-based ensemble learning method that has been effectively used for classification and regression. GBDT additively uses weak tree learners that are trained to fit a given model such that each learner will improve the performance of the previous one by minimizing the error of a differentiable loss function (e.g., mean squared error). GBDT is highly interpretable, adaptable, and can achieve good performance in both classification and regression tasks. Moreover, it is less prone to overfitting and can effectively model feature interactions and inherently perform feature selection (Ye et al., 2009). However, the interpretability and the model complexity become hard with increasing number of trees and paths. Therefore, several studies were proposed to perform parallelization and hyperparameters tuning (Ye et al., 2009; Chen and Guestrin, 2016). Many GBDT-based methods have been widely used in AD research to identify and diagnose AD and MCI from MRI, PET, and genetic data (Ahmed et al., 2022; Aghili et al., 2022; El-Sappagh et al., 2022; J.P.J.s.G.S.K.S. and R.w.S, 2022; Bharati et al., 2022; Khan and Zubair, 2022; Khan et al., 2022; Arnal Segura, 2022).

Support Vector Machine (SVM) is a well-known supervised non-probabilistic machine learning method that achieved superior performance in many classification and regression tasks. SVM was firstly proposed for binary classification (Cortes and Vapnik, 1995), however, it was later extended for multi-class problem (Hsu and Lin, 2002). SVM tries to find the best hyperplane (set of hyperplanes) that maximizes the distance to the nearest data point in the dimension space. This is typically achieved using a loss function that is regularized with a tunable hyperparameter (commonly named C). This works easily when the data points are linearly separable which is not always the case in the real world. Therefore, to handle the nonlinearity, SVM uses the kernel trick to map the low-dimension input space into high-dimension one so that the hyperplane(s) can be easily constructed. Several kernel functions can be used, yet, radius basis function remains the gold standard choice in most of cases. Generally, SVM is simple, computationally inexpensive, less sensitive to outliers, and effective in high-dimensional spaces. However, it is prone to overfitting when number of dimensions is much higher than the number of samples. Moreover, SVM decision is less interpretable as it does not provide probability estimate. In literature, SVM is widely used in the majority of conventional ML-based method for the diagnosis of AD and its prodromal stages (Eroglu et al., 2022; Aghili et al., 2022; Zhang et al., 2022; El-Sappagh et al., 2022; Lazli and Cham, 2022// 2022:; de Mendonça and Ferrari, 2023; Gao et al., 2022; Chen, 2022; Hernandez et al., 2022).

Naive Bayes (NB) is a family of supervised probabilistic machine learning classifiers that is fundamentally based on the conditional probability principle of Bayes theorem (Zhang et al., 2004). The basic idea of the simple NB classifier assumes there is no dependency among features (conditional independence) and they have an equivalent impact on the classification performance. Hence, using the chain rule, the classification problem can be formulated as a maximization of the posteriori probability. The assumption of distribution that the features follow specifies the type of the NB. For instance, if the features are assumed to follow Gaussian distribution, the classifier is called Gaussian NB. Generally, NB is less complex classifier, easy to implement, and can be efficiently trained using less data. NB has been used in several studies and achieved varying performance on AD diagnosis (Chang et al., 2021; Khan and Zubair, 2022; Shankar et al., 2022; Skolariki et al., 2021; Kaplan et al., 2021; Zandifar et al., 2020; Carvalho et al., 2020; Battineni, 2021; Kumari et al., 2022).

Multilayer Perceptron (MLP) is a common type of supervised feedforward neural networks that is widely used in classification (also regression). The MLP is an extension of the standard Perceptron which was proposed to perform binary classification with only a linear decision boundary (Freund and Schapire, 1998; Gareth et al., 2013). The MLP overcomes this limitation via a fully connected architecture that consists of input layer, hidden layer(s), and output layer. The inputs are typically associated with initial weights that are combined together via a weighted sum operation then mapped to an output through an activation function (e.g., sigmoid, tanh, Rectified Linear Unit, ReLU) and passed to the next layer. Using backpropagation and gradient decent methods, weights are continuously updated until minimizing the cost function. Generally, MLP is simple, easy to train, and can handle problems with complex (nonlinear) decision boundaries. Hence, many AD studies adopted the MLP either to perform the classification task or as baseline for comparisons from multimodal datasets (Raju and Gopi, 2021; Lin et al., 2021; Qiu, 2020; T.-H. Kung, T.-C. Chao, Y.-R. Xie, M.-C. Pai, Y.-M. Kuo, and G. G. C. Lee, "Neuroimage Biomarker Identification of the Conversion of Mild Cognitive Impairment to Alzheimer's Disease," Frontiers in Neuroscience, Original Research vol. 15, 2021; Perez-Valero et al., 2022; Alatrany et al., 2021// 2021:; AlSharabi et al., 2022; Liu et al., 2023).

Extreme learning machine (ELM) is another form of feedforward neural networks that can be used for classification, regression, and clustering. It was originally proposed by Huang et al. (Huang et al., 2006) to overcome the long time required for training the network due to the gradient-based backpropagation. Instead, ELM adopts the Penrose generalized inverse to set the network weights, hence it is very fast and easy to implement with good generalization performance. Nonetheless, ELM uses randomly generated weights which may yield unstable results. Several researchers adopted ELM in the literature for AD and MCI identification and achieved a promising performance (Nguyen et al., 2019; Lopez-Martin et al., 2020; Wang et al., 2021; Lama and Kwon, 2021; Zhou, 2021; Lama et al., xxx; Shaji et al., 2022; Lu et al., 2022; Lu et al., 2022).

4. Deep learning models

DL is a subcategory of artificial intelligence techniques that is based on the artificial neural networks. It uses a multiple structured layers to process and extract high-level features from the input data. As data is progressively passed from layer to another, non-linear transformation and abstract representation become conceivable. DL models have achieved an impressive performance in various applications and breakthroughs in solving intricate problems (LeCun et al., 2015). Unlike conventional ML models which require massive feature engineering from raw data, DL models have the potential to directly deal with raw data and automatically extract its internal representation. However, they are data-hungry and require high computational cost. Nevertheless, the recent advances graphics processing units and the open-source toolboxes have revolutionized the development of DL models to achieve the state-of-the-art performance in numerous fields. Based on learning strategy, DL models can be either supervised, semi-supervised, unsupervised, or reinforcement learning. Several DL architectures have been proposed in the literature and spanned a wide-range of scientific and industrial problems (Dong et al., 2021; xxxx; Zhang et al., 2022; Zhang et al., 2021; Alzubaidi, 2021). DL has been an active approach in AD diagnosis and detection in the recent years (Ebrahimighahnavieh et al., 2020; Fathi et al., 2022; Khojaste-Sarakhsi et al., 2022). Baseline architectures of most common DL models are shown Fig. 2 and summarized below.

Convolutional Neural Network (CNN, sometimes ConvNet) is the most commonly used supervised DL model in the literature. It has been extensively used in different image processing and computer vision tasks; including detection, segmentation, and recognition. Consequently, CNN attracted researchers' attention in the medical/biomedical field and it has been adopted to diagnose, detect, segment different diseases from different image and non-image data. Typically, CNN contains a set of stacked convolutional, pooling, and fully connected layers. Convolutional and pooling layers learn low- and high-level features, while the fully connected layer maps the learnt features into final output. CNN training is performed using backpropagation to minimize the cost/loss function via gradient descent optimization algorithm. A major problem with CNN-based models is the occurrence of overfitting, particularly when the size of the training set is small, which is common in medical imaging field. Continuous research is still going on to alleviate the effect of overfitting through regularization e.g., dropout, weight decay (e.g., L_2 regularization), batch normalization, and data augmentation (Yamashita et al., 2018). The major part of DL-based methods in literature for AD diagnosis is based on the CNN model and its variants, e.g., VGG (Simonyan and Zisserman, 2014); ResNet (He et al., 2016); InceptionNet/GoogLeNet (Szegedy, 2015); DenseNet (Huang et al., 2017); EfficientNet (Tan and Le, 2019). Lots of CNN models were proposed and achieved remarkable performance in AD diagnosis (Basher et al., 2021; Janghel and Rathore, 2021; Liu et al., 2021; Goenka and Tiwari, 2022; Chen et al., 2022; Liu et al., 2022; Ashtari-Majlan et al., 2022; Li et al., 2022; Qiao et al., 2022; Wu et al., 2022; Lim, 2022; Li, 2023).

Recurrent Neural Network (RNN) (Rumelhart et al., 1986) is another class of DL models with a directed-graph structure that allows the output of some nodes to be fed as input of those nodes, hence, creating a cycle/loop. In this sense, the output is dependent on the previous observations that can be thought as if it is stored in a memory. RNN can effectively handle temporal information and sequential data to extract useful patterns. However, the vanilla RNN only memorizes the very recent observation and loses information of long sequences. Moreover, it suffers from the problem of vanishing gradient, especially with increasing number of layers, as the gradients that are used to update the weights during the learning process are very small. Nevertheless, RNN was adopted in several studies for MCI and AD identification (Wang et al., 2018; Lee, 2019; Mehdipour Ghazi, 2019; Cui and Liu, 2019; Jo et al., 2019; Nguyen et al., 2020; Lei, 2022).

Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) is an RNN-based architecture that was mainly proposed to overcome the vanishing gradient problem. LSTM cell controls the information flow through the network by introducing a gating mechanism that decides which information to keep or forget. Typically, LSTM cell has three types of gates; forget gate, input gate, and output gate. The forget gate determines what information to keep or ignore from the cell via a sigmoid function. The input gate first updates the cell status using the current state and the previous hidden state using sigmoid function then passes them again to tanh function before they are eventually multiplied. Finally, the output gate decides the value of the next value of the hidden state by passing the current state and the previous hidden state to sigmoid function. The modified cell state is then passed to a tanh function whose output is multiplied by sigmoid function to maintain the information the hidden state shall keep. LSTM has attracted increasing attention useful in assessing the longitudinal data of AD patients and their clinical scores towards conversion estimation of the disease status.

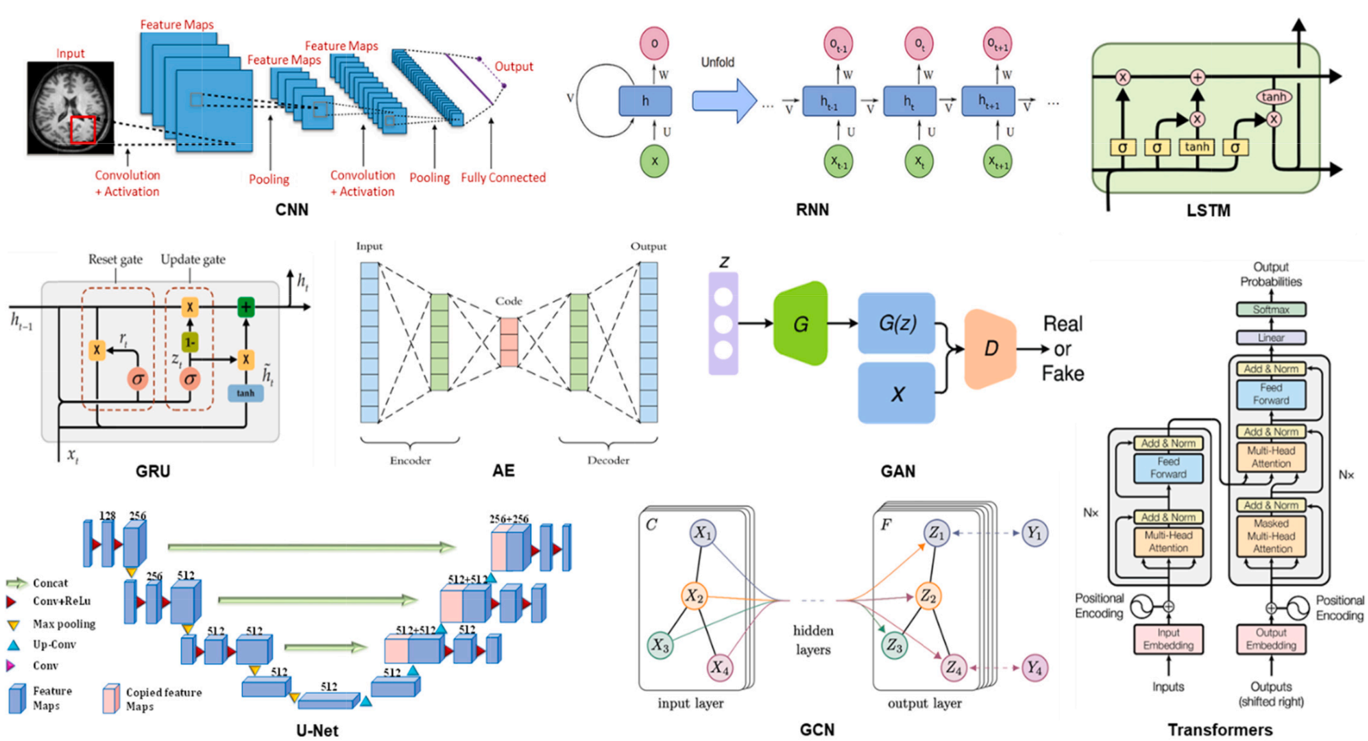

Fig. 2. Baseline architectures of the most common DL models.

Several studies adopted LSTM and stacked bidirectional versions of LSTM to effectively predict MCI and AD with remarkable performance (El-Sappagh et al., 2022; Feng, 2019; Li et al., 2020,; Wang et al., 2020; Tomassini et al., 2021; Jung et al., 2021; Alvi et al., 2023; xxxx; Ho et al., 2022).

Gated Recurrent Unit (GRU) (Cho, et al., 1406) is another RNN-based architecture that also uses gating mechanism like the LSTM but with simpler and fewer parameters. GRU uses the hidden state rather than the cell state to control the information transfer via only two gates; the reset gate and the forget gate. The former is responsible for the short-term memory and decides what information of the previous state to keep while the latter is responsible for the long-term memory and determines what past information needs to be incorporated in the new state. Due to its simpler architecture, GRU is preferably used over LSTM and has less computational complexity. It has been also reported to be effective in detecting AD-related biomarkers (Tabarestani, et al., 2019; Bakkouri et al., 2019; Hong et al., 2020; Huang et al., 2021; Ebrahimi et al., 2021).

Autoencoder (AE) is a feedforward unsupervised DL neural network that is widely used for dimensionality reduction, latent representation extraction, and patterns analysis in data (Hinton and Salakhutdinov, 2006). AE is primarily consisting of an encoder, code or latent representation, and a decoder. The encoder compresses the input data into a lower-dimensional representation (latent representation) while the decoder reconstructs the original data from the compressed representation. To avoid overfitting and learn effective feature representation, denoising AE was proposed by injecting random noise into the input (Vincent et al., 2008). Another regularization method is to impose sparsity on the latent representation as proposed in the sparse AE (Ng, 2011). Further, variational AE (VAE) (Kingma and Welling, 1312) was proposed to maintain the probability distribution of latent representation. In the literature, AE and its variants have been widely used to analyze neuroimaging data by extracting meaningful features to help early diagnosing and monitoring disease progression of MCI and AD (Mendoza-Léon et al., 2020; Martinez-Murcia et al., 2020; Ferri, 2021; Zhang et al., 2020; Cobbinah, 2022; Zhang et al., 2022; Martí-Juan et al., 2023; Chadebec et al., 2023; Shi, 2023).

Generative Adversarial Networks (GAN) are another DL architecture that have been recently used in the AD diagnosis research to generate synthetic brain images. GAN involves two neural networks that are simultaneously trained; a generator and a discriminator (Goodfellow, 2020). The generator learns to create synthetic images that resemble real images, while the discriminator learns to distinguish between real and synthetic images. The two networks are trained together in a process that involves the generator attempting to fool the discriminator and the discriminator attempting to correctly identify the real images. The most common application of GANs in AD research is the generation of synthetic brain images that can be used to augment the missing modality (e.g., PET) or the limited amount of available data for AD diagnosis and prediction (Wang, 2018; Yi et al., 2019; Kwon and Han, 2019; Islam and Zhang, 2020; Zhao et al., 2021; Zhang et al., 2022; Chen, 2022). Another application of GANs in AD research is the generation of images that highlight specific features of the brain that are important in AD diagnosis. These images can be used to aid in the development of new diagnostic techniques and treatments for AD (Kim et al., 2020; Zhou, et al., 2021; Gao et al., 2022; Yu, 2022; Yang, 2021). In addition, GANs can be adapted for direct classification tasks by training the discriminator to classify brain images instead of distinguishing between real and synthetic images (Li et al., 2021; Zhou, 2021; Yu et al., 2022; Hu et al., 2022).

Transformers are a recently proposed DL architecture that are primarily used in natural language processing tasks (Vaswani, 2017). Transformers are characterized by their self-attention mechanism, which allows them to consider the relationships between different parts of the input data. Although transformers are not specifically designed for image data, they can be adapted to process medical images, genetic data, and clinical data to extract meaningful features that help differentiate between healthy and diseased individuals. In AD research, Transformers can be applied to medical images by treating them as sequences of patches to capture local and global contextual information, which can then be used for feature extraction, image synthesis, and image classification (Dosovitskiy, et al., 2010; He et al., 2021; Dalmaz et al., 2022; Zhao, 2023). In addition, they can be used to integrate multimodal data sources, such as combining MRI scans with patients' clinical data or genetic information. By processing each data modality separately and then merging the extracted features, transformers can learn to capture complex relationships between different data types (Kushol et al., 2022; Kherchouche et al., 2022; Jang and Hwang, 2022; Xing et al., 2022; Hoang et al., 2023). Moreover, they can be also employed to analyze the temporal progression of AD by processing longitudinal MR images to capture the changes in brain structure and/or function over time, potentially allowing for early detection of AD and prediction of disease progression (Hu et al., 2023; Chen and Hong, 2302).

U-Net is a common DL architecture that was originally developed for medical image segmentation (Ronneberger et al., 2015). Yet, it has since been used in a variety of other applications and achieved superior performance. The U-Net architecture is based on the encoder-decoder approach. It consists of a contracting path, which captures context and reduces the spatial resolution, and an expanding path, which enables precise localization and segmentation. The contracting path is composed of convolutional and pooling layers, while the expanding path is composed of convolutional and upsampling layers. The two paths are connected by skip connections, which allow the expanding path to use information from earlier layers in the contracting path. U-Net is able to perform accurate segmentation even with limited amounts of training data and has an efficient use of memory, which enables it to process large images in real-time. Therefore, U-Net has been used for AD research, particularly for the segmentation of brain structures and the detection of abnormalities in MRI scans. Specifically, the hippocampus (left and right) which is one the early affected brain regions by AD, hence, many studies adopted U-Net-based architecture to segment the hippocampus to favor the early detection of AD (Hazarika et al., 2022; Sohail and Anwar, 2022; Helaly et al., 2022; Zhu et al., 2022). Similarly, U-Net-based models used the WM hyperintensities for the same purpose (Park et al., 2021; Carmo et al., 2021; Yu and Cham, 2021// 2021:; Røvang, et al., 2207; Duarte, et al., 2022). Moreover, it can also be used to directly classify and detect AD (Fan, 2021; Prabhu, et al., 2022; Qin et al., 2022).

Graph Convolutional Networks (GCNs) are a powerful neural network architecture designed to work directly with data structured as graphs. In traditional CNNs, data is assumed to be structured in a grid-like manner (e.g., images), which limits their applicability to graph-structured data. GCNs, on the contrary, are built specifically for graph data, allowing them to capture the complex relationships between nodes and edges more effectively. In a graph, data is represented as nodes (or vertices) and edges. The nodes represent entities (pixels, voxels, regions, or even subjects), while the edges represent relationships between these entities. GCNs work by applying a convolution operation directly on the graph. This involves aggregating the feature information from a node's neighborhood. This aggregation process involves weighting the features of the neighboring nodes according to their relative importance, and then combining these weighted features to generate a new feature representation for the node. Since Kipf and Welling have proposed GCNs in 2016 (Kipf and Welling, 2016), they have attracted immense attention in studying network neuroscience (Bessadok et al., 2022). GCNs-based methods can be used to analyze neuroimage data, helping to identify patterns and changes associated with AD (Zeng et al., 2022; Alorf and Khan, 2022; Li et al., 2023). In addition, they can be used for multimodal data fusion and address the data heterogeneity among different modalities, potentially providing a more holistic view of the disease (Song, 2021; Huang and Chung, 2022; Li, 2022; Lei, 2023). GCNs can also help in identifying key biomarkers for AD by analyzing the complex relationships and dependencies between different biological entities (such as genes or proteins) (Sügis, 2019; Miyazaki et al., 2020; Bi, et al., 2022; Zhu et al., 2022). Moreover, they can be used to analyze patient data and identify distinct brain regions affected by the disease and the interactions among those regions (Wang et al., 2021; Zhou et al., 2022; Zhang et al., 2023).

5. Model Selection, Implementation, and performance evaluation

Building ML or DL models involves the development of computational systems that can learn and make decisions or predictions from data. Although both ML and DL are subsets of artificial intelligence, DL models are generally more complex and computationally expensive. Building AD diagnosis and detection models typically depends on several factors such as quality and quantity of the available data, the level of data preprocessing required, the targeted task, and model interpretability. Regarding data quality and quantity, the availability of high-quality and sufficient data is crucial for training an accurate model. Ensuring that the dataset used for training is representative of the target population, contains diverse samples, and is properly labeled. In case of multimodal studies, the dataset should ideally include relevant features such as medical history, demographic information, cognitive assessments, neuroimaging data, and genetic information. The level of data preprocessing is another important factor while selecting among ML- or DL-based models. In fact, ML-based models tend to have extensive data preprocessing steps (e. g. data cleaning, handling missing values, and normalization) to ensure the model's robustness. On the contrary, DL-based models require less data processing, however they are data-hungry. In addition, training DL models requires a certain level of expertise and can be time-consuming. Hence, if computational resources are limited or lack expert personnel, simpler ML models may be more feasible. The complexity of the targeted task is also a key factor to determine which model to choose. Indeed, AD diagnosis involves complex biological processes and symptoms that can be difficult to understand. If the task is to identify simple patterns or make predictions based on a few key features, e.g., clinical scores, ML methods may suffice. However, if the task is to analyze complex data such as MRI or PET scans, DL methods can be better choices as they excel in tasks such as image recognition. The last important factor to consider is the model's and comprehensibility. AD diagnosis often requires interpretability to gain insights into the model's predictions and facilitate clinical decision-making. Traditional ML models often provide more interpretable results, which can be important in a clinical setting where understanding why a prediction was made can be as important as the prediction itself. DL models, on the other hand, are often seen as "black boxes" as their internal workings can be complex and opaque. However, recent DL methods adopted different interpretability methods for AD diagnosis (Qiu, 2020; Zhu et al., 2022; Lee et al., 2019; Qiu, 2022).

Implementing either ML or DL models for AD diagnosis involves developing and utilizing algorithms to analyze data and assist in identifying the presence or progression of the disease. As discussed in Sections 3 and 4, there plenty of models that can be used. There are several toolboxes that are commonly used to implement those models, see Table 3. When building the AD diagnosis model, the dataset is usually split into training data and testing data. The training data is used to train the model and adjust its parameters, while the testing data is used to evaluate the model's performance. This division helps to ensure that the model can generalize well to unseen data and is not just memorizing the training data (overfitting) (Ying, 2019). In addition to the training and testing data, sometimes a third set, called a validation set, is used. The validation set is used during the model training process to evaluate the performance of the model during the training phase and tune hyper-parameters. It serves as a kind of "mini test set" before the final evaluation on the actual test set. To further avoid overfitting and improve the robustness of performance evaluation, cross-validation is often used. In k-fold cross-validation, the training data is split into k subsets. The model is then trained k times, each time using a different subset as the test set and the remaining subsets as the training set (Wong and Yeh, 2019). The average performance across all trials is then used as the overall performance metric. It is noteworthy that, bias-variance trade-off should be considered. This concept reflects the balance that needs to be achieved between the model's ability to fit the training data (bias) and its ability to generalize to unseen data (variance) (Belkin et al., 2019). High bias can lead to underfitting (the model is too simple to capture the underlying pattern), while high variance can lead to over-fitting (the model is too complex and learns the noise in the training

Table 3
Commonly used DL libraries for building AD diagnosis models.

Name	Creator	Release Year	Software License	Backend	Website
TensorFlow	Google Brain	2015	Apache 2.0	C++, CUDA, Eigen	https://www.tensorflow.org/
Keras	François Chollet	2015	MIT License	TensorFlow, Theano, Microsoft Cognitive Toolkit	https://keras.io/
PyTorch	Facebook's AI Research lab	2016	BSD License	C++, CUDA, cuDNN	https://pytorch.org/
Caffe	Berkeley AI Research	2014	BSD 2-Clause "Simplified" License	C++, CUDA	https://caffe.berkeleyvision.org/
Theano	Montreal Institute for Learning Algorithms (MILA), University of Montreal	2007	BSD License	C++, CUDA	https://deeplearning.net/software/theano/
MXNet	Apache Software Foundation	2015	Apache 2.0	C++, CUDA, cuDNN	https://mxnet.apache.org/versions/1.9.1/
Chainer	Preferred Networks	2015	MIT License	Python, CUDA	https://chainer.org/
Deeplearning4j	Skymind (now Konduit)	2014	Apache 2.0	Java, Scala, C++	https://github.com/deeplearning4j
PaddlePaddle	Baidu	2016	Apache 2.0	C++, CUDA	https://github.com/PaddlePaddle/Paddle
CNTK	Microsoft	2016	MIT License	C++, CUDA	https://learn.microsoft.com/en-us/cognitive-toolkit/
DL4J	Deep Java Library	2014	Apache 2.0	Java, Scala	https://github.com/deeplearning4j
Torch	Ronan Collobert, Koray Kavukcuoglu, Clement Farabet	2002	BSD License	Lua, CUDA	https://torch.ch/
Fast.ai	Fast.ai	2016	Apache 2.0	Python	https://www.fast.ai/
OpenNN	Artelnics	2019	GNU Lesser General Public License	C++	https://www.opennn.net/
Lasagne	Lasagne contributors	2015	MIT License	Python, Theano	https://lasagne.readthedocs.io/en/latest/

data). Learning curves could be use plotted on both the training and the validation set as a function of training iterations. They can help analyze the learning procedure and identify whether the model is underfitting, overfitting, or has a normal learning.

Finally, performance evaluation of ML and DL models is a key step in any computer diagnosis system of AD. This process involves assessing how well the model has learned and making predictions or decisions. This assessment is typically done through various metrics and statistical measures that provide a quantitative understanding of the model's accuracy, precision, robustness, and generalization capabilities. The goal of performance evaluation is to understand how well the model will perform on unseen patients' data, to quantify its strengths and weaknesses, and to guide the improvement of the model. For instance, if a model has high accuracy but low recall, it might be necessary to adjust the model or its training process to increase the recall, depending on the target of the task. There are several metrics that can be used to evaluate the performance of the AD diagnosis and detection models. In case of disease classification, these metrics can include accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC). Another common way is to use the confusion matrix which is a table that describes the performance of a classification model. It outlines the number of true and false predictions made by the model, broken down by each class. From the confusion matrix, various performance metrics like precision, recall, F1-score can be calculated. In case of regression, other metrics such mean absolute error (MAE), mean squared error (MSE), and root mean square error (RMSE) could be adopted. The choice of metrics depends on the specific task as well as the trade-offs among different types of errors (Table 4).

Table 4
Different evaluation metrics that are used to assess AD diagnosis models.

Name	Usage	Equation
Accuracy (ACC)	Overall correctness of all predictions	$\frac{TP+TN}{FP+FN+TP+TN}$
Precision (PRE)	Correctness of positive predictions	$\frac{TP}{TP+FP}$
Recall (REC) Sensitivity (SEN)	Proportion of actual positives that are correctly identified	$\frac{TP}{TP+FN}$
Specificity (SPE)	Correctness of negative predictions	$\frac{TN}{TN+FP}$
F1 Score	Harmonic mean of Precision and Recall	$2\times\frac{1}{\frac{1}{PRE}+\frac{1}{REC}}$
Area Under ROC Curve (AUC)	Distinguish between classes	–
Mean Absolute Error (MAE)	Average absolute difference between predicted and actual values	$\frac{1}{N}\sum_{j=1}^{N}\left\|y_j-\widehat{y}_j\right\|$
Mean Squared Error (MSE)	Average of the squares of differences between predicted and actual values	$\frac{1}{N}\sum_{j=1}^{N}\left\|y_j-\widehat{y}_j\right\|^2$
Normalized Mean Squared Error (MSE)	Normalized version of MSE either the variance of the target's dataset	$\frac{MSE}{var(y)}\,or\,\frac{MSE}{(y_{max}-y_{min})^2}$
Root Mean Squared Error (RMSE)	Square root of the Mean Squared Error	$\sqrt{\frac{1}{N}\sum_{j=1}^{N}\left\|y_j-\widehat{y}_j\right\|^2}$
Root Mean Squared Logarithmic Error	Penalizes underestimates more than overestimates	$\sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(\log\left(y_j+1\right)-\log\left(\widehat{y}_j+1\right)\right)^2}$
Matthews Correlation Coefficient (MCC)	Measure of quality for binary classifications	$\frac{(TP\times TN)-(FP\times FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$
Balanced Accuracy (BAC)	Adjusted accuracy for imbalanced datasets	$\frac{1}{2}\left(\frac{TP}{TP+FN}+\frac{TN}{TN+FP}\right)$
Normalized Mutual Information (NMI)	Measure of mutual dependence between two variables	$\frac{2.MI(X,Y)}{H(X)+H(Y)}$ MI is mutual information, H is the entropy
Adjusted Rand Index (ARI)	Normalized Mutual Information	$\frac{Index-ExpectedIndex}{MaxIndex-ExpectedIndex}$
Correlation coefficient (CC)	Measure of the linear correlation between two variables	$\frac{n(\sum xy)-(\sum x)(\sum y)}{\sqrt{\left[n\sum x^2-(\sum x)^2\right]\left[n\sum y^2-(\sum y)^2\right]}}$
Weighted Correlation Coefficient	Considers the importance or weight of each point in the dataset	$\frac{\sum w_j\left(x_j-\overline{x}_w\right)-\left(y_j-\overline{y}_w\right)}{\sqrt{\sum w_j\left(x_j-\overline{x}_w\right)^2-\sum w_j\left(y_j-\overline{y}_w\right)^2}}$ w_j is the weight of each sample

6. Multimodal data fusion

Multimodal data fusion aims at integrating and combining multiple types/views of data from single or different sources/modalities to improve the accuracy and reliability of the AD diagnosis model. Traditional diagnostic methods often rely on a single modality, such as cognitive tests, medical imaging, or genetic markers. However, each of these modalities provides only a partial view of the disease, and combining multiple modalities can provide a more comprehensive and reliable diagnostic assessment. On the contrary, multimodal data fusion typically integrates data from different sources to capture various aspects of the disease and biomarkers associated with AD pathology. By combining these diverse data sources, multimodal data fusion approaches leverage the complementary information provided by each modality, enhance the spatial and temporal resolution in the characterization of brain processes, and utilize data from one modality to improve the data quality of another modality (Zhang, 2020). Thus, enabling clinicians and researchers to identify patterns, correlations, and abnormalities that may be indicative of AD or help differentiate it from other conditions. Multimodal studies have been extensively studied in the literature and attracted researchers' attention (Shi et al., 2019; Zhou et al., 2019; El-Sappagh et al., 2020; Bi et al., 2020; Aghili et al., 2022; El-Sappagh et al., 2022; Chen, 2022; Battineni, 2021; Liu et al., 2023; Martí-Juan et al., 2023; Gao et al., 2022; Qiu, 2022). Nevertheless, there is no consensus on the optimal data fusion method. Additionally, the ideal fusion method varies depending on the nature of the diagnostic task, the types of data being used, and the quantity and the quality of the data available.

Several data fusion techniques were proposed in the literature aiming at achieving the best utilization of the multimodal data, for more details see (Shoeibi, 2023; Meng et al., 2020; Stahlschmidt et al., 2022; Dimitri et al., 2022; Holzinger, 2022; Turrisi et al., 2023). Data fusion can be done on three different levels; raw data fusion or signal level (low-level or early fusion), feature-level, decision-level fusion (high-level fusion) (Meng et al., 2020; Azam, 2022). In addition, data fusion can depend on temporal and spatial space or knowledge-based. Below we discuss the different data fusion approaches used in the AD diagnosis research.

6.1. Signal-level fusion

In the signal level, sometimes known as early fusion or data-level, the data fusion takes place at the earliest stage where raw data from different sources (e.g., MRI, PET, …) are combined before any processing is done. It involves merging data at the sensor level. Fusing complementary information at this level can enhance the visualization and interpretation of anatomical and functional details. However, due to the indispensable existence of noise at this level, the model reliability and performance are usually affected. In addition, for image data fusion, registration is likely required to spatially align different image modalities to account for the differences in sensor characteristics. Several AD studies adopted the early fusion of multimodal data to boost the accuracy and robustness of the models. In particular, DL-based models adopted this level of fusion as they require less feature engineering (El-Sappagh et al., 2020; Zhang and Shi, 2020; Kong et al., 2022; Jia and Lao, 2022).

6.2. Feature-level fusion

For the feature-level fusion, also known as biomarker fusion, features extracted from the data sources are combined. This could involve extracting features of different brain regions from different modalities and then combining these features. Feature-level fusion is often performed by concatenating feature vectors from each data modalities. The advantage of this level of fusion is that, it can capture relationships among features from different data sources. This type of data fusion is very popular among ML-based methods used for AD diagnosis and detection. Many methods fused the extracted brain ROIs from MRI, PET, and SPECT as well as SNPs to improve the prediction accuracy of AD status (Lin et al., 2021; El-Sappagh et al., 2022; El-Sappagh et al., 2022,; Feng, 2019; Lei, 2020; Lei, 2020; Lin, 2020; Li et al., 2021; Sheng et al., 2022,; Zhang et al., 2021). However, this fusion commonly takes place in a linear way which is not always effective thus exploring different strategies to efficiently utilize the concatenated features became a research hotspot, recently.

6.3. Decision-level fusion

The decision-level fusion, sometimes known as late fusion, combines the results or decisions obtained from multiple classifiers or algorithms applied to different images or modalities. It aims to make a final decision by considering the outputs of individual classifiers, potentially improving the overall accuracy and robustness. Typically, each model makes its own decision based on the input data, and then a method such as majority voting, weighted voting, or *meta*-learning is used to combine these decisions. For example, in an ensemble of classifiers, each classifier could make a prediction of the disease status, and the final disease prediction could be the one that the majority of classifiers agree on. Another form of late fusion is based on separately processing each modality to extract deep feature using dedicated DL models and their predictions are combined at the decision level (Fang et al., 2020; Khan et al., 2022; Lazli and Cham, 2022// 2022:; Hernandez et al., 2022; Kumari et al., 2022; xxxx). This approach enables the models to learn complex relationships within each modality independently before fusing the predictions.

6.4. Spatial and temporal fusion

Spatial fusion means integrate information at the pixel or voxel level to create a fused image with improved spatial consistency. This can involve image registration to align and merge multiple images or spatial filtering methods. An example of this technique was proposed be Song et al. (Song et al., 2021) to generate a new fused modality that combines GM and metabolic characteristics from MRI and PET images, respectively. On the contrary, temporal fusion is used to combine information from multiple images acquired over time. This is particularly useful to track the progression of AD from longitudinal data, e.g., (Chen and Hong, 2302; Lei, 2020; Lee et al., 2019; Speiser, 2021). However, such techniques require image registration and temporal alignment, temporal averaging, or advanced methods like optical flow estimation or temporal filtering.

6.5. Knowledge-based fusion

Knowledge-based fusion involves prior knowledge together with other modalities into the fusion process; either at feature- or decision-level. In AD research prior knowledge includes information about subjects (e.g., age and sex), AD pathology, brain regions affected by AD, disease progression patterns, or clinical scores. This can include anatomical atlases, expert knowledge, or disease-specific models. Knowledge-based techniques can involve Bayesian networks, rule-based systems, or other knowledge-driven approaches that integrate prior knowledge to guide the fusion process and improve diagnostic accuracy. In many AD studies, the prior knowledge could guide the learning process and improve diagnostic accuracy (Shen, 2021; Chen, 2023; Ding, 2018).

6.6. Hybrid fusion

The above types of multimodal data fusion are not mutually exclusive and can be combined or adapted depending on the specific diagnostic requirements. Hybrid fusion could be used to combine multiple fusion techniques, such as feature-level, decision-level, and data-level, to maximize the benefits of each method. This fusion aims to leverage the strengths of different fusion strategies to improve the accuracy and robustness of AD diagnosis as in (El-Sappagh et al., 2020; Zhang and Shi, 2020). Note that, when combining different data fusion methods, a weighing technique might be required to control the effect of those methods (Dimitriadis et al., 2018).

7. Machine learning-based studies

In this paper, our primary focus is on contemporary literature published within the most recent five-year period, from 2019 through 2023. We targeted papers that incorporated keywords such as “Alzheimer’s disease”, “mild cognitive impairment”, “multitask”, “multimodality”, “fusion”, “machine learning”, “sparse learning”, “diagnosis”, “detection”, “prediction”, among others of similar meaning. We conducted an extensive search across various databases, encompassing Science Direct, IEEE Xplore, ACM digital library, Springer Nature, Wiley, Frontiers, ArXiv, and additional sources. Following PRISMA guidelines (Fleming et al., 2014), each study retrieved underwent a thorough assessment and selection process, based on the quality of the journal/conference, its relevance to the present study’s focus, and our interpretation of the work after reading their abstracts. Following this filtration process, we initially retrieved 97 papers out of them 55 papers were retained and analyzed. The key findings of selected papers and their data modalities’ distribution are given in Table 5 and Fig. 3, respectively. We further summarize their contents in the following subsections.

7.1. Single-modality studies

Majority of studies that only uses single modality is based on MRI data. For instance, Xiao et al. (Xiao, 2021) proposed a sparse logistic regression (SLR) to diagnose AD and predict the conversion of MCI to AD. This method adopted an $L_{1/2}$ regularization to impose a sparsity constraint on LR and effectively identify the most discriminative MRI biomarkers. Although this method achieved better performance than LR, it was tested on relatively small dataset. In another study, Cheng et al. (Cheng et al., 2022) introduced a multi-auxiliary domain transfer learning (MaDTL) method to diagnose the MCI conversion. This method proposed a sparse group correlation Lasso that comprises a group correlation Lasso penalty and correlation Lasso penalty using $L_{2,1}$ and $L_{1,1}$, respectively. Then, classification was performed using a multi-auxiliary domain transfer-based module via a linear weighting-based ensemble learning. To predict AD progression from longitudinal MRI data, Zhou et al. (Zhou et al., 2022) proposed a MTL framework with a novel penalty termed longitudinal stability adjustment (LSA) to adaptively

Table 5
Summary of the recent ML-based models indicating their used modalities, dataset size, performance, and other implementation details.

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
1	Liu et al. (Liu et al., 2019)	FGL-MTFL	MRI, COG Scores	ADNI dataset: 788 subjects (225 NC, 390 MCI, 173 CE)	cognitive scores prediction	ADAS: CC=0.668, nMSE=6.678 MMSE: CC=0.549, nMSE=6.678	Cross validation: 10-fold Code: https://bitbucket.org/XIAOLILIU/fgl-mtfl/src/master/
2	Wang et al. (Wang et al., 2018)	MMR	MRI, COG Scores	ADNI dataset: 804 subjects (225 NC, 393 MCI, 186 CE)	cognitive scores prediction	Fluency: RMSE=0.324, CC=0.543 RAVLT: RMSE=0.317, CC=0.887 TRAILS: RMSE=0.511, CC=0.580	Cross validation: 5-fold
3	Xiao et al. (Xiao et al., 2021)	SLR	MRI	ADNI dataset: 197 subjects (50 NC, 51 pMCI, 45 sMCI, 51 AD)	AD vs. NC MCI vs. NC pMCI vs sMCI	ACC=93.33, SEN=92.25, SPE=94.75, AUC=0.96 ACC=82.75, SEN=86.67, SPE=78.57, AUC=0.89 ACC=72.87, SEN=81.25, SPE=65.45, AUC=0.75	Cross validation: 10-fold
4	Cheng et al. (Cheng et al., 2022)	MaDTL	MRI, CSF	ADNI dataset: 409 subjects (112 NC, 89 pMCI, 105 sMCI, 102 CE)	AD vs. NC AD vs. pMCI sMCI vs. NC MCI to AD	ACC=96.45, AUC=0.991 ACC=73.34, AUC=0.815 ACC=77.85, AUC=0.821 ACC=80.37, SEN=76.61, SPE=83.39, AUC=0.882	Cross validation: 10-fold Code: cb729@nuaa.edu.cn.
5	Pan et al. (Pan et al., 2019)	LASSO-SVM*	FDG-PET	ADNI dataset: 272 subjects (90 NC, 44 pMCI, 44 sMCI, 94 AD)	AD vs. NC MCI vs. NC pMCI vs. sMCI	ACC=91.90, SEN=92.78, SPE=91.38, AUC=0.960 ACC=83.18, SEN=84.20, SPE=82.83, AUC=0.889 ACC=72.33, SEN=73.27, SPE=73.11, AUC=0.717	Nested cross validation: 10-fold
6	Wang et al. (Wang et al., 2019)	MTERL	Longitudinal MRI, COG Scores	ADNI dataset: 445 subjects (152 NC, 104 pMCI, 98 sMCI, 91 AD)	ADAS-Cog MMSE MCI (ADAS-Cog) MCI (MMSE)	nMSE=0.56, rMSE=0.76 nMSE=0.61, rMSE=0.79 ACC=71.98 SEN=80.39, SPE=65.04, AUC=0.812 ACC=73.04 SEN=78.29, SPE=68.89, AUC=0.818	Cross validation: 5-fold Code: https://ibrain.nuaa.edu.cn/code/
7	Pan et al. (Pan et al., 2019)	SSH and LSH + Ensemble	^{18}F-FDG PET	ADNI dataset: 741 subjects (246 NC, 123 pMCI, 125 sMCI, 247 CE)	AD vs. NC pMCI vs. sMCI AD vs. MCI vs. NC AD vs. pMCI vs. sMCI vs. NC	ACC=93.65, SEN=91.22, SPE=96.25, AUC=0.969 ACC=75.83, SEN=74.84, SPE=77.11, AUC=0.807 ACC=74.70 ACC=70.04	Cross validation: 10-fold Training (60 %), validation (20 %), and testing (20 %).
8	Lei et al. (Lei, 2020)	ASL	MRI, COG Scores	ADNI dataset: 814 subjects (220 NC, 146 pMCI, 256 sMCI, 192 CE)	NC vs. AD vs. MCI NC vs. AD vs. lMCI vs. sMCI	ACC: 90 ROIs = 75.80, 116 ROIs = 76.32, 200 ROIs = 73.08, multi-ROI=77.48 ACC: 90 ROIs = 64.23, 116 ROIs = 63.87, 200 ROIs = 61.66, multi-ROI=64.97	Cross validation: 10-fold
9	Zhou et al. (Zhou et al., 2019)	LRP*	MRI, PET, SNPs	ADNI dataset: 737 subjects (204 NC, 157pMCI, 205 sMCI, 171 CE)	NC vs. MCI vs. AD NC vs. sMCI vs. pMCI vs. AD sMCI vs. pMCI	Not reported (only shown in Figures) Not reported (only shown in Figures) ACC: 74.3	Cross validation: 5-fold
10	Peng et al. (Peng et al., 2019)	SSR-MKL*	MRI, PET, SNPs	ADNI (set 1) 189 subjects (47 NC, 93 MCI, 49 AD) ADNI (set 2) 360 subjects (90 NC, 185 MCI, 85 AD)	AD vs. NC MCI vs. NC AD vs. MCI	ACC=96.1, SEN=97.3, SPE=94.9, AUC=0.992 ACC=80.3, SEN=85.6, SPE=69.8, AUC=0.811 ACC=76.9, SEN=65.9, SPE=82.7, AUC=0.808	Cross validation: 10-fold
11	Han et al. (Han, 2022)	DSTC	MRI, PET	ADNI (set 1): 360 subjects (90 NC, 185 MCI, 85 AD)	AD Classification	ACC=90.14, SEN=91.34, SPE=87.17, YI=78.51, F1 = 91.0, PPV=90.68, NPV=88.16	Cross validation: 5-fold
12	Ning et al. (Ning et al., 2021)	RMM-SRL*	MRI, PET	ADNI-1: 392 subjects (99 NC, 121 sMCI, 79 pMCI, 93 AD) ADNI-1: 428 subjects (107 NC, 103 sMCI, 82 pMCI, 136 CE)	AD vs. NC pMCI vs. sMCI MCI vs. NC	ACC=96.9, SEN=95.6, SPE=97.8, AUC=0.976 ACC=84.5, SEN=74.8, SPE=90.9, AUC=0.840 ACC=82.6, SEN=91.0, SPE=65.7, AUC=0.820 ACC=96.8, SEN=95.7, SPE=98.2, AUC=0.977 ACC=85.9, SEN=80.8, SPE=90.0, AUC=0.856 ACC=81.5, SEN=83.1, SPE=81.5, AUC=0.831	Cross validation: 10-fold
13	Liang et al. (Liang et al., 2023)	BGP-MTFL	MRI	ADNI: 788 subjects (225 NC, 390 MCI, 173 CE)	ADAS MMSE	rMSE=6.65, CC=0.67, nMSE=3.92, wR=0.42 rMSE=2.19, CC=0.55, nMSE=3.92, wR=0.42	Cross validation: 10-fold
14	Yang et al. (Yang, 2021)	FSNPF	rs-fMRI	ADNI-2: 76 subjects (29 NC, 18 LMCI, 29 EMCI))	LMCI vs. NC EMCI vs. NC EMCI vs. LMCI	ACC=87.23, PRE=69.13, REC=75.58, AUC=0.923, BAC=87.55 ACC=82.76, PRE=79.38, REC=68.79, AUC=0.882, BAC=82.76 ACC=80.85, PRE=68.25, REC=71.06, AUC=0.849, BAC=75.00	leave-one-out cross-validation

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Table 5 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
15	Brand et al. (Brand et al., 2020)	JMMLRC	MRI, SNPs, demographic	ADNI-1: 412 subjects (147 NC, 110 MCI, 155 CE)	RAVLT regression AD classification MCI classification NC classification	RMSE=11.7, RMSE=18.3, RMSE=18.1 F1 = 56.6, BAC=58.4 F1 = 51.3, BAC=58.4 F1 = 68.3, BAC=58.4	Cross validation: 5-fold Code: https://github.com/minds-mines/jmmlrc
16	Tang et al. (Tang et al., 2022)	FAS-MTFL	MRI, COG Scores	ADNI: 788 subjects (225 NC, 390 MCI, 173 CE)	ADAS MMSE RAVLT TOTAL T30 RECOG	rMSE=6.68, nMSE=4.34, CC=0.663, wR=0.53 rMSE=2.19, nMSE=4.34, CC=0.543, wR=0.53 rMSE=9.68, nMSE=4.34, CC=0.518, wR=0.53 rMSE=3.34, nMSE=4.34, CC=0.515, wR=0.53 rMSE=3.61, nMSE=4.34, CC=0.419, wR=0.53	Nested cross validation: 5-fold
17	Wang et al. (Wang et al., 2019//2019:)	gk-MTSFS	rs-fMRI	ADNI-1: 412 subjects (50 NC, 43 LMCI, 55 EMCI)	LMCI vs. EMCI EMCI vs. NC	ACC=76.5, SEN=67.2, SPE=82.7, AUC=0.81 ACC=76.9, SEN=76.3, SPE=77.2, AUC=0.79	Cross validation: 10-fold
18	Hao et al. (Hao, 2020)	MFCC-MK-SVM	MRI, PET	ADNI-1: 202 subjects (52 NC, 56 sMCI, 43 pMCI, 51 AD) ADNI-2: 913 subjects (211 NC, 82 SMC, 187 sMCI, 273 pMCI, 160 CE)	AD vs. NC (1) MCI vs. NC (1) sMCI vs. pMCI (1) AD vs. NC (2) sMCI vs. NC (2) sMCI vs. pMCI	ACC=97.60, SEN=98.43, SPE=96.73, AUC=0.98 ACC=84.47, SEN=94.04, SPE=66.15, AUC=0.81 ACC=77.76, SEN=67.44, SPE=85.54, AUC=0.76 ACC=93.72, SEN=95.17, SPE=91.81, AUC=0.95 ACC=78.47, SEN=85.88, SPE=70.16, AUC=0.78 ACC=73.87, SEN=90.55, SPE=49.52, AUC=0.70	MATLAB toolbox Cross validation: 10-fold
19	Hao et al. (Hao, 2023)	MSLPL	MRI, PET, AV45, SNPs	ADNI-2: 913 subjects (211 NC, 82 SMC, 187 sMCI, 273 pMCI, 160 CE)	AD vs. NC sMCI vs. NC sMCI vs. pMCI	ACC=95.14, SEN=96.21, SPE=93.75, AUC=0.95 ACC=82.85, SEN=86.73, SPE=78.61, AUC=0.75 ACC=76.91, SEN=89.01, SPE=59.36, AUC=0.71	Cross validation: 10-fold
20	Chen et al. (Chen et al., 2023)	OLFG	MRI, PET	ADNI-2: 746 patients (283 NC, 234 sMCI, 85 pMCI 144 CE) OASIS-3: 358 (329 NC, 29)	AD vs. NC, pMCI vs. sMCI MCI vs. NC	ACC=94.7, SEN=89.0, SPE=98.2, AUC=0.970 ACC=80.2, SEN=42.5, SPE=95.3, AUC=0.814 ACC=67.1, SEN=69.7, SPE=64.0 AUC=0.719	Cross validation: 10-fold Code: https://github.com/chenz96/OLFG
21	Shen et al. (Shen, 2021)	HDF-SR*	MRI, PET, CSF	ADNI (1): 306 patients (99 NC, 55 pMCI, 59 sMCI 93 AD) ADNI (2): 162 patients (51 NC, 31 pMCI, 30 sMCI, 50 AD)	ADNI (1) pMCI vs. sMCI ADNI (2) pMCI vs. sMCI	ACC=65.5, SEN=67.9, SPE=63.20, AUC=0.642 ACC=62.30, SEN=68.20, SPE=56.2, AUC=0.402	Cross validation: 10-fold
22	Zhou et al. (Zhou et al., 2020)	MMLS-SVM*	MRI, PET	AND-1: 774 patients (226 NC, 157 pMCI, 205 sMCI, 186 CE)	MCI vs. NC MCI vs. AD sMCI vs. pMCI	Not reported (only shown in Figures) Not reported (only shown in Figures)	Cross validation: 10-fold
23	de Mondonca and Ferrari (de Mendonça and Ferrari, 2023)	Texture + SVM*	MRI	AND-1: 474 patients (200 NC, 153 MCI, 121 CE)	NC vs. AD NC vs. MCI MCI vs. AD	ACC=83.8, AUC=0.92 ACC=74.1, AUC=0.81 ACC=75.4, AUC=0.74	Code: https://github.com/lucasjome/alzheimer Cross validation: 10-fold
24	Zhang et al. (Zhang et al., 2023)	TMTL-GB	Longitudinal MRI	ADNI: 456 subjects (M48) (120 NC, 334 MCI, 2 AD)	MMSE regression ADAS regression	nRMSE=0.2552, rRMSE=1.9356 nRMSE=0.1582, rRMSE=4.2781	Cross validation:5-fold
25	Tabarestani et al. (Tabarestani, 2020)	DMMTL*	MRI, PET, COG Scores, CSF	ADNI: 1620 subjects (864 NC, 415 MCI, 336 CE, 5 MCI$\gg$AD)	MMSE regression	RMSE$_{BL}$=1.62, RMSE$_{06}$ = 1.78, RMSE$_{12}$ = 2.24, RMSE$_{24}$ = 2.38, RMSE$_{36}$ = 2.28, RMSE$_{48}$ = 2.19	Training (70 %), testing (30 %). Inner cross validation:5-fold
26	Fang et al. (Fang, 2020)	GDCA	MRI, PET	ADNI: 906 subjects (251 NC, 297 EMCI, 196 LMCI, 162 CE)	NC vs. EMCI EMCI vs. LMCI	ACC=79.25, SEN=79.31, SPE=79.17, AUC=0.796 ACC=83.33, SEN=82.35, SPE=89.66, AUC=0.895	Training (80 %), validation (10 %), testing (10 %). Cross validation:9-fold
27	Liu et al. (Liu et al., 2018)	MT-SGL	Longitudinal MRI	ADNI: 788 subjects (255 NC, 390 MCI, 173 CE)	MMSE regression ADAS regression RAVLT regression	RMSE=2.14, CC=0.52, nMSE=4.38, wR=0.53 RMSE=6.82, CC=0.64, nMSE=4.38, wR=0.53 RMSE=9.68, CC=0.53, nMSE=4.38, wR=0.53	Leave 5 % out cross validation Code: https://bitbucket.org/XIAOLILIU/mtl-sgl
28	Lin et al. (Lin et al., 2021)	LDA-ELM*	MRI, FDG-PET, CSF, SNPs	ADNI: 623 subjects (200 NC, 208 sMCI, 110 pMCI, 105 CE)	AD vs. MCI vs. NC AD vs. pMCI vs. sMCI vs. NC	ACC=66.7, F1 = 64.9 ACC=57.3, F1 = 55.7	Training (80 %), testing (20 %). Cross validation:5-fold
29	Chen et al. (Chen et al., 2022)	LSFSIL	Longitudinal MRI	ADNI: 814 subjects (227 NC, 397 MCI, 190 CE)	MMSE regression ADAS regression CDR-SB regression	RMSE$_{BL}$=0.076, RMSE$_{24}$ = 0.1426, CC$_{BL}$=0.529, CC$_{24}$ = 0.6236 RMSE$_{BL}$=0.1329, RMSE$_{24}$ = 0.1454, CC$_{BL}$=0.6272, CC$_{24}$ = 0.649	Cross validation:5-fold Code: https://github.com/chenz96/LSFSIL

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Table 5 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
						$RMSE_{BL}$=0.1730, $RMSE_{24}$ = 0.1655, CC_{BL}=0.5496, CC_{24} = 0.6142	
30	Zhang et al. (Zhang et al., 2021)	MMC-SVM*	MRI, PET, CSF	ADNI: 120 subjects (40 NC, 42 MCI, 38 AD)	AD vs. MCI AD vs. NC NC vs. MCI AD vs. MCI vs. NC	ACC=89.63, MAP=91.09 ACC=94.58, MAP=95.68 ACC=81.02, MAP=82.20 ACC=88.54, MAP=87.81	Cross validation:5-fold
31	Dong et al. (Dong et al., 2022)	MMLFF*	MRI, PET	ADNI: 156 subjects (52 NC, 52 MCI, 52 AD)	AD vs. NC MCI vs. NC AD vs. MCI	ACC=90.74, SEN=90.14, SPE=88.16, AUC=0.964 ACC=74.73, SEN=73.62, SPE=74.48, AUC=0.775 ACC=75.54, SEN=77.52, SPE=78.54, AUC=0.855	Cross validation: 10-fold MATLAB toolbox
32	Qin et al. (Qin, 2022)	HMC*	MRI, COG Scores, demographic	ADNI: 1670 subjects (731 NC, 318 EMCI, 341 LMCI, 280 CE)	NC vs. EMCI LMCI vs. (NC+EMCI) AD vs. (NC+EMCI+LMCI)	ACC=96.7, SEN=93.4, SPE=100, AUC=0.983, F1 = 96.5 ACC=82.9, SEN=89.5, SPE=76.5, AUC=0.894, F1 = 83.9 ACC=98.4, SEN=99.6, SPE=97.2, AUC=0.994, F1 = 98.3	Cross validation: 10-fold MATLAB toolbox
33	Li et al. (Li et al., 2022)	MMFS-AG*	MRI, PET, CSF	ADNI: 202 subjects (52 NC, 43 pMCI, 56 sMCI, 51 AD)	AD vs. NC MCI vs. NC AD vs. MCI	ACC=86.33, SEN=87.22, SPE=86.90, AUC=0.938 ACC=73.86, SEN=10.15, SPE=87.20, AUC=0.332 ACC=77.92, SEN=72.34, SPE=82.38, AUC=0.820	Cross validation: 10-fold
34	Huang et al. (Huang et al., 2021)	T-GSRAM	MRI, SNPs	ADNI: 404 subjects (140 NC, 181 MCI, 83 AD)	Disease progression Top 10 ROIs Top 10 SNPs	RMSE=0.15 AUC are shown in Figures	2 nested cross validation loops (each 5-fold) Code: https://github.com/Meiyan88/T-GSRAM
35	Wang et al. (Wang et al., 2022)	JSCCA	MRI, SNPs	ADNI: 386 subjects (113 NC, 186 EMCI, 62 LMCI, 25 AD)	MMSE regression	Not reported (only shown in Figures)	Cross validation: 5-fold
36	Zhou et al. (Zhou et al., 2022)	MTL-LSA	MRI	ADNI: 314 subjects	ADAS regression MMSE regression	$nMSE_{4M}$=0.46, wR_{4M}=0.74, $nMSE_{12M}$=0.48, wR_{12M}=0.74 $nMSE_{4M}$=0.57, wR_{4M}=0.67, $nMSE_{12M}$=0.61, wR_{12M}=0.66	MATLAB toolbox Cross validation: 5-fold
37	Lei et al. (Lei, 2021)	AWCMT	rs-fMRI, DTI	Clinical dataset: 214 subjects (67 NC, 89 MCI, 58 SCD) ADNI: 386 subjects (38 NC, 35 MCI, 58 SCD)	NC vs. SCD SCD vs. MCI NC vs. MCI NC vs. MCI (ADNI)	ACC=84.80, SEN=84.48, SPE=85.07, F1 = 84.03, AUC=0.940 ACC=89.12, SEN=88.76, SPE=89.66, F1 = 90.91, AUC=0.954 ACC=87.82, SEN=88.76, SPE=86.57, F1 = 89.50, AUC=0.939 ACC=97.78, SEN=94.29, SPE=89.47, F1 = 91.43, AUC=0.965	MATLAB toolbox leave-one-out cross-validation
38	Shi et al. (Shi, 2022)	ASMFS	MRI, PET	ADNI: 202 subjects (52 NC, 56 sMCI, 43 pMCI, 51 AD)	AD vs. NC MCI vs. NC sMCI vs. pMCI	ACC=96.76, SEN=96.1, SPE=97.47, F1 = 96.63, AUC=0.970 ACC=80.73, SEN=85.98, SPE=70.90, F1 = 85.30, AUC=0.788 ACC=69.41, SEN=65.30, SPE=72.83, F1 = 63.98, AUC=0.653	Cross validation: 10-fold
39	Bi et al. (X. a. Bi, Z. Xing, W. Zhou, L. Li, and L. Xu, , 2022)	WERF	rs-fMRI, SNPs	ADNI: 63 subjects (37 EMCI, 26 LMCI)	EMCI vs. LMCI	ACC=88.9	MATLAB toolbox
40	Shao et al. (Shao et al., 2020)	Hypergraph	MRI, PET	ADNI: 831 subjects (160 NC, 273 EMCI, 187 LMCI, 160 CE)	AD vs. NC LMCI vs. NC LMCI vs. EMCI	ACC=92.51, SEN=94.08, SPE=90.44, AUC=0.946 ACC=82.53, SEN=86.09, SPE=78.55, AUC=0.803 ACC=75.48, SEN=83.84, SPE=63.26, AUC=0.703	Cross validation: 10-fold Code: https://github.com//shaoweinuaa//HM2TFS
41	Lei et al. (Lei, 2020)	SCNC-MTL*	rs-fMRI, DTI	ADNI: 386 subjects (44 NC, 44 EMCI, 38 LMCI, 44 SMC)	NC vs. SMC NC vs. EMCI NC vs. LMCI SMC vs. EMCI SMC vs. LMCI EMCI vs. LMCI	ACC=82.95, SEN=88.64, SPE=77.27, AUC=0.898 ACC=85.23, SEN=86.36, SPE=84.09, AUC=0.935 ACC=87.80, SEN=84.21, SPE=90.91, AUC=0.989 ACC=84.09, SEN=81.82, SPE=86.36, AUC=0.913 ACC=90.24, SEN=89.47, SPE=90.91, AUC=0.967 ACC=81.71, SEN=78.95, SPE=84.09, AUC=0.921	MATLAB toolbox leave-one-out cross-validation

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Table 5 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
42	Zhu et al. (Zhu et al., 2019)	SPMRM	MRI, PET, CSF	ADNI-1: 202 subjects (53 NC, 99 MCI, 50 AD) ADNI-2: 913 subjects (211 NC, 82 SMC, 272 EMCI, 187 LMCI, 160 CE)	AD vs. NC (1) MCI vs. NC (1) EMCI vs. LMCI (1) AD vs. NC (2) MCI vs. NC (2)	ACC=85.46, SEN=96.33, SPE=94.57, AUC=0.966 ACC=83.54, SEN=95.00, SPE=62.86, AUC=0.782 ACC=78.89, SEN=74.40, SPE=83.99, AUC=0.769 ACC=88.02, SEN=94.14, SPE=80.00, AUC=0.972 ACC=84.14, SEN=94.31, SPE=55.26, AUC=0.891	Cross validation: 10-fold
43	Zhang et al. (Zhang, 2019)	SSGSR	rs-fMRI	ADNI-2: 104 subjects (52 NC, 52 MCI)	MCI vs. NC	ACC=88.5, SEN=86.2, SPE=90.40, AUC=0.965	leave-one-out cross-validation
44	Cao et al. (Cao et al., 2018)	SMKMTL	MRI, COG Scores	ADNI: 788 subjects (225 NC, 390 MCI, 173 CE)	ADAS regression MMSE regression AD vs. NC pMCI vs. sMCI	rMSE=6.8, CC=0.69 rMSE=2.37, CC=0.57 ACC=93, SEN=91, SPE=87 ACC=71, SEN=74, SPE=69	Cross validation: 10-fold
45	Jiao et al. (xxxx)	FC2FS	MRI, PET	ADNI: 244 subjects (73 NC, 53 EMCI, 49 LMCI, 69 AD)	NC vs. AD NC vs. LMCI NC vs. EMCI EMCI vs. LMCI	ACC=91.85, SEN=91.07, SPE=92.27, AUC=0.929, F1 = 91.81 ACC=85.33, SEN=85.38, SPE=84.85, AUC=0.851, F1 = 87.45 ACC=78.29, SEN=82.02, SPE=74.73, AUC=0.780, F1 = 81.00 ACC=77.76, SEN=77.72, SPE=78.94, AUC=0.776, F1 = 78.35	Cross validation: 10-fold
46	Xu et al. (Xu et al., 2019)	SFL_LF*	MRI	ADNI: 528 subjects (165 NC, 95 sMCI, 126 pMCI, 142 CE)	NC vs. AD NC vs. MCI pMCI vs. sMCI AD vs. NC vs. MCI	ACC=90.40, SEN=92.36, SPE=88.67, AUC=0.954 ACC=70.89, SEN=61.39, SPE=78.42, AUC=0.790 ACC=963.69, SEN=78.56, SPE=45.39, AUC=0.769 ACC=59.16	Cross validation: 10-fold
47	Zhu et al. (Zhu et al., 2019)	SRSR*	MRI	ADNI: 805 subjects (226 NC, 393 MCI, 186 CE)	NC vs. AD NC vs. MCI pMCI vs. sMCI AD vs. NC vs. MCI	ACC=90.3, SEN=91.5, SPE=81.9, AUC=0.912 ACC=72.2, SEN=68.8, SPE=85.0, AUC=0.790 ACC=71.3, SEN=68.1, SPE=75.5, AUC=0.781 ACC=63.9	Cross validation: 5-fold
48	Zhang et al. (Zhang et al., 2021)	CMC	MRI	ADNI-2: 1118 subjects (287 NC, 110 SMC, 307 EMCI, 79 MCI, 176 LMCI, 159 CE) ADNI-3: 1118 subjects (408 NC, 48 SMC, 74 EMCI, 143 MCI, 35 LMCI, 44 AD)	AD progression on ADNI-2 (12 views) AD progression on ADNI-3 (12 views)	ACC=34.07, NMI=0.0213, F1 = 32.16, ARI=0.0087 ACC=57.26, NMI=0.0260, F1 = 59.68, ARI=0.0126	2 MRI views were used: Axial & Sagittal
49	Xiao et al. (Xiao, 2021)	SLR_ElasticNet*	MRI	ADNI: 197 subjects (50 NC, 45 sMCI, 51 cMCI, 51 AD)	NC vs. AD NC vs. MCI pMCI vs. sMCI	ACC=96.10, SEN=94.21, SPE=96.12 ACC=84.67, SEN=90.75, SPE=82.61 ACC=75.87, SEN=73.71, SPE=77.51	Cross validation: 10-fold
50	Zhu et al. (Zhu et al., 2019)	OLRDR	MRI, 18F-FDG PET	ADNI: 396 subjects (101 NC, 202 MCI, 55 sMCI, 63 pMCI, 93 AD)	NC vs. AD NC vs. MCI pMCI vs. sMCI AD vs. MCI	ACC=91.7, SEN=91.8, SPE=91.6, AUC=0.943 ACC=74.5, SEN=42.8, SPE=90.3, AUC=0.701 ACC=72.6, SEN=84.8, SPE=58.5, AUC=0.735 ACC=78.9, SEN=46.3, SPE=95.5, AUC=0.787	Cross validation: 10-fold
51	Cui et al. (Cui, 2021)	PSO-ALLR	MRI	ADNI: 197 subjects (50 NC, 45 sMCI, 51 pMCI, 51 AD)	NC vs. AD NC vs. MCI pMCI vs. sMCI	ACC=96.27, SEN=93.33, SPE=95.78 ACC=84.81, SEN=90.00, SPE=85.71 ACC=76.13, SEN=78.00, SPE=86.50	Cross validation: 10-fold
52	Chen et al. (Chen et al., 2024)	SMJFS	MRI	ADNI-1: 814 subjects (227 NC, 397 MCI, 190 CE) ADNI-2: 786 subjects (296 NC, 342 MCI, 148 CE) OASIS: 392 subjects (83 NC, 26 AD)	ADNI-1 ADNI-2 OASIS	BACC=51.27, nRMSE=0.1119, CC=0.6110 BACC=54.93, nRMSE=0.0945, CC=0.5819 BACC=76.13, nRMSE=0.0952, CC=0.5577	Cross validation: 10-fold Code: https://github.com/chenz96/SMJFS
53	Zhang et al. (Zhang et al., 2024)	SSMFS	MRI, PET	ADNI-1: 202 subjects (52 NC, 43 MCI-C, 56 MCI-NC, 51 AD)	AD vs. NC MCI vs. NC MCI-C vs. MCI-NC	ACC=92.62, SEN=91.87, SPE=93.39, F1 = 92.49, AUC=0.941 ACC=75.26, SEN=82.35, SPE=61.64, F1 = 81.31, AUC=0.738 ACC=67.80, SEN=57.93, SPE=75.37, F1 = 60.42, AUC=0.630	Cross validation: 10-fold

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Table 5 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
54	Zhang et al. (Zhang et al., 2024)	GRMFS	MRI, PET	ADNI-1: 202 subjects (52 NC, 43 MCI-C, 56 MCI-NC, 51 AD)	AD vs. NC MCI vs. NC MCI-C vs. MCI-NC	ACC=97.92, SEN=97.23, SPE=98.66, F1 = 97.83, AUC=0.980 ACC=86.66, SEN=91.46, SPE=77.76, F1 = 89.99, AUC=0.840 ACC=81.85, SEN=78.40, SPE=84.50, F1 = 77.91, AUC=0.775	Cross validation: 10-fold
55	Sheng et al. (Sheng, 2024)	ILHHO-KELM	MRI, PET, CSF	ADNI dataset: 202 subjects (52 NC, 99 MCI, 51 AD)	AD vs. NC MCI vs. AD MCI vs. NC	ACC=99.20, SPE=100.00, PRE=100.00, MCC=0.985 ACC=92.00, SPE=80.30, PRE=91.10, MCC=0.826 ACC=94.60, SPE=88.70, PRE=95.50, MCC=0.887	Cross validation: 10-fold

* These are not standard names given by the referenced authors.

capture the intrinsic global temporal correlation among multiple time-points to utilize the longitudinal disease progression information. LSA was further combined with sparse group Lasso to identify disease-related biomarkers and predict disease progression.

To enhance the learning performance of cognitive score prediction, Liang et al. (Liang et al., 2023) proposed a bi-graph guided self-paced multi-task feature learning (BGP-MTFL) framework to explore the relationship among multiple tasks. This method was regularized using $L_{2,1}$ and self-paced learning scheme and optimized using alternating direction method of multipliers combined with accelerated proximal gradient to solve the non-smooth objective function. Liu et al. (Liu et al., 2018) proposed a multi-task sparse group lasso (MT-SGL) method for cognitive score modeling based on MR data. This method considers the structured sparsity of parameters with both coupling across tasks and group selection for individual tasks. For efficient optimization of MF-SGL, authors adopted two different approaches; proximal averaging that takes average the solutions from the proximal operator for the individual regularizers and proximal composition which uses the composition of the proximal operators for individual regularizers. Chen et al. (Chen et al., 2022) presented a low-rank sparse feature selection with incomplete labels (LSFSIL) model to predict AD progression with incomplete labeled data. LSFSIL decomposes the original target matrix into two parts. The first part represents the real cognitive scores with complete label and is regarded as regression target while the second part is an L_1-norm regularized matrix to characterize the recovery error. Additionally, a manifold regularized was adopted using $L_{2,1}$ into the objective function to attain most discriminative features.

Group sparse regularization can effectively perform feature selection and tackle the overfitting problem while optimizing the weights (Grave et al., 2011). Therefore, Zhu et al. (Zhu et al., 2019) proposed an SLR model with group sparsity to classify disease from MRI data by combining two different regularizations; task-oriented and a self-representation-oriented. The former learns the relative importance of features for representation of response variables by a linear combination of the input features in a supervised way. The latter, uses an unsupervised learning approach to consider the inherent information among neuroimaging features. The selected features were eventually fed to an SVM classifier for disease classification. Another work by Tang et al. (Tang et al., 2022) generalized the MTL framework by presenting a feature-aware sparsity-inducing norm (FAS-norm) penalty to incorporate a useful correlation information among ROI features. They also developed an alternating direction method of multipliers to optimize the non-smooth objective function and guarantee its convergence.

Regularizers based on different variants of L_p-norm have been widely used in the literature to select most discriminative disease-related features. For instance, Xu et al. (Xu et al., 2019) proposed a manifold regularized sparse feature regression model to identify disease status from MRI data. They adopted L2,1 $-$norm regularization term to jointly select disease-related features among the samples. Then, they developed a manifold regularization term that could be adjusted using the relative distance of the label information to maintain samples separability and compactness of inter-class and intra-class, respectively. Finally, the SVM was used to perform binary and multi-class classification tasks. Also, Zhang et al. (Zhang et al., 2021) proposed a consensus multi-view clustering (CMC) model that used a nonnegative matrix factorization to combine multi-dimension features in multiple views of MRI data to automatically learn a consensus representation matrix to assist in predicting the AD progression. It adopted L_1 and L_2 norms to maintain sparsity and utilized the neighborhood patterns and relational matrix to fuse the complementary information extracted from multiple views. For the same purpose, Cui et al. (Cui, 2021) used a two stage-based framework, an adaptive LASSO logistic regression model based on particle swarm optimization (PSO-ALLR), to diagnose AD from MRI data. In the first stage, the evolutionary particle swarm optimization algorithm was adopted for global search to remove redundant features and reduce the computational time for the later stage. In the second stage, the adaptive

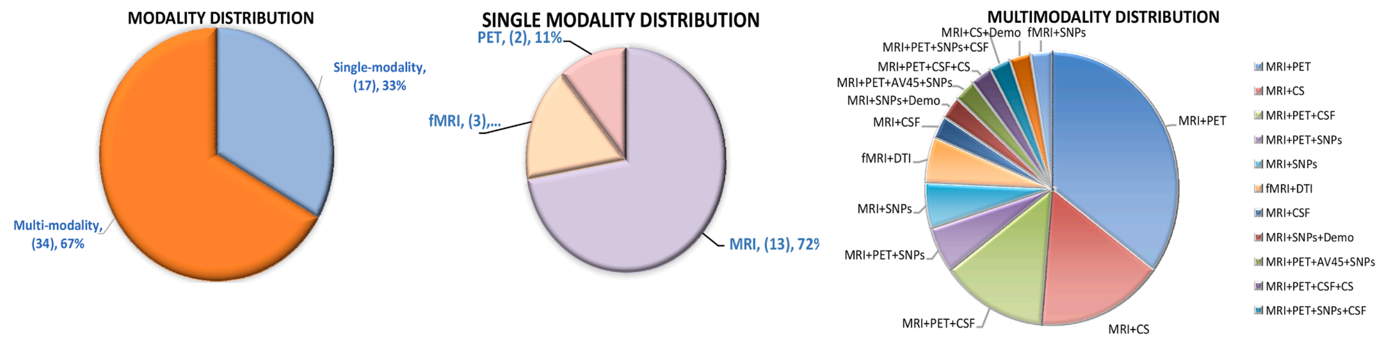

Fig. 3. Distribution of the modalities used in the ML-based studies.

LASSO was employed as a local search to select the most discriminative features for disease diagnosis. In another work, Xiao et al. (Xiao et al., 2021) presented an SLR method with the generalized elastic net to identify disease-related features from MRI data to improve diagnosis performance. The generalized elastic net combined L_p and L_2 regularizations to, respectively, produce sparse solutions and maintain the correlated brain ROIs in the solution. Recently, Chen et al. (Chen et al., 2024) introduced a shared manifold regularized joint feature selection (SMJFS) framework designed to perform both classification and regression for AD diagnosis. For classification, they developed an objective function that combines linear discriminant analysis and subspace sparsity regularization to acquire an informative feature subset. They also adaptively learned local data relationships based on samples' transformed distances to exploit the local data structure. For regression, the authors utilized the latent score space to address the issue of missing cognitive scores, capture correlations, and design a regression model with $L_{2,1}$-norm regularization, facilitating the feature selection.

Texture analysis approach has been used by de Mondonca and Ferrari (de Mendonça and Ferrari, 2023) to extract texture features from structural MRI using gray level co-occurrence matrix, the run length matrix, and the local binary patterns, as well as statistical moments and volume of multiple segmented brain regions. These features were used to build a graph that was then thresholded to remove irrelevant features and focus on the discriminative ones which were eventually classified using SVMs with different graph-kernels.

Since PET imaging modality could provide a rich information about brain functioning, several studies adopted it independently for AD diagnosis. For example, Pan et al. (Pan et al., 2019) proposed an ensemble method that used multi-level features from FDG-PET brain images to diagnose AD and MCI. They first extracted 3 levels of features of ROIs and their connectivities then applied a similarity-driven ranking method to perform feature dimension reduction. Finally, they trained different models using different types of features and the final decision was based on majority voting. In another study (Pan et al., 2019), instead of voxel-wise and ROI-wise features, they used a 2D Histogram of Oriented Gradient (HOG) descriptors to quantify spatial gradients of FDG-PET scans. Both small scale HOG and large-scale HOG features were extracted and then ranked. Eventually, an ensemble classification framework was designed for final prediction.

Other studies adopted resting-state fMRI (rs-fMRI) as single modality for detection and diagnosis of AD by utilizing their rich functional information. For instance, Yang et al. (Yang, 2021) utilized longitudinal rs-fMRI (baseline and year1) and proposed a fused sparse network model with parameter-free centralized learning (FSNPF) method to analyze MCI. This method performed feature selection via an MTL method to extract features from brain functional connectivity network which were fused and fed to an SVM for disease status prediction. In similar work, Zhang et al. (Zhang, 2019) presented a strength and similarity guided group sparse representation model (SSGSR) to integrates low- and high-level functional connectivity information from rs-fMRI data. This method estimated the high-order functional connectivity by measuring inter-subject low-order functional connectivity profile similarity as an additional source to guide the model. Moreover, this model was regularized by a novel term to encourage inter-subject brain network resemblance. Therefore, it could construct robust brain FCN that captures the inter-subject connectivity pattern (dis)similarity for better detection of group differences. Finally, Wang et al. (Wang et al., 2019// 2019:) presented a graph-kernel-based multi-task structured feature selection (gk-MTSFS) method that adopted the graph kernel to preserve the structural information of FCNs and utilized the MTL to attain the complementary information of multi-level thresholded FCNs. In this work, a graph-kernel based Laplacian regularizer was proposed to preserve the structural information of FCNs and an $L_{2,1}$-norm based group sparsity regularizer was used to select the discriminative features from multi-level FCNs for disease identification.

Cognitive score prediction was also studied in several works. For example, Liu et al. (Liu et al., 2019) proposed a fused group lasso regularized multi-task learning method (FGL-MTFL) via a non-smooth formulation and $L_{2,1}$-norm to consider the relatedness among features and tasks for cognitive scores prediction. Their method was optimized using an alternating direction method of multipliers to solve the proposed objective function. Similarly, Wang et al. (Wang et al., 2018) proposed a multi-layer multi-target regression (MMR) model for AD cognitive assessment. This method used a kernelized learning method to address the nonlinear relationships between the biomarkers and assessment scores. In addition, $L_{2,1}$ loss term and low-rank leaning were adopted to encode the inter-correlations among multiple assessment score. Another work by Wang et al. (Wang et al., 2019) introduced a multi-task exclusive relationship learning (MTERL) method to automatically capture the intrinsic relationship among tasks for disease progression prediction. It used an exclusive lasso regularization to encourage the partial group structure feature selection and a relationship induced regularization to capture the relationship among tasks. This study used longitudinal imaging data to estimate clinical measures and achieved good prediction performance. Lei et al. (Lei, 2020) proposed an adaptive sparse feature learning (ASL) framework using multiple brain parcellation atlases with various ROIs to diagnose neurodegenerative diseases. They extracted different features and fused them before performing feature selection. A least square regression model was constructed using different regularizers to predict the disease severity based on the cognitive scores.

7.2. Multi-modality studies

As mentioned earlier, multimodality studies tend to perform better than single modality counterparts. Therefore, several combinations of multimodalities have been widely explored in literature. MRI and PET neuroimaging data showed a promising combination of multimodal data. Zhu et al. (Zhu et al., 2019) utilized them and introduced an orthogonal low-rank dimensionality reduction (OLRDR) method for disease identification. This method first considers features variability by partitioning them into sub-classes using clustering to capture the multipeak distributional characteristics in the data. Then, it iteratively performs low-rank dimensionality reduction and orthogonal rotation in a sparse linear regression framework to select the disease-related features that are later classified using an SVM classifier. Hao et al. (Hao, 2020) proposed a multi-modal feature selection method with consistent metric constraint and a multi-kernel SVM method (MFCC-MK-SVM) for disease diagnosis by adopting the RF to calculate the similarity for each modality and extracting pairwise similarity measures from multiple modalities. In this method, group sparsity regularization term and sample similarity constraint regularization terms were used to constrain the objective function to attain robust features for better disease diagnosis. Recently, Zhang et al. (Zhang et al., 2024) proposed a self-paced semi-supervised multi-modal feature selection (SSMFS) method that projects MRI and PET data into a common label space with discriminative feature selection. Guided by prior multimodal similarity graphs, a unified graph is adaptively learned and embedded to preserve neighborhood structures. This method dynamically evaluates the discriminability and credibility of pseudo-labels and adaptively assigns a weight to each unlabeled sample through self-paced learning, reducing the negative influence of incorrect supervision. Finally, an MK-SVM is used to classify disease status. In another work (Zhang et al., 2024), they also proposed a multi-level graph-regularized robust multi-modal feature selection method (GRMFS) that performs noise-robust feature selection and adaptive multi-level similarity preservation, simultaneously. This method incorporates an ε-capped L_2-norm loss into the regression framework to enhance robustness against outliers. Additionally, GRMFS simultaneously learns intra-modal and inter-modal local similarities to explore and preserve the intrinsic MRI and PET local structures in the subspace, guiding the feature selection process effectively.

Shi et al. (Shi, 2022) proposed an adaptive-similarity-based multi-modality feature selection (ASMFS) to simultaneously perform similarity learning and feature selection from MRI and PET data. It learned the similarity matrix by jointly considering different modalities and conducted efficient feature selection by imposing group sparsity-inducing $L_{2,1}$-norm constraint. Therefore, ASMFS could attain the relationships across modalities and subjects through mining and fusing discriminative features from multimodal data for disease classification. In similar work, Peng et al. (Peng et al., 2019) developed a kernel-learning-based method for multimodal feature selection and integration for disease diagnosis. In this method, feature selection and multimodal data combination were integrated via a structured sparsity ($L_{1,p}$-norm) regularized kernel learning framework that performed both individual-level and group-level feature selection and fusion. Also, Ning et al. (Ning et al., 2021) proposed a relation-induced multi-modal shared representation learning method that integrated representation learning, dimension reduction, and classification into a unified framework. Bi-directional mapping between original space and shared space was used to obtain the shared representations of the data using several relational regularizers and auxiliary regularizers to encourage learning underlying inherent associations which were later projected into the target space for disease identification. Shao et al. (Shao et al., 2020) proposed a hypergraph based multi-task feature selection method to extract the complex relationships and the high-order structure in multimodal data (MRI and PET) for disease classification. They firstly performed feature selection on each modality as a single task and incorporated group-sparsity regularizer to jointly select common features across multiple modalities. Then, they introduced a hypergraph-based regularization term for the standard multi-task feature selection and adopted a multi-kernel SVM to fuse the selected features from different modalities to perform final prediction.

Learning in latent space has been used to build robust feature selection methods. For example, Chen et al. (Chen et al., 2023) proposed an orthogonal latent space learning with feature weighting and graph learning (OLFG) model that maps multiple modalities (MRI and PET) into a common latent space to capture the discriminative features for AD diagnosis. The authors devised a regularization term to preserve the data local structure in the latent space and integrate the graph construction into the learning processing. They also learned a joint graph to capture the correlations among modalities. Finally, the representations in the latent space are projected into the target space for disease diagnosis. Another work by Zhou et al. (Zhou et al., 2020) proposed a multi-modality latent space inducing ensemble SVM classifier for AD diagnosis. In this framework, the MRI and PET features were projected into the latent space to learn their latent representations. They also presented a complete multi-modality latent space and an incomplete multi-modality latent space learning models for complete and incomplete multi-modality data, respectively. Finally, disease classification is obtained using an ensemble SVM classifier. Dong et al. (Dong et al., 2022) presented a latent feature fusion method that utilizes MRI and PET data to learn a specific projection matrix for each modality by introducing a binary label matrix and local geometry constraints. Then, the original features of each modality are projected into a low-dimensional target space. Eventually, latent feature representations of different modalities are fused for disease identification via an SVM classifier. Han et al. (Han, 2022) presented a doubly supervised transfer classifier (DSTC) algorithm to exploit the knowledge among related MRI and PET data. DSTC integrated the SVM plus (SVM+) classifier and the low-rank representation into a unified framework. SVM+makes full use of the shared labels to guide the knowledge transfer between the paired data while the low-rank representation performs transfer learning between the paired data.

Jiao et al. (xxxx) developed feature correlation and feature structure fusion (FC2FS) that performs multi-modal (MRI and PET) feature selection for accurate disease classification. Firstly, the feature correlation regularization was constructed by combining a similarity matrix that reflected the relationships between different multi-modal features. Then, a manifold learning was employed to integrate the feature structure regularization based on the intrinsic structure of the features. These two regularization methods were embedded into the MTL model to effectively select the most relevant multi-modal features that were eventually fused and fed as input into an SVM for classification. Shen et al. (Shen, 2021) proposed a novel sparse regression method to fuse the auxiliary data with MRI and PET into the predictor data to classify MCI stages. This method jointly selected informative features from both auxiliary and predictor data via using L2,1-norm regularization to encourage row sparsity and utilized the features of AD and NC subjects as auxiliary information to improve feature selection for pMCI and sMCI classification. Moreover, they used subject's age to regress out the normal aging effect. For the same objective, Fang et al. (Fang, 2020) used MRI and PET data and proposed a supervised Gaussian discriminative component analysis (GDCA) method to capture the subtle changes of EMCI group in relation to the NC group. This method conducted a dimensionality reduction while considering the interclass information to define an optimal eigenspace that maximized the discriminability of selected eigenvectors. GDCA could be adopted with different classifiers and yield good classification performance.

Recent studies showed also that combining neuroimaging data with genetic could further boost disease identification and identify the disease-related gene variations. For Example, Zhou et al. (Zhou et al., 2019) proposed a joint latent-based representation learning method that utilized multimodality neuroimaging data (MRI, PET) and genetic data (SNPs), including samples with incomplete modality data. Then, the learned latent representation was projected for disease diagnosis. Also, Hao et al. (Hao, 2023) proposed a self-paced locality-preserving learning (MSLPL) framework to preserve the inherent structural relationships of multimodal data (MRI, PET, SNPs) and realize the sample selection process from "simple" to "complex". They adaptively estimated the contribution of each sample by weighting optimization during model training by self-paced learning. Additionally, they adopted a multi-kernel SVM to fuse the selected features for the final disease prediction.

Lin et al. (Lin et al., 2021) further added the CSF data to the previous modalities and proposed a method based on the linear discriminant analysis method to score and integrate multimodal data for efficient disease diagnosis. Specifically, they employed LASSO and principal component analysis to exclude irrelevant features then adopted linear discriminant analysis to individually calculate the score of the representative feature of each modality and the scores from different modalities, respectively. Finally, an ELM-based decision tree was utilized to perform multiclass diagnosis. Zhu et al. (Zhu et al., 2019) integrated multimodality neuroimaging data (MRI, PET) with CSF and proposed a self-paced sample weighting based multi-modal rank minimization (SPMRM) method that utilized multi-task feature selection to construct a more informative disease-related features. This method imposed a low-rank constraint on the weight matrix of the multimodal data to capture the intrinsic latent information across different modalities. Additionally, they adopted a self-paced learning to automatically estimate the importance of each sample. Finally, they used a multiple-kernel learning to classify the disease. Likewise, Li et al. (Li et al., 2022) proposed a multi-modal feature selection algorithm with anchor graph method to consider the inter-modal relationships among MRI, PET, and CSF and evaluate the importance of each modality as well as the local structure in the multimodal data. They first used the least square loss and $L_{2,1}-$ norm to estimate the weight of the feature under each modality. Then, a modal weight factor was embedded into the objective function to determine the significance of each modality. Finally, anchor graph method was presented to learn the local structure information in multimodal data. Using the same modalities, Zhang et al. (Zhang et al., 2021) proposed a multimodal multiclass classification framework based on SVM with embedding feature selection and fusion. It employed an L_{21}-norm regularization term combined with the multiclass hinge loss to select features across all the classes in each modality. It also adopted an L_p-norm regularization to perform multiple-kernel learning to effectively utilize complementary modalities. Authors also provided and verified a theorem to transform the minimization problem of the multiclass hinge loss with the L_{21}-norm and L_p-norm regularizations to a solvable optimization problem. In a recent work, Sheng et al. (Sheng, 2024) introduced a *meta*-heuristic ML model (ILHHO-KELM) that combines an enhanced Harris Hawks optimization algorithm (ILHHO), via iterative mapping to improve population diversity and local escaping operator to balance exploration–exploitation, with kernel extreme learning machine (KELM) classifier for simultaneous feature selection and classification of AD from MRI, PET, and CSF data. These enhancements are aimed at increasing population diversity and maintaining a balance between exploration and exploitation during the optimization process which is crucial for effectively extracting optimal diagnostic feature subsets.

Estimation of the cognitive status of the subjects and the underlying biological mechanisms is another significant task. Qin et al. (Qin, 2022) proposed a multi-class model that adopted the temporally constrained group sparse learning (tgLASSO) to select the optimal subset of features extracted from MRI data and patient' clinical scores. The selected features were classified using the AdaBoost algorithm to identify the key features of the disease. Brand et al. (Brand et al., 2020) proposed a joint multimodal longitudinal regression and classification (JMMLRC) method to identify the cognitive status of the subjects. They adopted several regularizers in the proposed objective function (non-smooth) and derived an efficient iterative algorithm to prove its convergence. Cao et al. (Cao et al., 2018) introduced a sparse multi-kernel based MTL (SMKMTL) framework with a joint sparsity-inducing regularization. This method was regularized using the $L_{2,1}$-L_1 norm to capture the nonlinear relationships between the MRI features and the cognitive scores. In addition, it projected the original feature vectors to a high-dimensional space using multiple non-linear mapping functions to perform regression task and perform the MTL in the multiple kernel space for modeling the disease's cognitive scores. In another work, Liu et al. (Liu et al., 2019) proposed a deep multi-task multi-channel learning (DM^{2}L) framework for simultaneous brain disease classification and clinical score regression. They extracted discriminative landmarks from MR images in a data-driven manner and the image patches around them. Then adopted a deep multi-task multi-channel CNN for joint disease classification and regression. Also, Tabarestani et al. (Tabarestani, 2020) proposed a distributed multimodal MTL method to predict AD progression. In this model, each individual task was defined as a regression model to estimate cognitive scores at a single time point and multitask regression models were developed to track the relationship between subsequent tasks. Then, they generalized the model by exploiting multiple the individual multitasks regression coefficient matrices of each modality and integrated them with complementary information to capture the most prevalent features. Eventually, the gradient boosting algorithm was adopted on the new feature space to improve the disease prediction.

Some studies also have investigated the fusion of structural neuroimages (MRI) with genetic data (SNPs) to either improve the diagnosis performance or identify the association between them. For instance, Wang et al. (Wang et al., 2022) utilized these two modalities and proposed two different regularization terms (fused pairwise group lasso and graph-guided pairwise group lasso) to build a joint sparse canonical correlation analysis (JSCCA) method that maintains chain relationship or graph network relationship within the same modality. They also utilized prior knowledge to construct a supervised bivariate learning model and used linear regression to select quantitative traits of images that are strongly correlated with the MMSE scores. Another study by Huang et al. (Huang et al., 2021) presented a temporal group sparsity regression and additive model (T-GSRAM) to identify associations between longitudinal imaging quantitative traits (QTs) and SNPs to detect potential AD biomarkers (MRI, SNPs). This model constructed a nonparametric regression model to analyze the nonlinear association between QTs and SNPs. Then, longitudinal QTs were used to identify the trajectory of imaging genetic patterns over time and incorporated group and individual levels information. The objective function of the T-GSRAM embedded three constraints (L_{21}, L_1, and L_2 norms) to analyze the changes in in single and all Qts across the brain as well as insure individual sparsity of the disease progression.

Fusing anatomical and functional neuroimage was also studied in the literature. Lei et al. (Lei, 2020) presented a low-rank self-calibrated network estimation method and joint MTL to diagnose MCI and its stages from rs-fMRI and DTI data. They developed a new functional brain network estimation method and introduced data quality indicators for self-calibration to improve data quality while completing brain network estimation. Then, a low-rank MTL method that was regularized via non-convex term to integrate and identify the most informative functional and structural features. Eventually, the SVM performed the classification task of the disease status. Later, they also presented an auto-weighted centralized multi-task (AWCMT) learning method to diagnose subjective cognitive decline (SCD) and MCI from structural and functional connectivity information (Lei, 2021). They separately and independently constructed two networks via a sparse and low-rank method to construct the functional brain network and a structural brain. The MTL could identify the most informative features by the integration of functional and structural connectivity and automatically learn each task's significance for accurate disease diagnosis.

Finally, another interesting study by Bi et al. (Bi et al., 2022) utilized the multi-omics data from fMRI and SNPs to explore the pathological mechanism of MCI development and provide accurate disease diagnosis. The authors proposed an ensemble learning model based on a weighted evolutionary random forest (WERF) to fuse multimodal data, perform feature selection, and diagnose MCI. They utilized the correlation analysis on sequence information extracted from ROIs and the digitalized gene sequences to construct the feature fusion of the samples.

8. Deep learning-based studies

In this Section, we present the most recent studies in the literature that adopted DL models, both single-and multi-modal data. We used the same PRISMA strategy as in the previous Section on the same sources but with DL-related keywords (Section 4). Following this filtration process, we initially retrieved 129 papers out of them 70 papers were retained and analyzed. Distribution of the modalities used in these papers as well as their DL models are shown in Figs. 4 and 5, respectively. The key findings of these studies are discussed below and summarized in Table 6.

8.1. Single-modality studies

The rich anatomical information of the MRI data and the accessibility of this modality made it widely available and used. Consequently, many DL-based models adopted it to diagnose, detect, and predict AD. Several CNN-based models have utilized it in the literature and achieved a remarkable performance. Wang et al. (Wang, 2019) presented an ensemble of 3D densely CNNs (3D-DenseNets) for disease diagnosis from MR images. They used dense connections to maximize the information flow through layers then adopted a probability-based fusion method to integrate the 3D-DenseNets with different architectures and varying hyperparameters. Likewise, Liu et al. (Liu, 2022) developed a 3D CNN model to distinguish subjects with AD and MCI from MRI data. Rashid et al. (Rashid et al., 2023) proposed a lightweight CNN-based model (Biceph-net) for AD diagnosis using 2D MRI scans rather than 3D volumes. This method could model both the intra-slice and inter-slice information as it has an inbuilt neighborhood-based model interpretation feature that could be beneficial to understand the classification decision taken by the model. Another study by Han et al. (Han et al., 2022) proposed a Broad Learning System (BLS) that used a feature mapping module to extract multi-scale informative features from MRI images and feature enhancement module that was adopted to extract abstract features. This model could integrate multi-scale convolution features and abstract features in an end-to-end learning to perform disease diagnosis with less computational cost.

Attention mechanisms could further improve model's efficiency in extracting informative features (Niu et al., 2021). This motivated Zhang et al. (Zhang et al., 2021) to propose a 3D connection-wise-attention-model-based densely connected CNN (CAM-CNN) to learn multi-scale multi-level features from MRI data. The connection-wise attention mechanism could attain compact high-level features from different convolution layers to be used for accurate disease classification. In similar work, Liu et al. (Liu et al., 2021) presented a shallow 3D attention residual network (3D ARNet) to perform 4-stages classification task of the AD. In this architecture, authors adopted an attention mechanism to attain the expressive information in brain MRI and apply the instance-batch normalization to highlight the global and local changes. Pei et al. (Pei et al., 2022) proposed a hierarchical pseudo-3D CNN model based on a kernel attention mechanism with a new global context block (PKG-Net). This method first extracted multi-scale features then an attention mechanism and global context blocks were applied to combine the features from different layers to hierarchically transform them into a compact high-level feature. It was trained using a joint loss function that integrates central loss and the angle margin loss. Also, Guan et al. (Guan et al., 2021) presented an unsupervised attention-guided deep domain adaptation (AD^2A) framework for multi-site MRI harmonization aiming at disease diagnosis from multi-site MRIs. The AD^2A consists of an MRI feature encoding module to extract representations of input images, an attention discovery module to automatically locate discriminative ROIs in each whole-brain MRI scan, and a domain transfer module trained with adversarial learning for knowledge transfer between the source and target domains. Liu et al. (Liu et al., 2023) presented a multiplane and multiscale feature-level fusion attention (MPS-FFA) network to identify the atrophy-affected regions due to the disease. This model captured and fused multiple pathological features from different MRI planes using feature encoder that adopted a multiscale feature extractor with hybrid attention layers. Then, a global attention classifier that combined clinical scores and two global attention layers was employed to evaluate the feature impact and balance the contributions of different feature blocks.

Finally, a feature similarity discriminator was used to minimize the feature similarities among heterogeneous labels to improve the network ability to discriminate atrophy features. Zhu et al. (Zhu, 2022) proposed a transformer-based model (BraInf) that utilized a multi-head self-attention block for representation learning to select the most representative features from the perspective of probability sparsity. Then, a structural distilling block was adopted to reduce the computational cost by maintaining the key features which were projected later into the target labels for disease diagnosis.

Qiao et al. (Qiao et al., 2021) developed a 3D CNN-based framework (Siamese network structure) with two types of contrastive losses to improve the diagnostic performance. Specifically, a group category (G-CAT) was proposed to learn the closer feature representation from images with the same classes and varying subject MMSE (S-MMSE) to explore subtle changes among individuals. Later, Li et al. (Li et al., 2022) proposed a multichannel contrastive learning using a 3D CNN based on the U-Net network to diagnose AD from brain 3D MRI images. They combined supervised and unsupervised contrastive learning to provide additional supervision information for the network to improve its performance and achieve accurate disease prediction. Recently, Qiao et al. (Qiao, 2024) developed a convolutional-squeeze-excitation-softmax-net (CSES-NET) model to effectively integrate various MRI feature for disease classification. The CSES-NET model combines voxel morphology, cortical features, and radiomics to create a comprehensive feature set from the MRI data of different MRI scales. Additionally, the model incorporates genetic data, facilitating a genome-wide association study to identify genetic variants associated with the disease. This combination of MRI features and genetic data enhances the understanding of the genetic underpinnings of the disease and improves classification performance.

Other studies have integrated the Fourier transform and Jacobian maps with CNN-based architectures. For instance, Zhang et al. (Zhang, et al., 2022) proposed a 3D global Fourier network (GF-Net) to utilize global frequency information that captures long-range dependency in the spatial domain from MRI modality. It contained a 3D discrete Fourier transform, an element-wise multiplication between frequency domain features and learnable global filters, and a 3D inverse Fourier transform. The GF-Net was trained by a multi-instance learning strategy to identify discriminative features. Abbas et al. (Qasim Abbas et al., 2023) proposed a novel 3D Jacobian domain CNN (JD-CNN) that calculated whole-brain Jacobian maps rather than ROIs from MRI data. The JD-CNN method could transform the brain's volumetric variations into meaningful transitions which were then exploited to identify disease related patterns without extracting any patches or ROIs. This method was trained based on the features generated by transforming the MRI from the spatial domain to the Jacobian domain and achieved a remarkable classification performance of AD. In another work, Liu et al. (Liu et al., 2023) presented a Monte Carlo Ensemble Neural Network (MCENN) that uses Monte Carlo sampling and an ensemble neural network. This model first utilized ResNet50 for initial feature extraction. MCENN was able to generate a large number of possible classification decisions via Monte Carlo sampling of feature importance within the combined slices. It could learn the 3D feature relevance across multiple slices from small number of 2D slices and minimal image processing.

Multiple DL models could also be combined with CNN for efficient feature learning. For example, Cao et al. (Cao, 2023) presented an end-to-end weekly-supervised automatic localization (AutoLoc) framework to jointly perform pathology detection and disease diagnosis using structured MRI data using CNN and LSTM. This method firstly used a pathology localization paradigm to predict the coordinate of the most disease-related ROIs rather than generating saliency maps slice. Then, they approximated the non-differentiable patch-cropping operation with the bilinear interpolation technique to enable the joint optimization of localization and diagnosis tasks. Hu et al. (Hu et al., 2023) combined CNN and Transformer models (VGG-TSwinformer) for short-term longitudinal study of MCI. They extracted low-level spatial

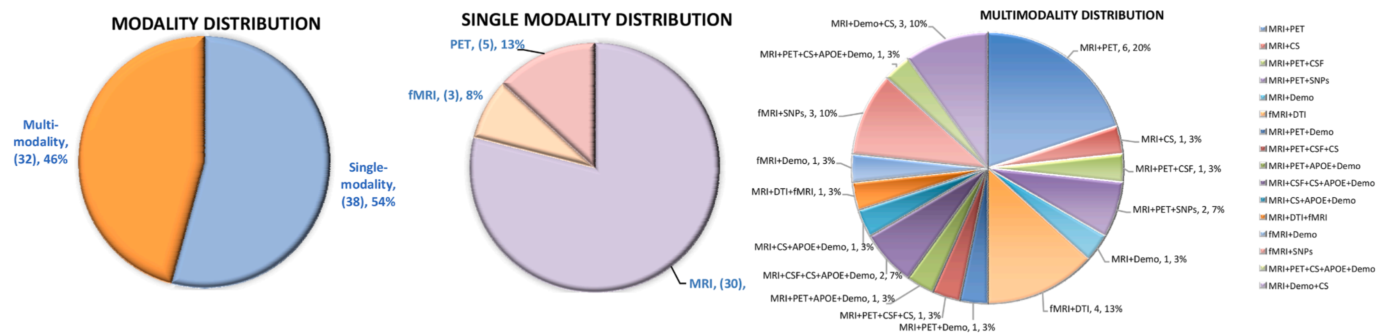

Fig. 4. Distribution of the modalities used in the DL-based studies.

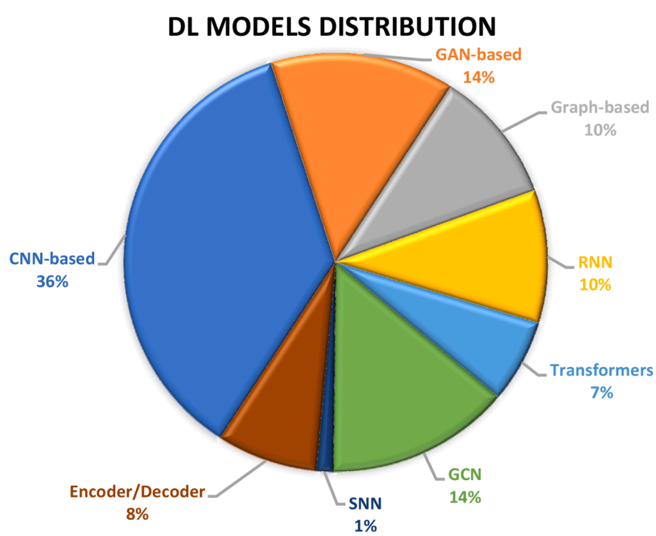

Fig. 5. Distribution of the various DL models used in the surveyed DL-based studies.

features of longitudinal structured MRI images using VGG-16 and mapped them to high-level feature representations. Then, a sliding-window attention and temporal attention were adopted for fine-grained fusion of spatially adjacent feature representations and linking longitudinal images to obtain information of brain structural changes in MCI patients. Lately, Khatri and Kwon (Khatri and Kwon, 2024) proposed a hybrid method that integrates convolution-attention mechanisms in transformer-based classifiers to enhance AD classification performance without excessive computing resources. The authors replaced the transformer's multi-head attention with a lightweight multi-head self-attention (LMHSA), utilized an inverted residual block, and presented a local feed-forward network.

To extract temporal information from longitudinal data, several studies adopted RNN-based models. For example, Ebrahimi et al. (Ebrahimi et al., 2021) analyzed LSTM, BiLSTM, and GRU models to use a sequence of features extracted by a pre-trained ResNet-18 to train a temporal convolutional network (TCN). This TCN could capture MRI features extracted from 2D MRI slices and understand the relationships among them to boost the disease classification.

Aiming at reducing the existing variations in multi-center raw MRI data, Cobbinah et al. (Cobbinah, 2022) presented a convolutional adversarial autoencoder (CAAE) that automatically aligned them into a common space. Then, they proposed a convolutional residual soft attention (CART) network to uncover the most disease-related ROIs for improved prediction performance. Later, Mulyadi et al. (Mulyadi et al., 2023) proposed an explainable AD likelihood map estimation (XADLiME) framework to model disease progression from MRI data. They established a set of topologically-aware prototypes using AD-spectrum-aware prototypical embedding network that adopted VAE with self-organized maps module and AD-spectrum-aware loss. In addition, they employed an estimating network to approximate the AD likelihood map which could improve model's explainability by revealing a thorough overview from clinical and morphological perspectives.

Other authors made use of GCN to attain better performance. For instance, Li et al. (Li et al., 2022) also proposed a dual interpretable (FSNet) to simultaneously select significant MRI features and samples aiming at improved disease diagnosis and interpretation. The FSNet network employed two modules that used $L_{2,1}$-norm to obtain feature and sample weight matrices for interpreting features and samples, respectively, and eventually a third module based on GCN was utilized for disease prediction. Zhu et al. (Zhu et al., 2022) coupled an interpretable feature learning module with another dynamic graph learning into a GCN architecture to be less influenced by the quality of the input graph. The former module selects the informative MRI features to provide interpretability for disease diagnosis and ignore redundant features while the latter adjusts the neighborhood relationship of every data point to yield robust node embedding. Ultimately, the GCN module outputs the diagnostic labels based on the learned inherent graph structure. Zhang et al. (Zhang et al., 2023) proposed a multi-relation reasoning network (MRN) to identify disease-related ROI through group difference analysis using graphs. They also designed a local reasoning module to better learn the pathological changes in disease-affected brain regions and their potential associations and interactions. The final disease classification was based on global reasoning method that could select discriminative ROI features based on structural MRI data. In another study, Xu et al. (Xu et al., 2024) introduced an interpretable AD diagnosis model called LA-GMF, which consists of two integrated modules. The first module, logits-constraint attention (LA), enhances diagnostic performance by directing the model's focus to critical areas and reducing inconsistencies between attention scores and class prediction probabilities. The second module, graph-based multi-scale fusion (GMF), combines GNN with a self-attention mechanism to explore correlations and complementarities between features at different scales, allowing for more comprehensive feature representation extraction. Together, these modules enable the LA-GMF framework to provide a more interpretable and effective analysis of AD by focusing on key diagnostic areas and leveraging multi-scale feature relationships.

Generative models have been extensively used in literature as potential solution for data scarcity and incomplete modality. Yu et al. (Yu, 2022) proposed a multidirectional perception GAN (MP-GAN) to visualize the morphological features from MRI data to measure disease severity at different stages. They devised a multidirectional mapping mechanism to capture the salient global features and delineate the subtle lesions via MR image transformations between the source domain and target domains. Also, they combined different loss terms (adversarial loss, classification loss, cycle consistency loss, and L_1 penalty) to effectively learn the class-discriminative maps for multiple classes. Bai et al. (Bai, 2022) presented a brain slice GAN for AD detection (BSGAN-

Table 6
Summary of the recent DL-based models indicating their used modalities, dataset size, performance, and other implementation details.

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
1	Pan et al. (Pan, 2021)	MiSePyNet (CNN-based)	^{18}F-FDG PET	ADNI: 1005 subjects (242 NC, 166 pMCI, 360 sMCI, 237 CE)	AD vs. NC pMCI vs. sMCI	ACC=93.13, SEN=90.32, SPE=95.49, AUC=0.971 ACC=83.05, SEN=72.12, SPE=88.06, AUC=0.868	Keras library + TensorFlow Training (60 %), validation (20 %), and testing (20 %).
2	Duan et al. (J. Duan, Y. Liu, H. Wu, J. Wang, L. Chen, and C. L. P. Chen, "Broad learning for early diagnosis of Alzheimer's disease using FDG-PET of the brain," Frontiers in Neuroscience, Original Research vol. 17, 2023)	BLADNet (CNN-based)	FDG PET	ADNI-1: 735 subjects (242 NC, 196 MCI, 297 CE) ADNI-2: 310 subjects (152 EMCI, 158 LMCI)	AD prediction MCI prediction CN prediction EMCI vs. LMCI	SEN=92.16, SPE=97.56, Pre = 94.00, F1 = 93.06 SEN=89.34, SPE=95.58, Pre = 91.60, F1 = 90.46 SEN=95.16, SPE=95.09, Pre = 91.47, F1 = 93.28 SEN=81.63, SPE=85.19, Pre = 83.33, F1 = 82.47	Radiologists' validation Code:https://github.com/YangLiuuuu/Extreme-Broad-Learning-System
3	Gao et al. (Gao et al., 2022)	TPA-GAN	MEI, PET	ADNI-1: 821 subjects (227 NC, 168 pMCI, 230 sMCI, 196 CE), 396 PET ADNI-2: 534 subjects (200 NC, 66 pMCI, 112 sMCI, 156 CE), PET 487	AD vs. CN pMCI vs. sMCI	ACC=92.0, SEN=89.1, SPE=94.0, AUC=0.956, F1 = 90.5 ACC=75.3, SEN=77.3, SPE=74.1, AUC=0.786, F1 = 69.9	PyTorch library
4	Zhang et al. (Zheng, 2022)	MMGL	MRI, PET, COG Scores, CSF	TADPOLE (ADNI): 603 subjects (211 NC, 275 sMCI, 45 pMCI, 72 AD)	AD vs. sMCI vs. NC sMCI vs. pMCI	ACC=92.31, AUC=0.939 ACC=92.30, SEN=93.26, SPE=91.50, AUC=0.924	Cross validation: 10-fold PyTorch library Code:https://github.com/SsGood/MMGL
5	Zhang et al. (Zhang et al., 2023)	MRN	MRI	ADNI-1, −2, −GO: 1662 subjects (459 NC, 768 MCI, 417 CE)	AD vs. CN MCI vs. CN AD vs. MCI vs. CN	ACC=92.57, SEN=83.33, SPE=96.69, AUC=0.976, F1 = 87.38 ACC=73.77, SEN=89.13, SPE=50.41, AUC=0.731, F1 = 80.39 ACC=63.23, SEN=59.34, SPE=78.13, AUC=0.751, F1 = 60.23	Cross validation: 5-fold PyTorch library
6	Zhu et al. (Zhu et al., 2022)	DGCN*	MRI	ADNI: 202 subjects (52 NC, 43 pMCI, 56 sMCI, 51 AD)	AD vs. NC AD vs. MCI NC vs. MCI sMCI vs. pMCI AD vs. NC vs. MCI AD vs. NC vs. sMCI vs. pMCI	Not reported (only shown in Figures)	Multiple data splitting
7	Zhu et al. (Zhu, 2022)	BraInf	MRI	ADNI: 959 subjects (324 NC, 319 MCI, 316 CE)	AD vs. CN MCI vs. CN	ACC=97.97, SEN=97.74, SPE=98.17, PRE=98.16 ACC=91.89, SEN=90.66, SPE=93.01, PRE=92.69	Cross validation: 10-fold PyTorch library
8	Cao et al. (Cao, 2023)	AutoLoc	MRI	ADNI: 1336 subjects (419 NC, 268 sMCI, 252 pMCI, 397 CE) AIBL: 103 subjects (48 NC, 55 AD)	AD vs. NC pMCI vs. sMCI AD vs. NC (AIBL)	ACC=93.38, SEN=92.94, SPE=93.80, AUC=0.967 ACC=81.12, SEN=83.69, SPE=78.71, AUC=0.873 ACC=91.46, SEN=87.73, SPE=95.79, AUC=0.971	Cross validation: 5-fold PyTorch library
9	Hu et al. (Hu et al., 2023)	VGG-TSwinformer	MRI	ADNI: 275 subjects (154 sMCI, 121 pMCI)	pMCI vs. sMCI	ACC=77.20, SEN=79.97, SPE=71.59, AUC=0.815	Training (65 %), Validation (20 %) and Testing (15 %)

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Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
10	Han et al. (Han et al., 2022)	BLS (CNN-based)	MRI	ADNI-1: 818 subjects (228 NC, 126 sMCI, 164 pMCI, 188 CE, 112 unMCI)	AD vs. NC pMCI vs. sMCI	ACC=91.83, SEN=90.47, SPE=93.00, PRE=91.50 ACC=75.52, SEN=74.96, SPE=76.18, PRE=77.04	Cross validation: 5-fold
11	Abdelaziz et al. (Abdelaziz et al., 2021)	CNN	MRI, PET, SNPs	ADNI-1: 817 subjects (226 NC, 126 sMCI, 167 pMCI, 186 CE, 112 unMCI)	NC vs. AD NC vs. sMCI NC vs. pMCI	ACC=98.22, SEN=97.78, SPE=98.76, PRE=98.99, F1 = 98.35 ACC=93.11, SEN=92.65, SPE=93.57, PRE=93.60, F1 = 93.07 ACC=97.35, SEN=97.82, SPE=96.71, PRE=97.58, F1 = 97.69	Keras library Cross validation: 10-fold
12	Abdelaziz et al. (Abdelaziz et al., 2022)	CNN, AE	MRI, PET	ADNI-1,2: 959 subjects (264 NC, 273 sMCI, 204 pMCI, and 218 CE)	AD vs. NC MCI vs. NC pMCI vs. sMCI	ACC=98.24, SEN=98.82, SPE=97.52, PRE=98.19, F1 = 98:43 ACC=94.95, SEN=90.26, SPE=96.98, PRE=94.95, F1 = 92.19 ACC=87.25, SEN=94.25, SPE=77.89, PRE=86.33, F1 = 89.49	Keras library Cross validation: 10-fold
13	Venugopalan et al. (Venugopalan et al., 2021)	Stacked denoising AE+CNN	MRI, health record, SNPs	ADNI: 2004 subjects (598 NC, 699 MCI, and 707 CE)	NC vs. MCI vs. AD	ACC=0.78, PRE=0.77, Recall = 0.78, F1 = 0.78	Cross validation: 10-fold
14	Liu et al. (Liu et al., 2023)	PDMML (CNN+LSTM)	MRI, PET, demographic	ADNI: 842 subjects (346 NC, 256 MCI, 240 CE)	AD vs. MCI vs. NC	ACC=80.8, PRE=81, REC=81, F1 = 81, CC=0.71, AUC=0.92	Cross validation: 5-fold
15	Zhu et al. (Zhu, 2023)	DMDIN	MRI, PET, CSF	ADNI-1: 202 subjects (52 NC, 43 pMCI, 56 sMCI, 51 AD) ADNI-2: 913 subjects (211 NC, 542 MCI, 160 CE)	AD vs. NC MCI vs. NC pMCI vs. sMCI AD vs. NC (ADNI2) MCI vs. NC (ADNI2)	ACC=96.75, SEN=98.57, SPE=95.50, AUC=0.984 ACC=86.56, SEN=93.34, SPE=72.38, AUC=0.837 ACC=84.38, SEN=80.54, SPE=86.90, AUC=0.773 ACC=92.45, SEN=94.27, SPE=90.44, AUC=0.963 ACC=86.75, SEN=95.15, SPE=67.86, AUC=0.912	Cross validation: 10-fold Code: https://github.com/xbl-NUAA/DMDIN.
16	Zhang et al. (Zhang, 2023)	LG-GNN	rs-fMRI, demographic	ADNI (1): 134 subjects (100 MCI, 34 AD) ADNI (2): 221 subjects (100 NC, 80 sMCI, 41 pMCI)	NC vs. MCI pMCI vs. sMCI	ACC=76.48, SEN=77.88, SPE=75.0, AUC=0.689, F1 = 78.23 ACC=79.23, SEN=87.50, SPE=63.0, AUC=0.670, F1 = 84.37	PyTorch library Cross validation: 10-fold
17	Li et al. (Li et al., 2022)	VAT-CNN	rs-fMRI	ADNI: 189 subjects (64 NC, 72 EMCI, 53 LMCI)	NC vs. EMCI EMCI vs. LMCI NC vs. EMCI vs. LMCI	ACC=90.9, SEN=90.4, SPE=91.4, AUC=0.967 ACC=89.8, SEN=87.6, SPE=91.4, AUC=0.940 ACC=82.7, SEN=82.1, AUC=0.866	Matlab + Pytorch Cross validation: 10-fold
18	Pei et al. (Pei et al., 2022)	PKG-Net	MRI	ADNI-1: 1237 (432 NC, 416 MCI, 389 CE) ADNI-2: 879 subjects (310 NC, 258 MCI, 311 CE)	AD vs. NC MCI vs. AD NC vs. MCI	ACC=97.28, REC=98.11, PRE=96.82 ACC=95.24, REC=93.98, PRE=96.54 ACC=96.93, REC=94.78, PRE=96.01	Cross validation: 10-fold
19	Zhang et al. (Zhang et al., 2021)	CAM-CNN	MRI	ADNI-1: 968 subjects (275 NC, 162 pMCI, 251 sMCI, 280 CE)	AD vs. NC pMCI vs. NC sMCI vs. pMCI	ACC=97.35, SEN=97.10, SPE=97.95, AUC=0.997 ACC=87.82, SEN=87.56, SPE=88.84, AUC=0.929 ACC=78.79, SEN=75.16, SPE=82.42, AUC=0.868	Training (70 %), Validation (15 %) and Testing (15 %) Keras library

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Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
20	Wang et al. (Wang, 2019)	3D-DenseNets	MRI	ADNI: 833 subjects (315 NC, 297 MCI, 221 CE)	AD vs. NC MCI vs. AD NC vs. MCI AD vs. NC vs. MCI	ACC=98.83, REC=98.70, PRE=98.70 ACC=93.61, REC=92.45, PRE=94.59 ACC=98.42, REC=98.34, PRE=98.37 ACC=97.52	Cross validation: 10-fold
21	Bai et al. (Bai, 2022)	BSGAN-ADD	MRI	ADNI: 818 subjects (271 NC, 303 MCI, 244 CE) OASIS: 416 subjects (316 NC, 100 CE)	AD vs. NC MCI vs. AD NC vs. MCI AD vs. NC vs. MCI	ACC=100, ACC=99.8 ACC=97.9, ACC=98.1 ACC=99.4, ACC=99.1 ACC=98.6, ACC=98.3	–
22	Liu et al. (Liu et al., 2019)	DM^2L	MRI, demographic	ADNI-1: 797 subjects (226 NC, 225sMCI, 165 pMCI, 181 CE) ADNI-2: 599 subjects (185 NC, 234sMCI, 37 pMCI, 143 CE) MIRIAD: 69 subjects (23 NC, 46 AD) AIBL: 519 subjects (447 NC, 72 AD)	ADNI-1≫ADNI-2 AD vs. pMCI vs. sMCI vs. NC Clinical scores Regression	$ACC=51.8, ACC_{NC}=60, ACC_{sMCI}=51.3, ACC_{pMCI}=24.3, ACC_{AD}=49$ CDRSB: CC=0.533, RMSE=1.666 ADAS11: CC=0.565, RMSE=6.2 ADAS13: CC=0.59, RMSE=8.537 MMSE: CC=0.567, RMSE=2.373	–
23	Qiao et al. (Qiao et al., 2021)	G-CAT S-MMSE	MRI	ADNI-1: 672 subjects (199 NC, 332 MCI, 141 CE) ADNI-2: 371 subjects (146 NC, 114 MCI, 111 CE) MIRIAD: 69 subjects (23 NC, 46 AD)	MIRIAD: AD vs. NC ADNI-1≫ADNI-2 AD vs. NC MCI vs. NC	ACC=92.8, SEN=94.3, SPE=94.8, AUC=0.971 ACC=93.5, SEN=96.6, SPE=94.9, AUC=0.978 ACC=89.3, SEN=89.5, SPE=88.7, AUC=0.942 ACC=89.6, SEN=90.9, SPE=83.3, AUC=0.949 ACC=75.8, SEN=77.2, SPE=65.9, AUC=0.825 ACC=78.1, SEN=67.5, SPE=74.0, AUC=0.823	Pytorch library Code: https://github.com/fengduqianhe/ADComparative
24	Ebrahimi et al. (Ebrahimi et al., 2021)	TCN	MRI	ADNI: 500 subjects (250 NC, 250 CE)	AD vs. NC	(TCN+ResNet-18): ACC=88, SEN=94, SPE=82 (TCN+4 residual blocks): ACC=91, SEN=94, SPE=88	MATLAB deep-learning toolbox
25	Guan et al. (Guan et al., 2021)	AD^2A	MRI	ADNI-1: 748 subjects (231 NC, 147 sMCI, 165 pMCI, 205 CE) ADNI-2: 708 subjects (205 NC, 253 sMCI, 88 pMCI, 162 CE) ADNI-3: 567 subjects (329 NC, 178 MCI, 60 AD) AIBL: 549 subjects (447 NC, 20 sMCI, 11 pMCI, 71 AD)	ADNI-1≫ADNI-2 AD vs. NC MCI vs. NC AD vs. MCI pMCI vs. sMCI ADNI-1 + 2≫ADNI-3 AD vs. MCI MCI vs. CN ADNI-1 + 2 ≫ AIBL AD vs. NC	ACC=89.92, SEN=87.65, SPE=91.70, AUC=0.940 ACC=67.03, SEN=63.64, SPE=72.68, AUC=0.703 ACC=75.55, SEN=46.29, SPE=89.44, AUC=0.777 ACC=78.01, SEN=53.41, SPE=86.56, AUC=0.788 ACC=82.35, SEN=50.00, SPE=93.25, AUC=0.776 ACC=71.60, SEN=44.94, SPE=86.02, AUC=0.692 ACC=90.35, SEN=87.32, SPE=90.83, AUC=0.954	Pytorch library Cross validation: 5-fold

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Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
26	Wang et al. (Wang et al., 2023)	HMGD	MRI, PET, SNPs	ADNI-2: 913 subjects (211 NC, 273 EMCI, 187 LMCI, 82 SMC, 160 CE)	AD vs. NC LMCI vs. NC EMCI vs. LMCI	ACC=96.48, SEN=97.83, SPE=94.68, AUC=0.97 ACC=81.35, SEN=88.46, SPE=73.16, AUC=0.81 ACC=76.38, SEN=93.36, SPE=52.15, AUC=0.72	Cross validation: 10-fold
27	Zuo et al. (Zuo et al., 2306)	HSCF	fMRI, DTI	ADNI: 216 subjects (72 NC, 72 EMCI, 72 LMCI)	NC vs. EMCI EMCI vs. LMCI	ACC=90.78, SEN=92.10, SPE=89.47, AUC=0.909 ACC=93.42, SEN=92.10, SPE=94.73, AUC=0.933	–
28	Tang et al. (Tang et al., 2023)	CsAGP	MRI, PET	ADNI: 766 subjects (326 NC, 226 MCI, 214 CE)	AD vs. CN AD vs. MCI CN vs. MCI AD vs. CN vs. MCI	ACC=99.04, SEN=97.96, SPE=99.54, AUC=0.998 ACC=97.43, SEN=94.25, SPE=98.81, AUC=0.992 ACC=98.57, SEN=98.52, SPE=98.61, AUC=0.998 ACC=98.72, SEN=98.65, SPE=99.34, AUC=0.999	Training (60 %), Validation (20 %) and Testing (20 %) Pytorch library
29	Aviles-Rivero et al. (Aviles-Rivero et al., 2022// 2022:)	MMHDP*	MRI, PET, demographic, APOE	ADNI: 500 subjects (NC, EMCI, LMCI, AD)	AD vs. NC EMCI vs. LMCI LMCI vs. NC AD vs. NC vs. MCI AD vs. NC vs. EMCI. Vs LMCI	ACC=92.11, SEN=92.80, PPV=91.33 ACC=85.22, SEN=86.40, PPV=84.02 ACC=82.01, SEN=84.01, PPV=81.80 Error rate = 16.25 Error rate = 18.31	–
30	Zhang et al. (Zhang, et al., 2022)	GF-Net	MRI	ADNI: 417 subjects (229 NC, 188 CE) ADNI: 382 subjects (320 NC, 62 AD)	AD vs. NC (ADNI) AD vs. NC (AIBL)	ACC=94.1, SEN=93.2, SPE=90.6, AUC=0.935, F1 = 91.8 ACC=93.2, SEN=93.3, SPE=94.6, AUC=0.938, F1 = 94.0	Training (60 %), Validation (20 %) and Testing (20 %) Pytorch library Code:https://github.com/qbmizsj/GFNet
31	Liu et al. (Liu et al., 2021// 2021:)	3D ARNET	MRI	ADNI-2: 200 EMCI ADNI-GO: 550 subjects (150 NC, 100 EMCI, 150 LMCI, 150 CE)	NC vs. EMCI vs. LMCI vs. AD	ACC=84.17	Training (80 %), Validation (20 %)
32	Narazani et al. (Narazani et al., 2022// 2022:)	3D ResNet	MRI, PET	ADNI: 1238 subjects (370 NC, 611 MCI, 257 CE)	AD vs. NC NC vs. MCI vs. AD	$BACC_{PET}$=90.5, $BACC_{MRI}$=86.6 $BACC_{early}$ = 88.5, $BACC_{middle}$ = 89.3, $BACC_{late}$ = 89.6 $BACC_{PET}$=54.1, $BACC_{MRI}$=53.6 $BACC_{early}$ = 57.3, $BACC_{middle}$ = 53.0, $BACC_{late}$ = 57.7	Training (65 %), Validation (15 %) and Testing (20 %)
33	Rahim et al. (Rahim et al., 2023)	3D CNN+BRNN	MRI, demographic, COG Scores	ADNI: 564 subjects (282 NC, 100 converted, 182 CE)	AD progression (BL) BL~M06 BL~M12	ACC=93, PRE=96, REC=88, mAUC=0.92 ACC=95, PRE=95, REC=92, mAUC=0.90 ACC=92, PRE=92, REC=92, mAUC=0.93	Pytorch library Stratified cross validation: 5-fold
34	Abbas et al. (Qasim Abbas et al., 2023)	JD-CNN	MRI	ADNI-1: 238 subjects (154 NC, 84 AD)	AD classification	ACC=96.61, SEN=97.83, SPE=95.92, AUC=0.983	Keras library Cross validation: 15-fold
35	Bi et al. (Bi, et al., 2022)	FAGCN	fMRI, SNPs	ADNI: 870 subjects (237 NC, 197 EMCI, 203 LMCI, 33 pEMCI)	NC vs. AD EMCI vs. LMCI LMCI vs. AD	ACC=89.36, SEN=88.46, SPE=90.48, BAC=89.47 ACC=85.00, SEN=86.36, SPE=83.33, BAC=84.85 ACC=86.36, SEN=85.71, SPE=86.67, BAC=86.19	Code: https://github.com/fmri123456/FAGCN.

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Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
36	Leng et al. (Leng, 2023)	MENet	MRI, FDG-PET	ADNI-1: 536 subjects (254 NC, 98 SMC, 184 CE)	AD vs. NC SMC vs. NC	ACC=97.67, SEN=98.21, SPE=98.55, AUC=0.982 ACC=81.63, SEN=72.15, SPE=85.29, AUC=0.986	Pytorch library Cross validation: 5-fold
37	Cobbinah et al. (Cobbinah, 2022)	CAAE+CRAT	MRI	ADNI-1, 2, GO: 1125 subjects (375 NC, 325 MCI, 325 CE)	AD vs. NC AD vs. MCI MCI vs. NC	ACC=93.10, SEN=95.40, SPE=96.11, AUC=0.947 ACC=91.41, SEN=92.32, SPE=93.12, AUC=0.922 ACC=89.26, SEN=89.45, SPE=91.13, AUC=0.923	Tensorflow library Stratified cross validation: 5-fold
38	Wang et al. (Wang et al., 2023)	MCNet	MRI, PET	ADNI-1: 543 subjects (165 NC, 144 sMCI, 116 pMCI, 118 CE) ADNI-2: 758 subjects (255 NC, 286 sMCI, 104 pMCI, 113 CE) ADNI-3: 86 subjects (73 sMCI, 13 pMCI) OASIS-3: 143 subjects (65NC, 48 sMCI, 23 pMCI, 7 AD)	MCI conversion (ADNI-1/2) MCI conversion (ADNI-3) OASIS	ACC=83.0, AUC=0.842, BAC=81.3 ACC=80.2, AUC=0.849, BAC=82.0 ACC=82.7, AUC=0.857, BAC=79.9	Pytorch library Code: https://github.com/Meiyan88/MCNET
39	Bi et al. (Bi et al., 2022)	IHGC-GAN	fMRI, SNPs	ADNI: 328 subjects (197 EMCI, 197 LMCI, 33 sEMCI, 33 pEMCI)	EMCI to LMCI	ACC=86.36, SEN=85.29, SPE=87.5, BAC=86.40	Code: https://github.com/fmri123456/IHGC-GAN.
40	Cui et al. (W. Cui et al., "BMNet: A New Region-Based Metric Learning Method for Early Alzheimer's Disease Identification With FDG-PET Images," Frontiers in Neuroscience, Original Research vol. 16, 2022)	BMNet	FDG-PET	ADNI: 898 subjects (263 NC, 290 EMCI, 147 198 CE)	AD vs. NC NC vs. LMCI LMCI vs. AD EMCI vs. LMCI	ACC=89.80, SEN=88.91, SPE=91.40, AUC=0.933, F1 = 91.8 ACC=81.46, SEN=84.96, SPE=76.05, AUC=0.810, F1 = 84.8 ACC=81.77, SEN=79.64, SPE=83.84, AUC=0.83, F1 = 78.57 ACC=79.40, SEN=83.22, SPE=72.43, AUC=0.771, F1 = 85.01	Pytorch library Cross validation: 5-fold
41	Li et al. (Li et al., 2022)	3D CNN-U-Net	MRI	ADNI: 928 subjects (330 NC, 299 MCI, 299 CE)	AD vs. NC MCI vs. NC	ACC=93.16, SEN=95.00, SPE=89.78, AUC=0.924, F1 = 94.72 ACC=80.44, SEN=83.18, SPE=78.59, AUC=0.809, F1 = 77.38	Training (60 %), Validation (20 %) and Testing (20 %) Cross validation: 3-fold
42	Gan et al. (Gan et al., 2021)	MGF*	rs-fMRI	ADNI: 107 subjects (48 NC, 59 AD)	AD vs. NC	ACC=88.84, SEN=89.55, SPE=88.25, AUC=0.902	Cross validation: 10-fold
43	Li et al. (Li et al., 2022)	FSNet	MRI	ADNI: 805 subjects (226 NC, 226 pMCI, 167 sMCI, 186 CE)	AD vs. NC AD vs. MCI NC vs. MCI EMCI vs. LMCI	ACC=84.4, SEN=83.6, SPE=85.9, AUC=0.843 ACC=73.6, SEN=74.4, SPE=72.5, AUC=0.749 ACC=71.8, SEN=68.4, SPE=73.7, AUC=0.720 ACC=70.2, SEN=72.0, SPE=67.4, AUC=0.705	Pytorch library Cross validation: 5-fold
44	Li et al. (Li, 2023)	QSGCN	PET	ADNI: 230 subjects (113 NC, 117 MCI)	NC vs. MCI	ACC=93.5, SEN=90.3, SPE=89.5, AUC=0.922	TensorFlow and Keras libraries Multiple cross validation
45	Liu et al. (Liu et al., 2023)	MCENN	MRI	ADNI: 2048 subjects (546 NC, 910 MCI, 592 CE) OASIS-3: 1088 subjects (712 NC, 102 MCI, 274 CE)	AD vs. NC (ADNI) NC vs. MCI (ADNI) AD vs. MCI (ADNI) AD vs. NC	ACC=90.0, SEN=83.5, SPE=96.9, AUC=0.914 ACC=68.7, SEN=71.5, SPE=63.3, AUC=0.691 ACC=73.5, SEN=69.5, SPE=75.7, AUC=0.778 ACC=82.6, SEN=74.3, SPE=84.4,	TensorFlow library Code: https://github.com/bieqa/Monte-Carlo-Ensemble-Neural-Network

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Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
				Clinical dataset: 239 subjects (104 NC, 85 MCI, 50 AD)	(OASIS-3) NC vs. MCI (OASIS-3) AD vs. MCI (OASIS-3) AD vs. NC (Clinical) NC vs. MCI (Clinical) AD vs. MCI (Clinical)	AUC=0.844 ACC=73.6, SEN=54.3, SPE=75.1, AUC=0.672 ACC=58.6, SEN=54.1, SPE=70.5, AUC=0.629 ACC=92.2, SEN=84.8, SPE=95.6, AUC=0.967 ACC=81.5, SEN=83.8, SPE=796, AUC=0.874 ACC=74.5, SEN=67.2, SPE=78.1, AUC=0.814	
46	Arco et al. (Arco et al., 2023)	SNN-SVM*	18F-FDG PET	ADNI: 241 subjects (68 NC, 39 pMCI, 64 sMCI, 70 AD)	AD vs. NC pMCI vs. NC sMCI vs. NC sMCI vs. pMCI	BACC=98.95, SEN=97.88, SPE=98.02, AUC=0.98, F1 = 98.67 BACC=94.01, SEN=96.92, SPE=91.10, AUC=0.98, F1 = 93.39 BACC=80.67, SEN=78.91, SPE=91.52, AUC=0.90, F1 = 82.03 BACC=78.41, SEN=96.38, SPE=61.43, AUC=0.91, F1 = 87.40	Torch, Numpy, Scikit-Learn Stratified cross validation: 5-fold
47	Zuo et al. (Zuo et al., 2305)	BSFL	DTI, rs-fMRI	ADNI: 241 subjects (82 NC, 82 SMC, 82 EMCI, 76 LMCI)	SMC vs. NC EMCI vs. NC LMCI vs. NC	ACC=94.30, SEN=93.42, SPE=95.12, AUC=0.981 ACC=90.85, SEN=93.90, SPE=87.80, AUC=0.979 ACC=85.97, SEN=96.34, SPE=75.61, AUC=0.939	TensorFlow library Cross validation:10-fold
48	Lei et al. (Lei, 2023)	MSE-GCN	DTI, rs-fMRI	ADNI: 184 subjects (67 NC, 77 EMCI, 40 LMCI)	EMCI vs. NC LMCI vs. NC LMCI vs. EMCI LMCI vs. EMCI vs. NC	ACC=85.42, SEN=86.57, SPE=84.42, F1 = 84.67 ACC=93.46, SEN=94.03, SPE=92.50, F1 = 94.74 ACC=92.31, SEN=93.51, SPE=90.00, F1 = 94.12 ACC=83.11, SEN=81.30, SPE=85.65, F1 = 83.62	TensorFlow library leave-one-out cross-validation
49	Yu et al. (Yu, 2022)	MP-GAN	MRI	ADNI-1: 417 subjects (213 NC, 204 CE) ADNI-GO: 158 subjects (16 NC, 142 EMCI) ADNI-2: 1118 subjects (271 NC, 121 SMC, 309 EMCI, 177 LMCI, 240 CE)	NC vs. SMC SMC vs. EMCI NC vs. EMCI EMCI vs. LMCI LMCI vs. AD	Not reported (only shown in Figures)	TensorFlow library
50	Yu et al. (Yu et al., 2022)	THS-GAN	MRI	ADNI: 833 subjects (315 NC, 297 MCI, 221 CE)	MCI vs. NC AD vs. MCI AD vs. NC	ACC=89.29, REC=96.67, AUC=0.887 ACC=85.71, REC=88.89, AUC=0.854 ACC=95.92, REC=95.83, AUC=0.959	TensorFlow library Training (80 %), Validation (10 %) and Testing (10 %)
51	Martí-Juan et al. (Martí-Juan et al., 2023)	MC-RVAE	MRI subcortical volume and thickness, COG Scores	ADNI: 897 subjects (229 NC, 324 MCI, 180 CE)	AD progression: Subcortical volume Subcortical volume Cognitive scores	MAE=0.61 MAE=0.63 MAE=0.14	Code: https://www.github.com/GerardMJuan/RNN-VAE
52	Mulyadi et al. (Mulyadi et al., 2023)	XADLiME	MRI	ADNI-1,2: 1540 subjects (433 NC, 497 sMCI, 251 pMCI, 359 CE)	AD vs. NC MCI vs. NC MCI vs. AD	BAC=91.83, F1 = 91.02, AUC=0.949 BAC=69.05, F1 = 77.47, AUC=0.752 BAC=0.722, F1 = 61.73, AUC=0.776	Code: https://github.com/ku-milab/XADLiME

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	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
				GARD (private): 745 subjects (261 NC, 375 MCI, 69 AD)	sMCI vs. pMCI CN/MCI/AD	BAC=67.85, F1 = 55.76, AUC=0.757 BAC=76.52, F1 = 61.03, AUC=0.765	
53	Huang et al. (Huang et al., 2022)	STA-BiGRU	rs-fMRI	ADNI: 382 (128 NC, 153 EMCI, 101 LMCI)	NC vs. EMCI EMCI vs. LMCI NC vs. EMCI vs. LMCI	ACC=90.2, SEN=92.1, SPE=88.0 ACC=90.0, SEN=89.7, SPE=90.1 ACC=81.5, ACC_{NC}=78.1, ACC_{EMCI}=87.8, ACC_{LMCI}=76.3	Cross validation: 5-fold Matlab, Pytorch library
54	Liu et al. (Liu et al., 2023)	MPS-FFA	MRI	ADNI-1,2,3: 1835 subjects (730 NC, 531 sMCI, 257 pMCI, 317 CE) AIBL: 239 subjects (85 NC, 69 sMCI, 11 pMCI, 74 AD)	AD vs. NC (ADNI) sMCI vs. pMCI (ADNI) AD vs. NC (AIBL)	ACC=97.7, SEN=96.8, SPE=98.5, AUC=0.977 ACC=88.3, SEN=84.0, SPE=94.4, AUC=0.892 ACC=94.9, SEN=92.9, SPE=97.2, AUC=0.951	Pytorch library https://github.com/LiuFei-AHU/MPSFFA
55	Tian et al. (Tian et al., 2023)	EH-GCN	MRI, DTI, fMRI	ADNI: 172 subjects (65 NC, 65 MCI, 42 AD) Clinical data: 346 subjects (102 NC, 95 MCI, 149 CE)	AD vs. NC AD vs. MCI MCI vs. NC	ACC=88.71, SEN=86.76, SPE=90.86, F1 = 88.94, AUC=0.888 ACC=82.71, SEN=83.33, SPE=81.85, F1 = 84.74, AUC=0.826 ACC=79.68, SEN=80.37, SPE=79.13, F1 = 77.94, AUC=0.798	Python language Cross validation: 5-fold
56	Rashid et al. (Rashid et al., 2023)	Biceph-net	MRI	ADNI+OASIS: 4095 subjects (1365 NC, 1365 MCI, 1365 CE)	AD vs. NC AD vs. MCI MCI vs. NC	ACC_{Axial} = 100, $ACC_{Coronal}$ = 99.45, $ACC_{Sagittal}$ = 97.80 ACC_{Axial} = 100, $ACC_{Coronal}$ = 99.45, $ACC_{Sagittal}$ = 97.80 ACC_{Axial} = 98.16, $ACC_{Coronal}$ = 97.80, $ACC_{Sagittal}$ = 97.31	Training (80 %), and Testing (20 %) Code:https://github.com/mtanveer1
57	El-Sappagh et al. (El-Sappagh et al., 2022)	Two-stage LSTM*	MRI, COG Scores, CSF, ApoE4, demographic	ADNI: 1371 subjects (419 NC, 473 sMCI, 140 pMCI, 339 CE)	4-class classification 3-class classification MCI conversion	ACC=82.80, PRE=78.74, RECALL=76.39, F1 = 82.80 ACC=91.22, PRE=91.28, RECALL=91.84, F1 = 91.22 MAE=0.1452, MSE=0.0568, RMSE=0.2382	Cross validation: 10-fold Code: https://github.com/hagersalehahmed/Alzheimer
58	Gao et al. (Gao et al., 2023)	BrainStatTrans-GAN	MRI	ADNI: 1054 subjects (431 NC, 264 MCI, 359 CE) AIBL: 340 subjects (261 NC, 79 AD) AIBL: 345 subjects (171 NC, 174 CE)	AD vs. NC MCI vs. AD	ACC=84.5, SEN=90.3, SPE=90.5, AUC=0.949, F1 = 89.2 ACC=78.7, SEN=78.1, SPE=79.2, AUC=0.860, F1 = 78.5	Pytorch library
59	Gao et al. (Gao et al., 2023)	MLG-GAN+Mul-T	MRI, PET	ADNI-1: 821 subjects (227 NC, 168 pMCI, 230 sMCI, 196 CE) ADNI-2: 534 subjects (200 NC, 66 pMCI, 112 sMCI, 156 CE) OASIS: 345 subjects (171 NC, 174 CE)	ADNI-2 > ADNI-1: AD vs. NC sMCI vs. pMCI OASIS: AD vs. NC	ACC=94.4, SEN=93.0, SPE=95.5, AUC=0.976 ACC=77.8, SEN=75.4, SPE=79.6, AUC=0.828 ACC=88.4, SEN=84.6, SPE=92.3, AUC=0.944	Cross validation: 5-fold Pytorch library
60	Qiang et al. (Qiang et al., 2023)	DANMLP	MRI, COG Scores, ApoE4, demographic	ADNI: 750 subjects (250 NC, 250 MCI, 250 CE)	AD vs. MCI MCI vs. NC	ACC=93.0, SEN=94.0, SPE=92.0, AUC=0.953 ACC=82.4, SEN=76.4, SPE=88.4, AUC=0.895	Cross validation: 5-fold Pytorch library
61	Zhang et al. (Cao, 2023)	IFDCGCN	MRI, PET, COG Scores, ApoE4, demographic	ADNI-1: 792 subjects (246 NC, 120 pMCI, 211 sMCI, 215 CE)	AD vs. NC sMCI vs. pMCI	ACC=96.68, SEN=99.19, SPE=94.49, AUC=0.968 ACC=78.00, SEN=54.96, SPE=89.37, AUC=0.721	Cross validation: 5-fold Training (70 %), Validation (15 %) and Testing (15 %)

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	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
62	Liu et al. (Liu, 2022)	3D CNN	MRI	ADNI: 2619 subjects (782 NC, 1089 MCI, 748 CE) NACC: 2025 subjects (1281 NC, 322 MCI, 422 CE)	NC classification MCI classification AD classification	AUC_{ADNI}=0.876, AUC_{NACC}=0.851 AUC_{ADNI}=0.626, AUC_{NACC}=0.625 AUC_{ADNI}=0.892, AUC_{NACC}=0.892	Code: https://github.com/NYUMedML/CNN_design_for_AD
63	Lei et al. (Lei, 2024)	FIL-DMGNN	Demographic, MRI, SNPs, CSF protein	ADNI dataset: 719 subjects (205 NC, 186 EMCI, 163 LMCI, 165 CE)	AD vs. NC EMCI vs. LMCI AD vs. MCI	ACC=93.24, PRE=93.27, REC=91.52, AUC=0.931 ACC=74.37, PRE=77.18, REC=64.24, AUC=0.737 ACC=82.69, PRE=76.80, REC=66.06, AUC=0.784	Cross validation: 5-fold Code: https://github.com/xiankantingqianxue/MIA-code.git
64	Khatri and Kwon (Khatri and Kwon, 2024)	LMHSA	MRI	ADNI dataset: 1075 subjects (390 NC, 370 MCI, 315 CE)	AD vs. MCI vs. NC AD vs. NC MCI vs. NC	ACC=94.31, SEN=97.14, SPE=94.11, F1 = 96.06 ACC=95.37, SEN=91.09, SPE=100, F1 = 0.889 ACC=92.15, SEN=89.92, SPE=94.56, F1 = 0.717	Training (70 %), validation (20 %), and testing (10 %).
65	Qiao et al. (Qiao, 2024)	CSES-MFF-NET	MRI, SNPs	ADNI-1: 421 subjects (224 NC, 197 CE) ADNI-2: 421 subjects (193 NC, 216 EMCI, 162 LMCI, 155 CE) ADNI-3: 392 subjects (329 NC, 63 AD)	AD vs. NC AD vs. LMCI EMCI vs. LMCI NC vs. LMCI NC vs. EMCI AD vs. EMCI	ACC=96.17, SEN=84.13, SPE=98.48, AUC=0.988 ACC=87.40, SEN=93.65, SPE=79.20, AUC=0.892 ACC=72.37, SEN=71.43, SPE=72.92, AUC=0.775 ACC=85.10, SEN=50.00, SPE=98.50, AUC=0.804 ACC=73.70, SEN=65.60, SPE=85.70, AUC=0.760 ACC=89.50, SEN=95.20, SPE=78.10, AUC=0.926	Cross validation: 10-fold
66	Xu et al. (Xu et al., 2024)	LA-GMF	MRI	ADNI-1: 684 subjects (214 NC, 286 MCI, 184 CE) ADNI-2: 651 subjects (279 NC, 232 MCI, 140 CE) AIBL: 531 subjects (456 NC, 75 AD)	ADNI: AD vs. NC AD vs. MCI NC vs. MCI AD vs. MCI vs. NC AIBL: AD vs. NC	ACC=93.02, F1 = 91.00, AUC=0.949 ACC=72.57, F1 = 59.48, AUC=0.732 ACC=70.33, F1 = 67.22, AUC=0.733 ACC=60.00, F1 = 59.07, AUC=0.763 ACC=92.24, F1 = 74.22, AUC=0.934	Cross validation: 5-fold Training (80 %), testing (20 %), Code: https://github.com/nollexu/LA-GMF
67	Zhu et al. (Zhu, 2024)	MCST-GCN	fMRI, DTI	ADNI-2,3: 227 subjects (82 NC, 76 SMC, 69 EMCI, 184 CE)	NC vs. SMV&EMCI NC vs. SMC NC vs. EMCI SMCI vs. EMCI	ACC=94.75, F1 = 92.32, AUC=0.902 ACC=91.71, F1 = 89.21, AUC=0.930 ACC=94.67, F1 = 93.32, AUC=0.916 ACC=90.52, F1 = 90.72, AUC=0.928	Training (90 %), testing (10 %). Cross validation: 10-fold Code: https://github.com/xbrainnet/OT-MCSTGCN
68	Han et al. (Han et al., 2024)	MMRN	MRI	ADNI-1: 757 subjects (229 NC, 175 pMCI, 165 sMCI, 188 CE) ADNI-2&3: 1193 subjects (538 NC, 109 pMCI, 323 sMCI, 223 CE) NACC: 1163 subjects (661 NC, 104 pMCI, 109 sMCI, 289 CE)	ADNI-2&3: AD vs. NC pMCI vs. sMCI NACC: AD vs. NC pMCI vs. sMCI	ACC=94.3, SEN=87.9, SPE=96.5, AUC=0.965 ACC=81.5, SEN=67.0, SPE=86.4, AUC=0.817 ACC=89.4, SEN=75.8, SPE=95.3, AUC=0.940 ACC=71.4, SEN=80.8, SPE=62.4, AUC=0.774	Cross validation: Stratified 5-fold

(*continued on next page*)

Table 6 (*continued*)

	Study	Method	Modality	Dataset details	Tasks	Performance	Remarks
69	Bi et al. (Bi, 2024)	CE-GAN	fMRI, SNPs	ADNI: 757 subjects (107 LMCI, 48 pMCI, 48 sMCI, 107 CE)	AD risk prediction	ACC=91.67, SEN=89.67, SPE=94.74, AUC=0.918	Cross validation: 5-fold Code: https://github.com/fmri123456/CE-GAN
70	Wu et al. Y (Wu et al., 2024)	DS-FSH	MRI, SNP, FDG-PET, AV45-PET	ADNI-1: 618 subjects (193 NC, 213 EMCI, 131 LMCI, 81 AD) ADNI-2: 913 subjects (211 NC, 82 SMC, 273 EMCI, 187 LMCI, 160 CE)	NC classification SMC classification EMCI classification LMCI classification AD classification	ACC=90.90, SEN=89.40, SPE=91.30 ACC=89.50, SEN=70.00, SPE=95.30 ACC=85.90, SEN=84.70, SPE=86.20 ACC=88.10, SEN=84.10, SPE=86.20 ACC=92.80, SEN=91.60, SPE=93.00	Cross validation: 5-fold Training ADNI-2, testing ADNI-1

* These are not standard names given by the referenced authors.

ADD) based on CNN aiming at extracting low dimensional high-level features of the disease status. During the training of BSGAN-ADD, the generator learns to integrate the disease status feedbacks from classifier into 2D-brain slice image reconstruction process for image enhancement. Then, a stacked CNN layers in the generator were used to extract high-level brain features from category-enhanced 2D-brain slices which are passed to the classifier in the prediction phase. Yu et al. (Yu et al., 2022) proposed a semi-supervised learning-based GAN (THS-GAN) based on a three-player cooperative game-based framework that was tensorized to utilize brain structural information for improved disease diagnosis. They also incorporated a high-order pooling scheme into the classifier to effectively utilize second-order statistics and capture long-range dependences between slices of different directions so that attain more informative features in a self-attention manner. Recently, Gao et al. (Gao et al., 2023) developed a brain status transferring GAN model (BrainStatTrans-GAN) to generate corresponding healthy images of patients to decode brain atrophy differences in AD subjects. It consists of generator, discriminator, and status discriminator. In addition to the generator and discriminators of the vanilla GAN, another a status discriminator was designed in the training process to produce normal brain images. The residual between the generated and input images was estimated to quantify the anatomical changes in the brain. This method was further leveraged by a residual-based multi-level fusion network for more accurate disease diagnosis. Lately, Han et al. (Han et al., 2024) developed a multi-template *meta*-information regularized network (MMRN) to account for diagnostic variations arising from different spatial transformations of various brain templates. They introduced a weakly supervised *meta*-information learning approach based on Info-GAN (Chen et al., 2016) to extract *meta*-information from latent embeddings obtained through self-supervised learning, thereby maintaining the discriminability of the encoder. Additionally, they employed mutual information minimization to enhance disentanglement, ensuring that class-related representations remain as *meta*-information agnostic as possible.

The PET imaging was adopted as a single modality by some authors to predict and diagnose AD using DL methods. For instance, Pan et al. (Pan, 2021) used 18F-FDG PET brain images and proposed a CNN-based model, multi-view separable pyramid network (MiSePyNet), to predict the conversion of MCI to AD and classification of AD from NC subjects. This architecture adopted the factorized convolution idea and deployed with separable CNNs, slice-, and spatial-wise CNNs, for each view. This model jointly considered information from different views (axial, coronal, and sagittal) without losing spatial information and achieved high conversion prediction accuracy. Also, Cui et al. (W. Cui et al., "BMNet: A New Region-Based Metric Learning Method for Early Alzheimer's Disease Identification With FDG-PET Images," Frontiers in Neuroscience, Original Research vol. 16, 2022) introduced a bilinear pooling and metric learning network (BMNet) to diagnose AD from FDG-PET images. It first constructed a shallow neural network as the baseline model then adopted a bilinear pooling module into the network to capture the inter-region representation features. It also used a deep metric learning losses in the loss function to help discriminate difficult samples in the embedding space. Likewise, Duan et al. (J. Duan, Y. Liu, H. Wu, J. Wang, L. Chen, and C. L. P. Chen, "Broad learning for early diagnosis of Alzheimer's disease using FDG-PET of the brain," Frontiers in Neuroscience, Original Research vol. 17, 2023) also used brain FDG-PET images and presented a broad learning network-based model for early diagnosis of AD called BLADNet. In BLADNet, authors employed a broad neural network to enhance the features of FDG-PET extracted via 2D CNN. This method adopted a broad learning system block to search for information over a broad space without retraining the whole network. Recently, Li et al. (Li, 2023) integrated a quadruple Siamese network and GCN with self-attention pooling module (QSGCN) to accurately identify MCI. They firstly constructed a multiple protein features network and higher-order protein features network from PET images then incorporated a pooling operation with self-attention mechanism into GCN to extract robust features. Another study by Arco et al. (Arco et al., 2023) presented a Siamese neural network (SNN) that received each pair of brain ROIs extracted from PET images and evaluated the functional differences as a symmetry measure which were used to assess disease classification. Then, they passed features extracted from the final layer of the SNN branches to an SVM classifier for final prediction.

Functional MRI was also exploited in recent studies to diagnose AD stages by analyzing the bountiful information of functional connectivity information. For instance, Gan et al. (Gan et al., 2021) presented a multi-graph fusion framework to explore both the common and the complementary information between a fully-connected FCN and a 1-nearest neighbor FCN. This framework conducted graph fusion at first to represent the rs-fMRI data with high discriminative ability and then utilized the L_1-SVM to jointly select ROIs and perform disease diagnosis. Huang et al. (Huang et al., 2022) proposed a spatio-temporal attention-based bidirectional GRU (STA-BiGRU) network to extract inherent spatio-temporal information from dynamic adaptive functional connectivity (dAFC) network for MCI diagnosis. This method adopted a group lasso-based Kalman filter algorithm to obtain accurate dAFC network and spatial and temporal attention modules to extract discriminative disease-related spatio-temporal information. Eventually, they used spatio-temporal regularizations to enhance the robustness of the model. In similar work, Li et al. (Li et al., 2022) proposed a virtual adversarial training CNN (VAT-CNN) to extract the local features of dynamic effective connectivity that was reconstructed using a group constrained Kalman filter from rs-fMRI data to identify MCI. To aggregate the dynamic effective connectivity features, they presented a high-order connectivity weight-guided graph attention networks (cwGAT) that incorporated the weight information of high-order connectivity into the attention mechanism. Recently, Zhang et al. (Zhang, 2023) proposed a local-to-global graph neural network (LG-GNN) to analyze and identify disease-related ROIs from rs-fMRI and the non-imaging data through designing a local ROI-GNN and a global Subject-GNN. The former could learn feature embeddings of local brain regions using a self-attention based pooling module while the latter learn the relationship among the subjects an adaptive weight aggregation block.

8.2. Multi-modality studies

Comparable to ML-based multimodal studies, plenty of DL-based studies exploited multimodal data for better diagnosis and analysis of AD and it stages. For instance, Abdelaziz et al. (Abdelaziz et al., 2021) adopted CNN to jointly classify disease status and regress clinical scores from multimodal fusion of neuroimaging and genetic data. To address the missing multimodalities, they adopted linear interpolation then the independently learned multimodal features were combined to identify target labels and clinical scores. Later, to achieve better performance, they utilized convolutional AE and CNN to extract original and synthetic image features from MRI and PET. Then, the extracted features were later fused and passed to fully connected layer for final disease classification (Abdelaziz et al., 2022). In similar study, Venugopalan et al. (Venugopalan et al., 2021) independently trained a stacked de-noising AE for feature learning of patients' health record and SNPs while adopting a 3D CNNs for MRI data. Narazani et al. (Narazani et al., 2022// 2022:) adopted the 3D ResNet architecture to evaluate the efficacy of single- and multi-modal data under different modes of data fusion; early, middle, and late fusion. The authors investigated the feasibility of multi-modal data by performing ablation studies using MRI and FDG-PET images. Interestingly, this study concluded that, FDG-PET alone was sufficient for disease diagnosis and could provide useful insights on the disease biomarkers. In similar work, Leng et al. (Leng, 2023) presented a lightweight multimodal cross enhanced fusion network (MENet) which was based on patch-based 3D CNN to adaptively recalibrate the multiscale long-range receptive field to localize discriminative brain ROIs for AD and SMC diagnosis. It could extract the intensity and location responses between MRI and FDG-PET neuroimages as an enhancement fusion for better diagnosis. This model requires fewer training parameters while attaining superior performance, thus it is computationally efficient.

CNN-based architectures could be further combined with recurrent models to capture longitudinal features and assess disease progression. For example, Rahim et al. (Rahim et al., 2023) proposed a hybrid longitudinal multimodal deep-learning framework that combines 3D CNN to extract the intra-slice features from MRI data and a bidirectional RNN (BRNN) to learn the inter-sequence patterns related to the disease progression. They also fused the longitudinal MRI data with demographic and cognitive scores. Moreover, they presented an explainability approach to help domain experts and practitioners understand the outcome of their model. In similar work, they presented a two-stage deep learning framework for AD progression detection and pMCI conversion time prediction based on information fusion of several longitudinal multimodal data (MRI, COG Scores, CSF, ApoE4, demographic) (El-Sappagh et al., 2022). In this work, they firstly used an LSTM model to perform multiclass classification task then adopted an LSTM regression task to predict the conversion time of MCI subjects. Another study by Wang et al. (Wang et al., 2023) introduced an end-to-end multi-view imputation and cross-attention network (MCNet) using minimal RNN-based network and adversarial learning to impute the missing data in longitudinal data (MRI and PET). In addition, they presented two cross-attention blocks to make full use of the potential associations in longitudinal and multimodal data. Then, they established an MTL model for data imputation, longitudinal classification, and MCI conversion prediction. Finally, Liu et al. (Liu et al., 2023) presented a patch-based deep multi-modal learning (PDMML) framework that adopted a discriminative location discovery approach to select normal ROIs without prior knowledge from MRI, PET, and demographic data. It learned the patch-level representation and global representation via CNN and Bi-LSTM networks, respectively. Specifically, the multimodality imaging features were fused at the patch level to get the multi-view representation of the brain then jointly learned the local patches to prevent the loss of spatial information. In a different study, Martí-Juan et al. (Martí-Juan et al., 2023) proposed a multi-channel model based on recurrent VAE (MC-RVAE) to capture spatio-temporal formation from multimodal data (MRI subcortical volume and thickness, COG Scores) to predict disease progression. They jointly modeled the dependencies across the channels of the network via a variational recurrent block to capture the temporal dependencies alongside every channel. It could capture disease progression, relationships among modalities, and reconstruct missing modalities over time.

Other studies adopted MLP with CNN architecture to capture more discriminative disease-related features. For instance, Qiang et al. (Qiang et al., 2023) developed a dual attention network and MLP (DANMLP) framework to integrate and diagnose AD from MRI, clinical data, and genetic data. It utilized a Patch-CNN to learn image features from each patch and adopted a position self-attention block to capture the dependencies between features within a patch. Then, it devised a channel self-attention block to capture the dependencies of inter-patch features and finally used two MLP networks for extracting the clinical features and outputting the disease labels. In similar work, Zhu et al. (Zhu, 2023) introduced a deep multi-modal discriminative and interpretability network (DMDIN) to combine samples in a discriminative common space and to identify potential ROIs from MRI, PET, and CSF data. They reconstructed each modality with a hierarchical representation using a deep encoder MLP and adopted a generalized canonical correlation analysis to congregate or separate samples based on their category. Eventually, they utilized knowledge distillation to reproduce coordinated representations and improve the interpretability of the DMDIN.

Generative models have demonstrated their effectiveness as a suitable solution to address the issue of multimodal data incompleteness. Gao et al. (Gao et al., 2022) proposed a task-induced pyramid and attention GAN (TPA-GAN) with a pathwise transfer dense convolution network. TPA-GAN integrated a pyramid convolution, attention module, and disease classification task into GAN to generate the missing PET from their MRI. Then, a dense convolution network with pathwise transfer blocks were used to learn and combine the multimodal features for final disease classification. Later, they developed a multi-level guided GAN (MLG-GAN) and multimodal transformer framework to, respectively, perform image generation for the incomplete data and classify disease status using both MRI and PET data (Gao et al., 2023). The MLG-GAN was guided by multi-level information from voxels, features, and tasks. It also used a feature-level auto-regression branch to embed the features of target images for an accurate image synthesis. Guided by the synthesized images, the multimodal transformer combined the global and local features and modeled the latent interactions and correlations from one modality to another with the cross-modal attention mechanism to classify the disease.

Generative models could be further integrated with graph learning methods. For instance, Bi et al. (Bi et al., 2022) proposed an influence hypergraph convolutional GAN (IHGC-GAN) to detect disease risk by constructing a hypergraph with genetic and fMRI data. Then, they built an influence transmission model to learn the associations between nodes and the transmission rule of disease information. It combined the GCN and GAN where the former was used as the generator for the latter to spread and update the lesion information of nodes in the hypergraph. In another study (Bi, et al., 2022), they also proposed a feature aggregation GCN (FAGCN) in which the interactions between brain regions and genes were modeled as node-to-node feature aggregation in a brain region–gene network. Additionally, they adopted a weight matrix based on correlation analysis to measure the similarity rather than the input from the adjacency matrix so that attain broader associations of nodes. Finally, a full-gradient saliency graph mechanism was adopted to score and extract the pathogenetic disease-related ROI and risk genes. Similarly, using fMRI and SNP data, Bi et al. (Bi, 2024) proposed an AD risk prediction method by developing a mathematical model to describe functional degeneration through community evolution driven by entropy information propagation. They introduced an interpretable community evolutionary GAN (CE-GAN) to predict disease risk by capturing the evolutionary patterns of the disease.

Many graph learning models were proposed to handle data heterogeneity and capture their complex relationships. For instance, Zhang et al. (Zheng, 2022) introduced an end-to-end multi-modal graph learning framework (MMGL) that was able to capture the shared and multimodal complementary information (MRI, PET, COG Scores, CSF); the modality-specified representation and the modality-shared representation considering inter-modal correlations. They also proposed a lightweight adaptive graph learning to reveal the intrinsic relations among subjects aiming at constructing an optimal graph structure for efficient disease prediction. Aviles-Rivero et al. (Aviles-Rivero et al., 2022// 2022:) proposed a semi-supervised hypergraph learning framework to extract higher-level relations among multimodal imaging and non-imaging data (MRI, PET, demographic, APOE). To preserve data semantics, they introduced a dual embedding strategy for constructing a robust hypergraph by enforcing perturbation invariance at the image and graph levels using a contrastive based mechanism. Then, a dynamically adjusted hypergraph diffusion model was adopted via a semi-explicit flow to improve the predictive uncertainty. Wang et al. (Wang et al., 2023) proposed a hypergraph-regularized multimodal learning by graph diffusion (HMGD) that used a graph diffusion method to measure the similarity among subjects with multimodal data (MRI, PET, and SNPs) and integrated them into a unified graph. Then, they employed the unified graph to represent the high-order similarity relationships among subjects and finally adopted a multi-kernel SVM (MK-SVM) to fuse the selected discriminative features for the disease diagnosis. Zuo et al. (Zuo et al., 2306) presented a hierarchical structural–functional connectivity fusing (HSCF) model to construct brain structural–functional connectivity matrices and predict abnormal brain connections based on fMRI and DTI modalities. They used the GCN to incorporate prior knowledge into the separators for disentangling the information of each modality. Also, they designed a hierarchical representation fusion module to maximize the combination of relevant and effective features between modalities for robust functional connectivity analysis. Zhang et al. (Cao, 2023) developed a population-based multimodal GNN to early predict AD from image features (MRI and PET) and phenotypic information. This method constructed brain networks and combined imaging data with phenotypic data to represent the relationships between features and subjects. This method further adopted a late fusion strategy in the multimodal GNN to combine the preliminary decisions made in each branch for final disease diagnosis. Similar study fused the structural and functional information from DTI and rs-fMRI data was proposed by Lei et al. (Lei, 2023) using a multi-scale enhanced GCN (MSE-GCN) for MCI diagnosis. Local weighted clustering coefficients was devised to combine brain connective networks as the feature vector as well as age and gender to construct a sparse population graph. They further devised various parallel GCN layers with multiple inputs using random walk embeddings in the GCN to explore the high order similarity features which are eventually fused for disease diagnosis. Lately, Zhu et al. (Zhu, 2024) utilized optimal transport theory to capture the topology evolution of dynamic brain networks and developed a multi-channel spatio-temporal GCN (MCST-GCN) to collaboratively extract temporal and spatial features from the evolution networks from fMRI and DTI modalities. The proposed GCN incorporates an attention mechanism to enhance the extraction of intrinsic temporal and spatial topology information. Ultimately, an MLP was employed to classify the dynamic brain networks. Likewise, Tian et al. (Tian et al., 2023) proposed an extensible hierarchical GCN (EH-GCN) that used multi-modal structural and functional features (MRI, DTI, fMRI) via multi-branch ResNet. This method embedded the image features with the internal brain connectivity features into a spatial population-based GCN to enhance the disease prediction performance. Further, Lei et al. (Lei, 2024) presented a framework (FIL-DMGNN) that fuses MRI, SNPs, protein, and clinical data using a feature induction learning module to alleviate data heterogeneity impact. Moreover, they presented a dual multilevel GNN to extract local and global salient multi-modal feature interaction information at different levels. Recently, Wu et al. (Wu et al., 2024) developed a deep self-reconstruction fusion similarity hashing (DS-FSH) method, regularized with a random walk graph, to learn nonlinear relationships and local similarities between samples. This approach effectively captures disease biomarkers from multi-modal data, including SNP, MRI, FDG-PET, and AV45-PET by mapping the reconstruction matrix into a common Hamming space using the FSH method. A deep sparse autoencoder classifier is then employed to fuse the learned multi-modal features for the final classification.

Recently, vision transformer achieved a remarkable performance in several applications which attracted researchers' attention in AD research. One of these studies was presented by Tang et al. (Tang et al., 2023) which developed a dual-branch vision transformer with cross-attention and graph pooling (CsAGP) method to capture multi-level feature interactions between MRI and PET modalities to extract a shared feature representation. They designed a cross-attention fusion module to attain discriminative features using two independent branches then fused them to enhance each other. Eventually, they developed a concise graph pooling algorithm to reduce token redundancy for improved disease diagnosis. Also, Zuo et al. (Zuo et al., 2305) proposed a brain structure–function fusing-representation learning (BSFL) model to integrate representative features from DTI and rs-fMRI modalities for MCI analysis. They designed a knowledge-aware Transformer to extract local and global connectivity features and a variational graph AE to decompose the feature space into unique and uniform spaces. Also, they devised a uniform-unique contrastive loss to further improve the effectiveness the structural–functional complementary features.

It is noteworthy that several DL models in Table 6 (e.g., (Cheng et al., 2022); (Peng et al., 2019); (Qin, 2022); (Zhang et al., 2021); (Bai, 2022), (Tang et al., 2023); (Rashid et al., 2023); (Cao, 2023) have achieved impressive performance metrics, exceeding 99 %. However, despite this high performance, reliability remains a concern. Although cross-validation is widely adopted in the surveyed studies, it might not be sufficient on its own to thoroughly evaluate the performance of different metrics. Instead, incorporating additional practices can provide a more comprehensive and reliable evaluation of a model's performance, ensuring that reported results are not only impressive but also robust and statistically significant. One such technique is bootstrapping, which can be used to estimate the variability of performance metrics and assess their statistical significance. This method involves resampling the data with replacement to create numerous simulated samples, thereby enabling the estimation of confidence intervals and variability of the metrics. Another approach is to evaluate the models on independent external datasets to confirm their generalizability beyond the training data. This helps to ensure that the model performs well not only on the data it was trained on but also on new, unseen data. Additionally, applying statistical tests such as paired *t*-tests or Wilcoxon signed-rank tests can compare performance metrics between models, ensuring that observed improvements are statistically significant and not due to random chance. Finally, encouraging the adoption of standardized reporting guidelines and data harmonization can further improve the transparency and reproducibility of DL studies in AD detection and diagnosis. These practices will help to build a more reliable foundation for transitioning these models from research environments to real-world clinical settings.

9. Discussion

Several ML and DL models have achieved remarkable performance in the literature. However, the state-of-the-art models for AD diagnosis leverage advanced DL techniques, particularly CNNs for analyzing MRI and PET images (Han et al., 2022; Pei et al., 2022; Zhu, 2022; Qasim Abbas et al., 2023; J. Duan, Y. Liu, H. Wu, J. Wang, L. Chen, and C. L. P. Chen, "Broad learning for early diagnosis of Alzheimer's disease using FDG-PET of the brain," Frontiers in Neuroscience, Original Research vol. 17, 2023; Leng, 2023; Tang et al., 2023), and RNNs or LSTMs for handling longitudinal data (El-Sappagh et al., 2022,; Liu et al., 2023; Lei, 2022; Ho et al., 2022). These models often incorporate attention mechanisms and transformers to improve feature extraction from complex multimodal data. Additionally, GANs are used to generate synthetic data to augment training datasets, especially in multimodal studies (Yu, 2022; Yu et al., 2022; Gao et al., 2023; Bi, 2024). Multimodal fusion models, such as multimodal autoencoders and GCNs (Lei, 2023; Cao, 2023; Zhu, 2024; Tian et al., 2023), could effectively integrate data from various sources like imaging, cerebrospinal fluid, and genetic data to enhance diagnostic accuracy and robustness. These cutting-edge models represent the forefront of research in AD diagnosis.

Although DL models have achieved state-of-the-art performance, researching traditional ML and shallow neural network models remains useful in certain cases. Traditional ML models, such as logistic regression, decision trees, and SVM, offer high interpretability, allowing researchers and clinicians to understand and trust the decision-making process. They are also less data-intensive compared to DL models, making them suitable for smaller datasets where DL models might overfit. Additionally, traditional ML models and shallow neural networks require less computational power and training time, making them accessible for researchers with limited resources. These models can serve as strong baselines for comparison with more complex models, ensuring that any observed performance improvements are genuinely due to advanced techniques rather than differences in model complexity. Lastly, in some cases, traditional ML models can achieve comparable performance to DL models, particularly when the problem is well-understood and the data is relatively simple.

There are still several challenges in recent studies that need to be addressed, which can be broadly categorized as data-based or model-based. Data-based challenges include data availability, modality heterogeneity, missing longitudinal and/or multimodal data, and data imbalance. Model-based challenges encompass model interpretability and explainability, reproducibility and applicability, and comparability and reliability. Below, we elaborate on these challenges and their potential solutions.

9.1. Data-based challenges

9.1.1. Data availability

Access to clinical data, including patient medical records, cognitive assessments, and neuroimaging scans, can be restricted due to privacy concerns and regulatory constraints. Researchers often face difficulties in obtaining sufficient and diverse AD data to train and validate their models. Besides, the quality of available data can vary, widely. Inconsistent data collection methods, missing data, and noise in clinical records can impact the performance of machine learning algorithms. In fact, high-quality and well-curated datasets are essential for robust model development. However, AD studies must adhere to strict ethical guidelines and informed consent protocols from patients. These ethical considerations can limit data availability and make data collection more challenging. Potentially, the aforementioned issues could be addressed via collaboration among multiple institutions and organizations to pool their data resources and create larger and more diverse datasets, like ADNI. Synthesizing data techniques can be used to augment limited datasets, creating additional samples that mimic real data while preserving privacy and confidentiality such as using AE-based (Mendoza-Léon et al., 2020; Tuan et al., 2021; Ferri, 2021; Zhang et al., 2020; Cobbinah, 2022; Martí-Juan et al., 2023; Chadebec et al., 2023; Shi, 2023) or GAN-based models (Islam and Zhang, 2020,; Zhang et al., 2022; Gao et al., 2022; Yu, 2022; Bai, 2022; Bi et al., 2022; Lan et al., 2021; Jung et al., 2023). Rigorous data preprocessing techniques can also help mitigate data quality issues, such as missing values and noise. Moreover, transfer learning techniques allow models trained on larger datasets to be fine-tuned on smaller collected AD datasets, which can help leverage available data more effectively (Cheng et al., 2022,; Mehmood, 2021; Sharma, 2022).

9.1.2. Missing multimodal and longitudinal data

Longitudinal and multimodal data are crucial for understanding the disease's progression, tracking patients' cognitive decline, and biomarker changes. However, gathering and maintaining such data can be resource-intensive. In addition, patients exhibit varying symptoms and undergo different diagnostic tests based on individual circumstances, leading to inherently fragmented data collection. Moreover, disparities in access to healthcare services mean that some individuals do not receive the same level or frequency of diagnostic attention. Finally, limited access to advanced diagnostic tools (like MRI or PET scanners) surely leads to inconsistent data collection practices. There are several strategies that might alleviate the above issues. Data imputation could be effective way by using statistical methods to estimate missing values based on observed data or using ML-based imputation to predict missing information using patterns learned from available data. Also, synthesizing missing modalities became a common approach to generate missing modalities from available ones, e.g. PET from MRI (Wang, 2018; Hu et al., 2022). Federated learning could mitigate the multicenter data issue by training different models across multiple institutions without needing to share raw data, thus preserving privacy while allowing for a more comprehensive training dataset, as in (Peng et al., 2022). Finally, knowledge transfer and transfer learning could be adopted to address the missing modalities as discussed earlier.

9.1.3. Data heterogeneity

Data heterogeneity poses a substantial challenge on building an accurate AD diagnostic model from multimodal data. In multimodal studies, data from patients' records, genomic, neuroimaging, and clinical assessment scores are typically collected. However, these data types are often collected from various sources, in different formats and with different methodologies, making consistent and unified analysis laborious. Therefore, there are always difficulties with inconsistency across different sources, integration of data in a meaningful way, deriving a biologically and clinically relevant insights from the integrated data, and ensuring that analyses across heterogeneous data are reproducible and verifiable. Adopting comprehensive protocols can alleviate the data heterogeneity challenge by adhering to common standards for data collection and recording. In addition, data harmonization techniques could be adopted to handle the potential variances due to different acquisition equipment and protocols as in (Fortin, 2018; Bilgel, 2023; Chen et al., 2023; Zhao, 2023). Likewise, algorithms that minimize batch effects and align data from different sources to a common scale or format could be helpful, e.g. ComBat algorithm (Johnson et al., 2007). Stage-learning strategy could also alleviate the effect of data heterogeneity as proposed in (Zhou et al., 2019).

9.1.4. Data imbalance

Data imbalance is another critical issue that commonly exists when building AD diagnostic models. This imbalance typically occurs when the number of observations in one class significantly outweighs the observations in another. For instance, there might be far fewer cases of confirmed AD in the dataset compared to healthy controls or individuals with other neurological conditions. This can severely compromise the learning process, leading to biased models that may over-predict the majority class and under-predict the minority, resulting in poor generalization performance on new or unseen data. This could mean the model might predict fewer instances of AD than there actually are, potentially leading to misdiagnosis or delayed treatment for patients. The simplest way to address this issue is oversampling the minority class using SMOTE algorithm (Chawla et al., 2002) or under sampling the majority class reducing the number of instances in the majority class to balance out the ratio. Another potential solution for data imbalance problem is adopting advanced ensemble methods like RF and boosting, particularly those designed to handle imbalance by weighting the classes differently during the learning process. Finally, SNN or models incorporating attention mechanisms, might be more robust against class imbalance (Arco et al., 2023; Liu, 2019; xxxx). They can be particularly useful when dealing with medical images where they focus on critical features irrespective of class distribution.

9.2. Model-based challenges

9.2.1. Model interpretability and explainability

Model interpretability and explainability are related concepts, however they are slightly different. Interpretability means that the model's decision-making process is inherently clear and could be understood without requiring additional or external explanation. It is more about the transparency of the system and has a significant concern to understand and explain the decisions. Explainability is the extent to which an external observer (a human) can understand why a model made a specific decision. It often involves the use of additional tools or techniques to provide insights into the model's decision-making process. Explainable AI tries to make black-box models more transparent by creating interfaces that help humans understand the model's decisions or by using surrogate models that approximate the black-box model's decisions in an interpretable way. The issue of interpretability and explainability is particularly sensitive in healthcare, where model decisions can significantly impact patient lives, healthcare provider trust, and the acceptance of such technology in clinical practices. Although it has been recently explored in some studies (Hernandez et al., 2022; Qiu, 2020; Li et al., 2023; Zhou et al., 2022; Lee et al., 2019; Zhang et al., 2023; Mulyadi et al., 2023; Rahim et al., 2023; Zhu, 2023), there are some issues in this research. AD progression and symptoms can vary greatly among individuals as the disease has a complex interplay of genetic, environmental, and lifestyle factors, which can be difficult to fully capture and interpret by the models. If a model identifies patterns not clearly understood by medical professionals, it is challenging to trust the model's decisions without interpretable reasoning. Indeed, Clinicians not only need to trust the AI's diagnosis (trust often gained through interpretability), but they also need to be able to understand the 'why' behind that diagnosis (explainability) to combine this information with their clinical judgment. For instance, DL-based models could provide high accuracy, as shown in Table 6, however, they are not easy to understand for humans. One solution to solve this problem is using simpler models (e.g., linear regression or decision trees) which might be preferable if it offers sufficient accuracy with far greater interpretability than more complex models. Another solution is developing techniques to highlight which input features (e.g., ROIs or particular SNPs) that are most influencing the model's decisions. This insight can sometimes provide a form of interpretability. Finally, developing models that come with built-in explainability while maintaining the performance, rather than only providing an output, would also provide reasoning in a form that humans can understand.

9.2.2. Reproducibility and applicability

A "reproducible model" refers to a model whose results can be consistently duplicated using the same data and methods. Reproducibility is a fundamental principle as it lends credibility to the findings and allows for the progression of studies based on consistent and reliable foundations. However, reproducing some models can be resource-prohibitive and challenging due to the data and software availability. For instance, we only see from Tables 5 and 6 that, there are only 20 % and 25.7 % of the codes shared by authors for ML- and DL-models, respectively. Moreover, there is often a lack of standardization in how data is collected, processed, and used for training. In additional, many models might be initialized with random weights, or data might be shuffled randomly during training. Finally, detailed reporting on the methodology used to develop and train the model can often be insufficient for full reproduction due to various reasons, including page limitations in scientific publications or proprietary interests.

To the best of our knowledge, no model for AD detection and diagnosis has been put into clinical use. Instead, many models have been deployed in clinical settings and imaging equipment to enhance image quality through noise removal and bias field correction, assisting physicists. Similarly, registration techniques are used clinically to align multimodal and longitudinal images. Transitioning AD diagnosis and detection models from research environments, where they are developed and trained, to real-world clinical settings involves several non-trivial considerations and hurdles, making the process complex and challenging. The major problem is that, the model developed in a research setting is typically trained on a specific dataset that may not represent the broader diversity of the real world. For instance, a model might be trained on data from a specific population, and it is unknown whether this model will perform as well with different demographics (varying by ethnicity, age, socioeconomic status, etc.), which can be quite common in diverse settings like hospitals. Another problem is the integration with the clinical workflow. For a model to be adopted and used effectively, it must fit seamlessly within the existing clinical workflow. This means it must work with the current information technology systems, follow the clinic's operational practices, and not disrupt the healthcare providers' work. This integration can be technically challenging and might meet resistance from staff who are wary of adopting new technologies in diagnosis. Finally, ethical and regulatory approvals are necessary to ensure that the system complies with ethical implications as well as meets minimum safety and efficacy standards. Addressing the previous problems requires a concerted effort across multiple domains, including ML researchers, clinicians, software developers, regulatory experts, and ethical committees. This multidisciplinary approach is essential to ensuring that these models can transition successfully from research prototypes into effective, safe, and reliable clinical tools. Yet, it is still undergoing work.

9.2.3. Comparability and reliability

Ensuring the comparability of AD diagnosis models is crucial for the validity and applicability of their research findings. Although most models in Tables 5 and 6 utilized the ADNI dataset as a benchmark and employed similar evaluation metrics, comparability remains challenging due to variations in the number of samples and modalities used, different preprocessing procedures, and the complexity of the models. For instance, in the AD vs. NC task, the model proposed by Shi et al. (Shi, 2022) achieved the highest accuracy among ML models (96.76 %), while de Mondonca and Ferrari (de Mendonça and Ferrari, 2023) achieved the lowest accuracy (83.80 %). This challenge is further exacerbated when comparing traditional ML models with DL models, given their differences in complexity, training time, resources, and data requirements. Additionally, unlike DL models, the interpretability of ML models is often overlooked when evaluations are based solely on performance metrics. Therefore, using standardized datasets, consistent evaluation metrics, and uniform preprocessing techniques is necessary to achieve meaningful comparability.

Performance reliability of AD diagnosis models is another critical issue to guarantee their effectiveness and generalizability in clinical applications. While several AD diagnosis models have made significant strides and shown promising performance, their reliability is not yet fully established. In fact, reliable performance requires robust validation techniques. Although cross-validation is widely adopted in most surveyed studies, it might not be sufficient on its own to thoroughly evaluate the performance of different metrics (Gorriz et al., 2401). For instance, the models in (Cheng et al., 2022,; Peng et al., 2019; Qin, 2022; Rashid et al., 2023; Zhang et al., 2021; Cao, 2023; Bai, 2022; Tang et al., 2023) have achieved impressive performance metrics, exceeding 99 %. Nevertheless, they often lack statistical significance testing. Indeed, incorporating additional practices can provide a more comprehensive and reliable evaluation of a model's performance, ensuring that reported results are not only impressive but also robust and statistically significant. One such technique is bootstrapping, which can be used to estimate the variability of performance metrics and assess their statistical significance. This method involves resampling the data with replacement to create numerous simulated samples, thereby enabling the estimation of confidence intervals and variability of the metrics. Another approach is to evaluate the models on independent external datasets to confirm their generalizability beyond the training data. This helps to ensure that the model performs well not only on the data it was trained on but also on new, unseen data. Additionally, applying statistical tests such as paired t-tests or Wilcoxon signed-rank tests can compare performance metrics between models, ensuring that observed improvements are statistically significant and not due to random chance. Furthermore, as suggested in (Jimenez-Mesa et al., 2023), non-parametric statistical inference could be useful to analyze the prediction certainty of different models and compare their level of significance. Finally, encouraging the adoption of standardized reporting guidelines and data harmonization can further improve the transparency and reproducibility of AD detection and diagnosis models. These practices will help build a more reliable foundation for transitioning these models from research environments to real-world clinical settings.

10. Conclusions

Alzheimer's Disease is a progressive neurodegenerative disorder that poses a significant challenge for healthcare systems due to its devastating impact on individuals, their families, and society. Early and accurate diagnosis of AD is critical for patient care and the development of new treatments. In this paper, we presented the different modalities used in AD detection and diagnosis followed by various ML and DL models adopted for this purpose. Then, we discussed the different criteria to consider to build ML or DL models and discussed the widely used multimodal data fusion approaches. Next, we presented the recent ML and DL models in the literature and highlighted their methodologies, data used, performances, and main findings. We finally discussed the major challenges in this field and the potential solutions to address them.

From this work we conclude the following points. 1) Current research is leveraging multimodal data sources, including imaging data (MRI, PET scans), genetic information, cognitive tests, and biological markers, to create a more comprehensive and holistic view of a patient's status. Studies have found that models using integrated data could predict AD with higher accuracy than those based on single data types. 2) Deep learning is showing promise in identifying intricate patterns in complex datasets. Advanced CNN-based models with attention modules and the RNN-based models could detect subtle neurological changes before AD symptoms appear. These techniques have significantly improved the sensitivity and specificity of early AD diagnosis. 3) The predictive models for detecting disease progression are crucial for patient care planning, setting up clinical trials, and developing new therapies. Using longitudinal data to train sophisticated algorithms can predict how, when, and where the disease will progress, which could be revolutionary for early intervention strategies. 4) Novel biomarkers discovery is another critical area to identify new potential biological markers that were not previously associated with AD. These biomarkers can significantly enhance the accuracy of early diagnosis and help monitor disease progression and response to treatment. 5) Interpretable and explainable models are of high significance to promote the diagnosis understanding and the applicability in clinical usage. 6) Finally, multicenter studies with federated learning techniques could be a key solution to data collection and ethical and data privacy considerations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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